

**REPRODUCIBLE RAMSEY ANALYSIS ACROSS
ASYMPTOTIC, FINITE, AND ALGORITHMIC REGIMES:
RETAINED SPINES, GAUSSIAN RESIDUALS,
PROOF-CARRYING BARRIERS, AND QUANTUM
SEARCH**

CHENG XIN

ABSTRACT. We present a unified reproducible account of four logically independent results in Ramsey theory. First, relative to version-pinned regularization, parameterized-book, and spine-compatibility interfaces and a locally certified six-stage off-diagonal rate theorem, an exact-diagonal correlation certificate yields

$$R(k, k) \leq (3.780685290)^{k+o(k)},$$

with certified unrounded base below 3.780685288379640114. The compact inequalities are enclosed by 512-bit ball arithmetic and replayed by four separately structured implementations sharing the same interval-arithmetic trust dependency. The positive-moment and tail transfer used there are proved locally. Second, after fixing the absolute source constant K , under pinned lower-interface items (S1), (S2), (S4), and (S5), with weighted propagation (S3) proved locally, a deterministic nonexchangeable triangular residual ledger supplies, for every sufficiently large fixed C , a positive source-relative term

$$\hat{H}_*(C) = \frac{1 + o_{C \rightarrow \infty}(1)}{64 \log C}$$

in the fixed-ratio lower-bound exponent. The threshold $C_0(K)$ is existential. The coefficient $1/64$ is a cap only within the specified scalar residual class, and no comparison across incompatible terminal normalizations is asserted. Third, for a content-addressed 100-vertex $R(3, 18)$ near miss, semantic CNF reconstruction and six checked DRAT refutations prove that its one-sided deletion repair radius under free additions is at least seven. The branch-1 exact-seven refinement proves four non-vacuous singleton consequences by four further checked DRAT refutations, excluding 87.8832136551...% of raw residual supports when combined with a degree cap. A checked common-relaxation model contains an independent 18-set. The exact-seven analysis further proves a maximal-triangle-free normal form and combines it with the complete four-bank mask union in a byte-audited CNF; its single bounded call returned only UNKNOWN. The exact-seven layer remains UNKNOWN; this local theorem does not imply $R(3, 18) \geq 101$. Fourth, coherent edge-oracle access to any graph on at least 4^{K-1} vertices permits a scale-aware quantum implementation of the constructive majority recursion which returns a K -clique or K -vertex independent set with error at most η using

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

queries. A random-Painter argument gives the classical randomized lower bound $\Omega(2^{(2-\sqrt{2})K})$, which does not prove a quantum-classical separation. An estimation-free size-biased recursion separately gives $O(2^K K^2 \log K \log(1/\eta))$ queries, admits a direct multicolour extension, and supplies a simple survival lemma checked by exact finite split-tree dynamic programs. Direct parameter substitution also exposes an unresolved conflict with a published $N^{1-o(1)}$ quantum-lower-bound remark; author clarification remains a submission gate. This fourth result changes the complexity of finding a guaranteed witness, not a Ramsey-number bound. The common contribution is an auditable separation of imported interfaces, analytic arguments, interval certificates, proof-carrying finite computations, and unresolved searches, with content-addressed artifacts and tiered reproduction entry points.

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1. INTRODUCTION AND MAIN RESULTS

For positive integers k, ℓ , the Ramsey number $R(k, \ell)$ is the least n such that every red–blue colouring of K_n contains a red K_k or a blue K_ℓ . The classical upper and lower exponential scales originate in the work of Erdős–Szekeres and Erdős [ES35, Erd47]. Modern improvements combine delicate analytic inequalities, asymptotic reductions, and computer-assisted certificates. In the finite regime, SAT solvers can explore highly constrained graph spaces, but a solver outcome becomes a theorem only after the graph-to-CNF reduction and a proof trace have both been checked.

This article places four results developed here in one manuscript. They concern different regimes and use different mathematics; their unity is a common claim-to-evidence discipline rather than a claimed common proof mechanism. The first two give conditional, source-relative refinements of asymptotic ledgers, the third establishes a local finite obstruction without improving a tabulated Ramsey number, and the fourth gives a quantum query algorithm for constructive Ramsey search without changing an extremal bound.

1.1. Diagonal upper bound. Part I combines a certified six-stage off-diagonal rate, an exact-diagonal correlation inequality, and the version-pinned regularization, parameterized-book, and spine-compatibility interfaces of Yang and Mao [YM26]. The positive-moment and tail transfer is proved locally. Under the source interfaces stated in source interface 4.2 and the certified rate theorem Theorem 5.1, it proves

$$R(k, k) \leq (3.780685290)^{k+o(k)},$$

with certified unrounded base less than 3.780685288379640114. This is an asymptotic, source-relative, computer-assisted theorem. It supplies neither an effective finite- k threshold nor a global optimum or priority claim.

1.2. Fixed-ratio lower bound. Part II assumes the version-pinned items (S1), (S2), (S4), and (S5) imported from the Gaussian construction of Hunter, Milojević and Sudakov and the exact cumulant input of Lin and Niu [HMS25, LN26]. The weighted reverse-propagation rule (S3) is proved locally from the indicator kernel, projection identities, and the tower property. After fixing the absolute source constant K , those four source items give an existential threshold $C_0 = C_0(K) > 1$ such that, for each fixed $C \geq C_0$,

$$\liminf_{\ell \rightarrow \infty} \frac{1}{\ell} \log R(\ell, \lfloor C\ell \rfloor) \geq -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C),$$

where

$$\hat{H}_*(C) = \frac{1 + o_{C \rightarrow \infty}(1)}{64 \log C} > 0.$$

The order of limits is part of the statement and proof: after fixing K and C , one takes $\ell \rightarrow \infty$, then sends the auxiliary cutoff $w \downarrow \omega_0$, and only afterwards studies the leading behaviour as $C \rightarrow \infty$. The threshold $C_0(K)$ is non-effective, and the theorem does not assert a strict comparison with every published final constant under a different normalization.

1.3. Finite proof-carrying barrier. Part III starts from the public 99-vertex construction of Nagda, Raghavan, and Thakurta [NRT26] and fixes a labelled, content-addressed one-vertex extension H on 100 vertices with 827 edges, no independent set of order 18, and one triangle. If additions outside $E(H)$ are free and only deletions from $E(H)$ are charged, semantic reconstruction and six DRAT refutations prove that every graph F on the same labelled vertex set that is triangle-free and satisfies $\alpha(F) < 18$ obeys $|E(H) \setminus E(F)| \geq 7$. Exact seven remains UNKNOWN. Within branch 1, four additional proof-checked singleton consequences force four named seed edges to be preserved. Together with a degree cap they exclude $87.8832136551 \dots \%$ of raw residual supports. A checked satisfying model proves that the common relaxation is nonempty, but the model contains an independent 18-set and is not a repair. The theorem is strengthened methodologically by a maximal-triangle-free existence normal form and a complete four-bank mask union. Their combined, byte-audited branch-1 formula has (639290) variables and (1972360) clauses; its one bounded call returned only UNKNOWN. The theorem is local to this seed and metric: it neither constructs a 100-vertex $(3, 18)$ -Ramsey graph nor implies $R(3, 18) \geq 101$.

1.4. Quantum constructive search. Part IV quantizes the elementary majority-neighbourhood recursion. For a graph on at least 4^{K-1} vertices, capped uniform search on implicit candidate predicates, conditional concentration, and a scale-aware error schedule find a K -clique or K -vertex independent set using

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

coherent edge queries with failure probability at most η . At $N = 2^n$ this finds order $\lfloor n/2 \rfloor + 1$ using $O(\sqrt{N} \log N \log(\log N/\eta))$ queries. A random-Painter/Yao argument gives a classical randomized lower bound $\Omega(2^{(2-\sqrt{2})K})$, still below the quantum upper bound. Accordingly, we claim a near-quadratic improvement over the standard explicit recursion, not a separation from every classical algorithm.

A second, estimation-free algorithm samples one uniform candidate and follows its edge colour, thereby choosing a branch in proportion to its size. Its survival induction gives $O(2^K K^2 \log K \log(1/\eta))$ queries and extends to every number of edge colours. Although its two-colour polynomial factor is

weaker than the scale-aware theorem, it isolates the simplest mechanism and shows that the method is not binary-specific.

The exact-model prior-art audit finds an unresolved incompatibility with the $N^{1-o(1)}$ quantum lower bound printed by Jain, Li, Robere, and Xun: substitution into their Ramsey reduction and the cited multi-collision theorem gives exponent at most $1/24$. The manuscript records the calculation but does not present it as an author-confirmed correction.

1.5. Organization. Section 2 states the evidence taxonomy and trust boundaries used throughout. Parts I–IV give the complete arguments and their result-specific verification records. The unified reproduction contract appears in Section 33; full technical records, proof ledgers, and artifact tables are retained in the appendices. Symbols introduced inside a Part are local to that Part unless explicitly declared otherwise. This convention prevents unrelated analytic functions, Gaussian potentials, and finite graphs from being conflated.

2. SOURCE INTERFACES AND EVIDENCE TAXONOMY

The four results have different logical dependencies. We therefore attach an evidence type to every principal statement instead of using “computer-assisted” as a single undifferentiated label.

Definition 2.1 (Evidence types). The manuscript uses the following five categories.

- (i) A *source interface* is either a version-pinned statement imported from an external or frozen local source or a separately identified extra hypothesis. Its justification is outside the current logical kernel unless reproduced in an appendix.
- (ii) An *analytic theorem* follows from written mathematical arguments once its named source interfaces are granted.
- (iii) An *interval-certified theorem* additionally depends on outward-rounded enclosures covering a stated continuum.
- (iv) A *proof-carrying SAT theorem* depends on a proved encoding, exhaustive case coverage, and independently replayed proof traces.
- (v) A *bounded observation* records a terminated computation without promoting SAT, UNSAT, or UNKNOWN beyond what its certificate establishes.

Table 1: Claims, evidence, and non-implications.

ID	Claim	Evidence and dependency	Required boundary
U0	Six-stage off-diagonal rate $U = F_6$, with $e^{U(1)} = 3.780698427796 \dots$	Repaired book induction and continuum interval certificates.	A locally replayed computer-assisted rate theorem, not a proof-assistant formalization.
U1	Exact-diagonal correlation certificate at the stated rational parameters.	Analytic ratio, separator and tail arguments plus 512-bit compact enclosures.	Search selected the point; the frozen checker, not the search grid, proves it.
U2	Conditional diagonal base 3.780685290.	U0, U1, and the pinned retained-spine interfaces.	Asymptotic and source-relative; no effective threshold, global optimum, or world-best claim.
L0– L3	Exact-merge rigidity and the controlled-residual addition $\hat{H}_*(C) = (1 + o(1))/(64 \log C)$.	Written argument conditional on pinned items (S1), (S2), (S4), (S5), with weighted propagation (S3) proved locally; arithmetic replay checks formulas only and is not the induction.	The local S3 proof uses the indicator kernel and tower property. Fix the source constant K and sufficiently large C , then take $\ell \rightarrow \infty$, $w \downarrow \omega_0$, and only afterwards $C \rightarrow \infty$; $1/64$ is a cap only in the specified scalar residual class.
F0– F1	The fixed seed has the stated graph properties and deletion repair radius at least seven.	Direct bitset checks, branch completeness, CNF semantics, and six replayed DRAT proofs.	Local seed and one-sided metric; no implication $R(3, 18) \geq 101$.
F2	The branch-1 degree cap excludes $\binom{808}{6} / \binom{826}{6} = 0.875813 \dots$ of raw residual deletion sets.	Neighbourhood-independence proof and pinned seed degrees.	A necessary local pruning condition, not a runtime ratio or Ramsey-number bound.

ID	Claim	Evidence and dependency	Required boundary
F3	The first three exact-seven runs and sixteen deletion-first follow-ups ended UNKNOWN.	Bounded solver logs, an independently checked oracle-order reconstruction, and a sanitized endpoint summary.	Reverse ordering removed two apparent separator stalls, but seven repaired gates then reached their global walls while still in the master under the tested configurations; neither existence nor nonexistence evidence, and two solver-trusted fixed-deletion UNSAT endpoints have no checked trace.
F4	Four branch-1 singleton consequences force four named seed edges to be preserved and, with F2, exclude 87.8832136551...% of raw residual supports.	Four independently reconstructed singleton CNFs and replayed DRAT traces, plus a checked satisfying model of the common relaxation.	The SAT model contains an explicit independent 18-set; the consequences are non-vacuous necessary conditions, not branch closure, an exact-seven resolution, or a global Ramsey-number bound.
Q0	A scale-aware implicit-set quantum algorithm finds a homogeneous K -set in every graph on at least 4^{K-1} vertices using $O(2^K K \log(K/\eta))$ coherent edge queries.	Written oracle-algorithm proof: capped symmetric search, conditional Hoeffding, an exact affine recurrence, and a geometric query sum.	A worst-case query upper bound, not a numerical Ramsey bound, physical implementation, or separation from all randomized classical algorithms.
Q1	An estimation-free size-biased recursion gives $O(2^K K^2 \log K \log(1/\eta))$ queries and extends to q colours.	Written survival induction, exact nesting, fixed-cap amplification, and finite exact split-tree diagnostics.	The diagnostics check the formula but do not replace its proof; the multicolour statement is algorithmic rather than a competitive extremal bound.

ID	Claim	Evidence and dependency	Required boundary
Q2	Randomized classical query complexity is at least $\Omega(2^{(2-\sqrt{2})K})$ at $M = 4^{K-1}$.	Yao’s principle applied to the version-cited random-Painter theorem, with a written treatment of unqueried output edges.	The exponent is below 1/2 and therefore proves no quantum–classical separation.

Shared trust is not independence. The upper-bound replays use separately structured implementations but share Arb or python-flint containment semantics and the same mathematical specification. The finite proof traces are checked independently of the SAT solver, but their theorem still depends on the encoding and case-cover proofs. The lower arithmetic checker replays identities and asymptotic margins; it does not replace the reverse-induction and extraction proof.

Version and hash discipline. All imported source snapshots, frozen parameter files, graph matrices, CNF instances, proof traces, and checkers used as evidence are identified by version or SHA-256 digest. A digest authenticates bytes, not mathematics; the relevant semantic implication is stated separately in each Part.

Part 1. A retained-spine diagonal upper bound

3. DIAGONAL UPPER BOUND: THEOREM AND PROOF ARCHITECTURE

For positive integers k, ℓ , let $R(k, \ell)$ be the least n such that every red–blue colouring of K_n contains either a red K_k or a blue K_ℓ . The classical argument of Erdős and Szekeres gives $R(k, k) \leq 4^{k+o(k)}$ [ES35]. Campos, Griffiths, Morris and Sahasrabudhe obtained the first exponential improvement of this base [CGMS23]. Gupta, Ndiaye, Norin and Wei subsequently reorganized the book method and stated the concrete bound $R(k, k) \leq 3.8^{k+o(k)}$ [GNNW24]. Yang and Mao introduced a retained-spine refinement in a multicolour setting [YM26].

This part combines two ingredients. The first is a certified, symmetric off-diagonal rate

$$\log R(k, \ell) \leq kU(\ell/k) + o(k), \quad 1 \leq \ell \leq k, \quad (\text{U1.1})$$

where $U = F_6$ is given explicitly in Section 5. The second is a new correlation certificate designed for the retained-spine transfer. The certificate is analytic except on compact intervals, where every inequality is proved by outward-rounded Arb balls [Joh17].

3.1. Main result. We state the result with its imported interfaces visible. The exact interfaces are recorded in Section 4; they are theorem statements from Yang–Mao together with the reviewed rate theorem (U1.1), rather than unstructured assumptions about an implementation.

Theorem 3.1 (Conditional main theorem). *Assume the version-pinned regularization, parameterized-book, and spine-compatibility interfaces stated in source interface 4.2, and the certified rate theorem Theorem 5.1. The positive-moment and tail implication used to establish the correlation input are proved locally in Section 6. Then*

$$R(k, k) \leq \exp\left((U(1) - 3.4754 \cdot 10^{-6})k + o(k)\right) \leq (3.780685290)^{k+o(k)}.$$

The certified unrounded upper base is less than 3.780685288379640114.

This part is a source-relative proof package: every imported statement is named, every new inequality is proved here, and every numerical step is replayable by separately structured implementations. Those implementations share Arb’s interval-containment semantics as a trust dependency. No finite value of the asymptotic threshold is claimed.

3.2. The exact-diagonal mechanism. Fix

$$u_0 = \frac{1235783}{500000}, \quad L = u_0^3, \quad a = e^{-u_0},$$

and define

$$E(z) = \sum_{n \geq 0} \frac{z^n}{(3n)!}, \quad H(z) = 1 + aE(z)^2, \quad G(z) = \frac{H(z) - H(-z)}{2}.$$

The correlation form is

$$F(x, y) = G(x)H(y) + G(y)H(x).$$

Our central observation is that the complete two-dimensional problem reduces to the exact identity

$$F(z, z) = H(z)^2 - H(z)H(-z), \quad z \geq 0,$$

not merely to the larger square difference $H(z)^2 - H(-z)^2$. This improves the compact envelope while preserving a simple analytic tail.

3.3. Proof architecture. The proof has four logically separate layers.

- (i) Section 5 states the off-diagonal rate and records its exact polynomial and verification boundary.
- (ii) Section 6 proves the ratio/seperator, the two-dimensional reduction, the exact diagonal envelope and the moment/tail budget.
- (iii) Section 7 proves the retained-spine max–min transfer and inserts the certified parameters.
- (iv) Section 8 describes continuum coverage, independent implementations and the content-addressed evidence records.

The complete repaired book induction and the frozen rate record are included in the appendices. Search scripts are not part of the proof; they only chose the exact rational data subsequently verified from scratch.

4. RAMSEY RATES AND IMPORTED INTERFACES

4.1. Symmetric homogeneous rate. Let $U : [0, 1] \rightarrow \mathbb{R}$ be continuous with $U(0) = 0$. Define its symmetric homogeneous extension by

$$\hat{U}(x, y) = \begin{cases} \max\{x, y\} U\left(\frac{\min\{x, y\}}{\max\{x, y\}}\right), & \max\{x, y\} > 0, \\ 0, & x = y = 0. \end{cases} \quad (\text{U2.1})$$

For residual problems only, we extend the notation by $R(a, b) = 1$ when $\min\{a, b\} \leq 0$; this means that no further page vertices are needed in at least one colour. It is not a new convention for positive Ramsey numbers.

Lemma 4.1 (Uniform homogeneous extension). *Assume in addition that $U \geq 0$. Suppose (U1.1) holds uniformly for $1 \leq \ell \leq k$, and the same bound is available after exchanging the two colours. Then, uniformly for $x, y \in [0, 1]$,*

$$\log R(\lfloor xk \rfloor, \lfloor yk \rfloor) \leq k\hat{U}(x, y) + o(k). \quad (\text{U2.2})$$

Proof. The extension $\hat{U}$ is continuous on the square (continuity at the origin uses $U(0) = 0$), hence uniformly continuous. Fix $\epsilon > 0$, and choose K_0 from the uniform epsilon formulation of (U1.1). For $x \geq y$, put $k' = \lfloor xk \rfloor$ and $\ell' = \lfloor yk \rfloor$. If $k' \geq K_0$ and $\ell' \geq 1$, then

$$\log R(k', \ell') \leq k'U(\ell'/k') + \epsilon k' = k\hat{U}(k'/k, \ell'/k) + \epsilon k.$$

Uniform continuity and the $O(k^{-1})$ perturbation of the normalized coordinates replace the last $\hat{U}$ by $\hat{U}(x, y) + o(1)$, uniformly. If $k' < K_0$, only finitely many positive Ramsey numbers occur, so their logarithms are $O_{K_0}(1) = o(k)$; nonnegativity of $\hat{U}$ permits the same upper bound. If one floored coordinate is zero, the residual convention gives logarithmic cost zero. The case $y \geq x$ follows by symmetry. Since ϵ is arbitrary, these cases give one uniform $o(k)$. $\square$

4.2. Yang–Mao interfaces. The following package is extracted from the v1 source of Yang–Mao [YM26]. Its source member and line map are frozen in Appendix A. We state the complete specialization used later, so that the conditional theorem is falsifiable without guessing what the words “book interface” mean.

For completeness, we fix the precise meaning of the correlation input. We write $\mathcal{G}_2^{(3)}(\beta, C)$ when the following assertion holds: for every finite set $\mathcal{X}$, independent identically distributed $U, U' \in \mathcal{X}$, real Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, and maps $\sigma_i : \mathcal{X} \rightarrow \mathcal{H}_i$, if $Z_i = \langle \sigma_i(U), \sigma_i(U') \rangle$, then there are $i \in \{1, 2\}$ and $\lambda \geq -1$ such that

$$\mathbb{P}(Z_i \geq \lambda, Z_{3-i} \geq -1) \geq \beta \exp\left(-C(\lambda + 1)^{1/3}\right). \quad (\text{U2.2a})$$

For $0 < \beta \leq 1$, $C > 0$, and parameters $p, \delta, \lambda_0 > 0$, put

$$L_\delta = \log(1/\delta), \quad L_p = \log(1/p), \quad \rho_\beta = \log(2/\beta),$$

$$\Pi = \frac{3\delta}{p} + \frac{6L_\delta L_p}{\lambda_0}, \quad q = L_p + \Pi, \quad (\text{U2.3})$$

and

$$\Xi = 2\rho_\beta + 4C\lambda_0^{1/3} + \frac{12CL_\delta}{\lambda_0^{2/3}}. \quad (\text{U2.4})$$

Source interface 4.2 (Retained-spine source interface). Fix $0 < \eta$, $0 < p < 1/2 - \eta$, $0 < \delta \leq \min\{p/4, 1/4\}$, $\lambda_0 \geq \max\{2, 6\log(1/\delta)\}$, and $\tau > 0$. Suppose the correlation property $\mathcal{G}_2^{(3)}(\beta, C)$ holds. The version-pinned Yang–Mao specialization supplies the following conclusions for all sufficiently large k .

- (a) For every positive integer N , every red–blue colouring of K_N admits pairwise disjoint sets S_R, S_B, W such that S_R is a red clique, S_B is a blue clique, every edge from S_R to W is red, and every edge from S_B to W is blue. Writing $s_i = |S_i|$, one has

$$|W| \geq \left(\frac{1+\eta}{2}\right)^{s_R+s_B} N \quad (\text{U2.5})$$

and every $w \in W$ has at least $(1/2 - \eta)|W| - 1$ red neighbours and at least that many blue neighbours inside W .

- (b) Let t, m be positive integers with $t \geq \lambda_0 \delta^{-1/2}$. If every $w \in W$ has at least $p|W|$ neighbours of each colour inside W , and

$$|W| \geq p^{-t} e^{\Pi t} m, \quad |W| \geq 4t e^{2t\Xi}, \quad (\text{U2.6})$$

then the parameterized-book theorem, applied with $X = Y_R = Y_B = W$, returns a colour $i \in \{R, B\}$, a monochromatic t -clique $T \subseteq W$ of colour i , and $P_0 \subseteq W \setminus T$, $|P_0| \geq m$, such that every edge from T to P_0 has colour i .

- (c) The compatibility conclusion is the following exact residual statement. If $i = R$, then a red K_{k-s_R-t} in P_0 (when this order is positive) extends with $S_R \cup T$, while a blue K_{k-s_B} extends with S_B . If $i = B$, interchange the colours, so the two residual orders are $k - s_R$ and $k - s_B - t$. A nonpositive residual order means that the retained spines already contain the required K_k .

For the proof below the exact correlation input is

$$\beta = \frac{330867}{500000}, \quad C = \frac{12366348252219}{12500000000000}. \quad (\text{U2.7})$$

The point of stating source interface 4.2 is auditability. The correlation property in (U2.7) is proved in this paper; the regularization and book compatibility are imported. The source line mapping and pinned archive hash are included in the reproducibility record.

4.3. **A general retained-spine transfer.** Put

$$c_\eta = \log \frac{2}{1 + \eta}.$$

For a normalized spine vector $\sigma = (\sigma_R, \sigma_B) \in [0, 1]^2$, define

$$D(\sigma) = c_\eta(\sigma_R + \sigma_B) + \widehat{U}(1 - \sigma_R, 1 - \sigma_B). \quad (\text{U2.8})$$

Here q and Ξ are the explicit quantities in (U2.3)–(U2.4). Define

$$M_\tau(\sigma) = \max\{\widehat{U}((1 - \sigma_R - \tau)_+, 1 - \sigma_B), \widehat{U}(1 - \sigma_R, (1 - \sigma_B - \tau)_+)\}, \quad (\text{U2.9})$$

$$P(\sigma) = c_\eta(\sigma_R + \sigma_B) + \tau q + M_\tau(\sigma), \quad (\text{U2.10})$$

$$Q(\sigma) = c_\eta(\sigma_R + \sigma_B) + 2\tau\Xi, \quad (\text{U2.11})$$

$$B(\sigma) = \max\{P(\sigma), Q(\sigma)\}. \quad (\text{U2.12})$$

Finally set

$$C_* = \max_{\sigma \in [0, 1]^2} \min\{D(\sigma), B(\sigma)\}. \quad (\text{U2.13})$$

Theorem 4.3 (Retained-spine transfer). *Under source interface 4.2 and the uniform rate (U2.2),*

$$R(k, k) \leq \exp(C_* k + o(k)).$$

Proof. Let $N = \lceil e^{(C_* + \varepsilon)k} \rceil$ and apply regularization. If either preliminary spine already has order at least k , there is nothing to prove. Otherwise write $s_i = \sigma_i k + O(1)$. From (U2.5),

$$\log |W| \geq (C_* + \varepsilon)k - c_\eta(s_R + s_B) + o(k). \quad (\text{U2.14})$$

At this σ , choose the smaller of the direct and book branches.

If $D(\sigma) \leq B(\sigma)$, then $D(\sigma) \leq C_*$. Equations (U2.2), (U2.8) and (U2.14) imply $|W| \geq R(k - s_R, k - s_B)$. The resulting red or blue clique extends with the corresponding preliminary spine.

If $B(\sigma) < D(\sigma)$, then both $P(\sigma) \leq C_*$ and $Q(\sigma) \leq C_*$. The former supplies the page size needed for either possible book colour, using (U2.2) and (U2.9); the latter supplies the common reservoir threshold. More explicitly, with $t = \lfloor \tau k \rfloor$ and m the larger of the two residual Ramsey thresholds, the two strict exponent inequalities give exactly (U2.6). Since $p < 1/2 - \eta$ and the reservoir is exponential, the additive -1 in the degree conclusion is absorbed; fixed $\tau > 0$ also gives $t \geq \lambda_0 \delta^{-1/2}$. The parameterized-book interface produces the t -spine and a page set of size at least m , and compatibility extends the appropriate page clique with the retained spines to a monochromatic K_k . The fixed ε -slack absorbs every uniform $o(k)$ term. $\square$

5. THE CERTIFIED OFF-DIAGONAL RATE

Set

$$h(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda$$

and

$$\begin{aligned} P_6(\lambda) = & -0.2500000000000000\lambda + 0.019891840059418\lambda^2 \\ & - 0.012275954247190\lambda^3 + 0.144110816398083\lambda^4 \\ & + 0.006277913420654\lambda^5 - 0.066101623491668\lambda^6 \\ & - 0.002287675993602\lambda^7 - 0.058238059505066\lambda^8 \\ & + 0.030675864198693\lambda^9 + 0.052472201910597\lambda^{10} \\ & + 0.043300454861853\lambda^{11} - 0.042529975074653\lambda^{12} \\ & - 0.055263639426635\lambda^{13} + 0.036695886931268\lambda^{14}. \end{aligned} \quad (\text{U3.1})$$

All displayed decimals in (U3.1) are exact. Define

$$U(\lambda) = F_6(\lambda) = h(\lambda) + P_6(\lambda)e^{-\lambda}. \quad (\text{U3.2})$$

Theorem 5.1 (Certified rate, conditional on the stated trust boundary).
For $1 \leq \ell \leq k$, uniformly in the ratio $\lambda = \ell/k$,

$$\log R(k, \ell) \leq kU(\lambda) + o(k), \quad (\text{U3.3})$$

and the same statement holds after exchanging k and ℓ . Moreover

$$e^{U(1)} = 3.7806984277961236987518816497522162\dots \quad (\text{U3.4})$$

Proof. The certificate schema, all continuum obligations, and the theorem-to-rate implication are stated and proved in Proposition C.3. The exact-linked replay accepts all six certificate files, proves the elementary first prior in Lemma C.2, and verifies exact target-to-prior equality at the other five handoffs. Its sixth report gives $F_6(1) = 1.3299087618219930723187812632\dots$; outward-rounded exponentiation gives (U3.4). Outside the version-pinned Yang–Mao interfaces in source interface 4.2, the repaired BookCor induction is proved locally in Appendix B; the remaining machine trust is Arb’s interval-containment semantics. $\square$

The proof is an exact-linked six-stage descent built on the repaired book-induction lemma. At each stage an interval certificate verifies the prior concavity, both Ramsey-region orientations, the main exponent inequality and the complete small-ratio interval. The first three stages use 65,536 cells; the last three use 131,072 cells. The sixth-stage minimum main slack is $4.81847 \cdot 10^{-5}$, while an independently implemented direct exponent check gives a minimum standard-orientation margin $1.06281 \cdot 10^{-6}$.

For the reader who wishes to separate mathematics from computation, the logical content is as follows.

- (i) The combinatorial book induction converts a strict interior point of the asymptotic Ramsey region into a one-step rate improvement.

- (ii) The analytic descent lemma is degree-independent: it requires only a concave prior rate and strict inequalities on the continuum.
- (iii) Every prior polynomial equals the preceding target polynomial as an exact decimal tuple; no refitting occurs between stages.
- (iv) Arb intervals enclose the continuum, not a sampled mesh. Search meshes are used only to propose a target.

The complete standalone book proof, certificate-correctness theorem, and stage-six record appear in Appendices B and C. The six JSON certificates and their hashes are part of the machine-readable evidence package.

6. THE EXACT-DIAGONAL CORRELATION CERTIFICATE

We now prove the new analytic input. Recall

$$u_0 = \frac{1235783}{500000}, \quad L = u_0^3, \quad a = e^{-u_0},$$

and

$$E(z) = \sum_{n \geq 0} \frac{z^n}{(3n)!}, \quad H(z) = 1 + aE(z)^2, \quad G(z) = \frac{H(z) - H(-z)}{2},$$

$$F(x, y) = G(x)H(y) + G(y)H(x).$$

6.1. The bad-event separator. For $u \geq 0$, define the cubic filters

$$P(u) = \frac{e^u + 2e^{-u/2} \cos(\sqrt{3}u/2)}{3}, \quad (\text{U4.1})$$

$$N(u) = \frac{e^{-u} + 2e^{u/2} \cos(\sqrt{3}u/2)}{3}. \quad (\text{U4.2})$$

Then $E(u^3) = P(u)$ and $E(-u^3) = N(u)$. Put

$$\sigma_0 = 10^{-8}, \quad T = 2 + 2\sigma_0, \quad \sigma_* = \frac{1 + 2\sigma_0}{2} = \frac{50000001}{100000000}. \quad (\text{U4.3})$$

Lemma 6.1 (Half-line ratio). *For every $u \geq u_0$,*

$$\frac{1 + aP(u)^2}{1 + aN(u)^2} > T. \quad (\text{U4.4})$$

Proof. Let $R(u) = (1 + aP(u)^2)/(1 + aN(u)^2)$. Its derivative has the sign of

$$W(u) = PP'(1 + aN^2) - NN'(1 + aP^2). \quad (\text{U4.5})$$

An outward-rounded enclosure proves $R(u_0) - T > 2.6595 \cdot 10^{-7}$ and $W > 0$ on 65,536 rational cells covering $[u_0, 2.9]$.

For $u \geq 2.9$, use

$$P(u) \geq \frac{e^u - 2e^{-u/2}}{3}, \quad |N(u)| \leq \frac{2e^{u/2} + e^{-u}}{3}.$$

After clearing the positive denominator, it suffices to prove positivity of

$$B_T(u) = e^{2u} - 4Te^u - 4e^{u/2} - 4Te^{-u/2} + 4e^{-u} - Te^{-2u}.$$

At $u = 2.9$, interval arithmetic proves $aB_T - 9(T - 1) > 0$ and the separate base gate

$$D(2.9) > 506.45 > 0.$$

Moreover

$$B'_T(u) \geq D(u) := 2e^u(e^u - 2T) - 2e^{u/2} - 4e^{-u},$$

and

$$D'(u) = 4e^{2u} - 4Te^u - e^{u/2} + 4e^{-u} > 0.$$

Indeed, for $u \geq 2.9$ one has $e^u - T > 1$, and therefore

$$D'(u) > 4e^u - e^{u/2} > 0.$$

Thus $D(u) > 0$ and then $B'_T(u) > 0$ throughout $[2.9, \infty)$. Together with the endpoint gate, this proves that B_T , and therefore $R - T$, remain positive. $\square$

Lemma 6.2 (Separator). *If one of A_1, A_2 is less than -1 , then*

$$F(LA_1, LA_2) < -\sigma_*. \quad (\text{U4.6})$$

Proof. Let $x = -t$ with $t \geq L$, put $A = H(-t)$, and set $R = H(t)/H(-t)$. The exact identity

$$F(-t, y) = \frac{A}{2}((2 - R)H(y) - H(-y)) \quad (\text{U4.7})$$

holds for every real y . By Lemma 6.1, $R > 2 + 2\sigma_0$, whereas $A, H(y), H(-y) \geq 1$. Thus the right side of (U4.7) is less than $-(1 + 2\sigma_0)/2 = -\sigma_*$. Exchange the coordinates if necessary. $\square$

6.2. Complete two-dimensional reduction.

Lemma 6.3 (Diagonal reduction). *If $x, y \geq -L$ and $m = \max\{x, y, 0\}$, then*

$$F(x, y) \leq F(m, m). \quad (\text{U4.8})$$

Proof. The series for H has nonnegative coefficients, so H , and therefore its odd part G , are nondecreasing on the positive half-line. If $x = -v^3 \in [-L, 0]$, the root-filter formula gives

$$|E(-v^3)| \leq e^{v/2}, \quad H(x) \leq 1 + e^{-u_0}e^v \leq 2. \quad (\text{U4.9})$$

Assume $x \leq y$. If $y < 0$, both odd parts are negative and hence $F(x, y) < 0 = F(0, 0)$. If $x < 0 \leq y$, discard the nonpositive summand and use (U4.9):

$$F(x, y) \leq G(y)H(x) \leq 2G(y) \leq 2G(y)H(y) = F(y, y).$$

If $0 \leq x \leq y$, monotonicity bounds both summands by $G(y)H(y)$. This exhausts all sign patterns and their boundaries. $\square$

On the diagonal we retain the exact identity

$$F(z, z) = 2G(z)H(z) = H(z)^2 - H(z)H(-z), \quad z \geq 0. \quad (\text{U4.10})$$

6.3. Compact interval and analytic tail. Let $z = u^3$ and $w = (u^3 + u_0^3)^{1/3}$. Define

$$D_0 = \frac{88053}{10^9}.$$

Lemma 6.4 (Exact diagonal envelope). *For every $u \geq 0$,*

$$\frac{H(u^3)^2 - H(u^3)H(-u^3)}{e^{4w}} < D_0. \quad (\text{U4.11})$$

Proof. The compact interval $[0, 20]$ is covered by 131,072 exact rational cells. On each cell a 512-bit Arb evaluation encloses the left side of (U4.11); the minimum slack is greater than $4.2687 \cdot 10^{-6}$.

For $u \geq 20$, use $H(-u^3) \geq 1$ and write $H(u^3) = 1 + aP(u)^2$. Then

$$H(u^3)^2 - H(u^3)H(-u^3) \leq aP(u)^2 + a^2P(u)^4.$$

Since

$$|P(u)| \leq \frac{e^u}{3}(1 + 2e^{-3u/2})$$

and $w \geq u$, the normalized expression is at most

$$\frac{a}{9}e^{-2u}(1 + 2e^{-3u/2})^2 + \frac{a^2}{81}(1 + 2e^{-3u/2})^4. \quad (\text{U4.12})$$

Both terms decrease. At $u = 20$, (U4.12) is below $8.805216326983 \cdot 10^{-5}$, leaving strict slack greater than $8.3673 \cdot 10^{-10}$; its limiting value $a^2/81$ has the same strict ordering. This proves the entire half-line. $\square$

Combining Lemmas 6.3 and 6.4, if $A_1, A_2 \geq -1$ and $M = \max(A_1, A_2)$, then

$$F(LA_1, LA_2) < D_0 \exp(4u_0(M + 1)^{1/3}). \quad (\text{U4.13})$$

If $M \leq 0$, the left side is nonpositive by the sign case in Lemma 6.3, whereas the right side is positive. If $M > 0$, take $u = u_0 M^{1/3}$; then the diagonal-reduction parameter is $m = LM = u^3$ and the w in Lemma 6.4 equals $u_0(M + 1)^{1/3}$, proving the displayed inequality.

6.4. Exact moment/tail budget. The final passage from the separator to a correlation tail is short but proof-critical, so we give it in full rather than treating it as an unnamed interface.

Lemma 6.5 (Positive-moment expectation). *For any finite set $\mathcal{X}$, independent identically distributed $U, U' \in \mathcal{X}$, real Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, and maps $\sigma_i : \mathcal{X} \rightarrow \mathcal{H}_i$, put $Z_i = \langle \sigma_i(U), \sigma_i(U') \rangle$. Then*

$$\mathbb{E}F(LZ_1, LZ_2) \geq 0. \quad (\text{U4.14})$$

Proof. The Taylor coefficients of E and H are nonnegative, and taking the odd part of H merely retains its odd-degree coefficients. Hence every bivariate Taylor coefficient of $F(x, y) = G(x)H(y) + G(y)H(x)$ is nonnegative. For nonnegative integers r, s , tensorization gives

$$\mathbb{E}(Z_1^r Z_2^s) = \|\mathbb{E}(\sigma_1(U)^{\otimes r} \otimes \sigma_2(U)^{\otimes s})\|^2 \geq 0.$$

Because $\mathcal{X}$ is finite, (Z_1, Z_2) has bounded support; the entire Taylor series is therefore absolutely integrable and may be averaged term by term. Summing the nonnegative terms proves (U4.14). $\square$

Take

$$\beta = \frac{330867}{500000}, \quad \varepsilon = \frac{6893}{10^7}.$$

Exact rational arithmetic gives

$$\sigma_*(1 - \beta) - D_0\beta(1 + 2/\varepsilon) = \frac{39437634699447}{3446500000000000000} > 10^{-5}. \quad (\text{U4.15})$$

Proposition 6.6 (Certified correlation tail). *The exact separator, envelope, and budget above imply*

$$\mathcal{G}_2^{(3)}\left(\frac{330867}{500000}, 4u_0(1 + \varepsilon)\right).$$

Proof. Fix data from the definition (U2.2a), put $\mathcal{E} = \{Z_1 \geq -1, Z_2 \geq -1\}$, $M = \max\{Z_1, Z_2\}$, $Y = ((M+1)_+)^{1/3}$, and $c = 4u_0$. By Lemmas 6.2 and 6.5 and (U4.14), while (U4.13) controls the good event,

$$0 \leq \mathbb{E}F(LZ_1, LZ_2) < D_0\mathbb{E}(e^{cY}\mathbf{1}_{\mathcal{E}}) - \sigma_*\mathbb{P}(\mathcal{E}^c). \quad (\text{U4.16})$$

If $\mathbb{P}(\mathcal{E}) \geq \beta$, choose $\lambda = -1$ and either coordinate in (U2.2a). Otherwise suppose, for a contradiction, that the claimed correlation conclusion fails for both coordinates at every $\lambda \geq -1$. For $v \geq 0$, taking $\lambda = v^3 - 1$ and using the union bound gives

$$\mathbb{P}(\mathcal{E} \cap \{Y \geq v\}) < 2\beta e^{-c(1+\varepsilon)v}. \quad (\text{U4.17})$$

The layer-cake identity and Tonelli's theorem now give

$$\begin{aligned} \mathbb{E}(e^{cY}\mathbf{1}_{\mathcal{E}}) &= \mathbb{P}(\mathcal{E}) + c \int_0^\infty e^{cv} \mathbb{P}(\mathcal{E} \cap \{Y > v\}) dv \\ &< \beta + 2\beta c \int_0^\infty e^{-c\varepsilon v} dv = \beta(1 + 2/\varepsilon). \end{aligned}$$

Substitution into (U4.16) makes its right-hand side at most $D_0\beta(1 + 2/\varepsilon) - \sigma_*(1 - \beta) < 0$, by (U4.15). This contradiction proves the proposition. $\square$

Finally,

$$4u_0(1 + \varepsilon) = \frac{12366348252219}{1250000000000} = 9.8930786017752,$$

which is the exact handoff asserted in (U2.7).

7. THE RETAINED-SPINE CERTIFICATE

The exact outer tuple is

$$\begin{aligned} \eta &= 0.02868896, & p &= 0.47130887, & \delta &= 0.000053863, \\ \lambda_0 &= 13233, & \tau &= 0.00069386, & \Delta &= 0.0000034754. \end{aligned} \quad (\text{U5.1})$$

Every terminating decimal in (U5.1) is interpreted as a rational number.

The functions D, B and C_* are those of Theorem 4.3. Direct interval evaluation of the full square would introduce unacceptable variable dependence. The certificate instead uses the geometry of the ordered wedge. Concavity of U fixes the monotonicity of each page branch; the red and blue page expressions reduce to one-dimensional boundary intervals, and the diagonal reservoir reduces to an endpoint inequality. Every reduction is analytic before numerical evaluation.

Proposition 7.1 (Outer certificate). *For the exact tuple (U5.1) and the correlation input proved in Section 6,*

$$C_* \leq U(1) - \Delta. \quad (\text{U5.2})$$

The certified lower margins are as follows.

<i>Inequality</i>	<i>lower margin</i>
<i>degree gate</i>	$1.5056 \cdot 10^{-9}$
<i>red page</i>	$5.1006 \cdot 10^{-9}$
<i>blue page</i>	$3.4825 \cdot 10^{-6}$
<i>diagonal reservoir</i>	$6.0468 \cdot 10^{-6}$

Proof. Write $x = \sigma_R$, $y = \sigma_B$, and by symmetry first assume $0 \leq x \leq y \leq 1$. Put

$$A = U(1) - 2c_\eta, \quad E = U'(1) - c_\eta.$$

The certificate proves $0 < A < E$. At $x = y = 1$, one has $U(1) - D(1, 1) = A > \Delta$ because $r_d = \Delta/A < 1$, so the direct branch already applies. At every other point of the ordered triangle, $a = 1 - x \geq b = 1 - y$ has $a > 0$, and concavity at one gives

$$aU(b/a) \leq aU(1) - (a - b)U'(1),$$

and hence

$$U(1) - D(x, y) \geq xA + (y - x)E. \quad (\text{U5.3})$$

Thus the direct branch applies outside the closed triangle

$$\mathcal{W} = \{0 \leq x \leq y : xA + (y - x)E \leq \Delta\}.$$

Set

$$r_a = \Delta/E, \quad r_d = \Delta/A, \quad s = 1 - A/E.$$

The certified order $0 < r_a < \tau < r_d < 1$, $r_d + \tau < 1$, and $0 < s < 1$ shows that $\mathcal{W}$ has vertices $(0, 0), (0, r_a), (r_d, r_d)$, with sloping side $y = r_a + sx$. In particular, throughout $\mathcal{W}$,

$$0 \leq x \leq y \leq r_d, \quad 0 \leq y - x \leq r_a < \tau. \quad (\text{U5.3a})$$

It remains to bound both possible page colours and the reservoir on $\mathcal{W}$. For the first page orientation put

$$z = \frac{1 - \tau - x}{1 - y}, \quad P_R = c_\eta(x + y) + \tau q + (1 - y)U(z).$$

The order above and $r_d + \tau < 1$ give the complete derivative domain

$$0 < 1 - \frac{\tau}{1 - r_d} \leq z \leq \frac{1 - \tau}{1 - r_a} < 1. \quad (\text{U5.3b})$$

Since $U'' < 0$ on $(0, 1]$, $U'(z) \geq U'(1) = E + c_\eta > c_\eta$. Also, for $\phi(z) = U(z) - zU'(z)$, $\phi'(z) = -zU''(z) > 0$, and hence $\phi(z) \leq \phi(1) = c_\eta + A - E < c_\eta$. Therefore its coordinate derivatives on all of $\mathcal{W}$ are

$$\partial_x P_R = c_\eta - U'(z) < 0, \quad \partial_y P_R = c_\eta - U(z) + zU'(z) > 0.$$

It therefore suffices to follow the sloping side. Its derivative is

$$\mathcal{B}(z) = c_\eta - U'(z) + s\{c_\eta - U(z) + zU'(z)\},$$

where $\mathcal{B}'(z) = (sz - 1)U''(z) > 0$, because $0 < s, z < 1$. Along the side, z decreases from $z_a = (1 - \tau)/(1 - r_a)$; the 512-bit enclosure gives $\mathcal{B}(z_a) < -3.6952685 \cdot 10^{-4}$. Hence P_R is maximized at $(0, r_a)$, where

$$P_R(0, r_a) = c_\eta r_a + \tau q + (1 - r_a)U\left(\frac{1 - \tau}{1 - r_a}\right). \quad (\text{U5.4})$$

For the other page orientation, put

$$w = \frac{1 - \tau - y}{1 - x}, \quad P_B(x, y) = c_\eta(x + y) + \tau q + (1 - x)U(w).$$

Here $0 < w < 1$ on $\mathcal{W}$: positivity follows from $y \leq r_d < 1 - \tau$, and $w < 1$ follows from $y \geq x$ and $\tau > 0$. Thus $\partial_y P_B = c_\eta - U'(w) < 0$, so the maximum reduces to $y = x$. There it equals

$$P_B(x, x) = 2c_\eta x + \tau q + (1 - x)U\left(1 - \frac{\tau}{1 - x}\right).$$

For $0 \leq x \leq r_d$, the variable $v = \tau/(1 - x)$ lies in

$$\tau \leq v \leq \frac{\tau}{1 - r_d} < 1,$$

so every derivative below is evaluated with $1 - v \in (0, 1)$. The diagonal derivative is

$$\Gamma(v) = 2c_\eta - U(1 - v) - vU'(1 - v), \quad \Gamma'(v) = vU''(1 - v) < 0.$$

Thus the maximum occurs at $x = 0$; the enclosure gives $\Gamma(\tau) < -1.8476200 \cdot 10^{-4}$ and

$$P_B(0, 0) = \tau q + U(1 - \tau). \quad (\text{U5.5})$$

Finally, $Q(x, y) = c_\eta(x + y) + 2\tau\Xi$ is coordinatewise increasing, so its maximum on $\mathcal{W}$ is

$$Q(r_d, r_d) = 2c_\eta r_d + 2\tau\Xi. \quad (\text{U5.6})$$

Subtracting (U5.4)–(U5.6) from $U(1) - \Delta$ gives, respectively, the red-page, blue-page, and reservoir margins in the table. The symmetric triangle $y \leq x$ exchanges the two colours and adds no new inequality. The same run verifies the scalar source gates, including $\tau(1/2 - \eta - p) > 1.5056 \cdot 10^{-9}$, and the exact equality of the inner and outer correlation constants. Exact parameters, interval outputs, and checker identities are recorded in Appendix A. $\square$

Proof of Theorem 3.1. By Theorem 4.3 and Proposition 7.1,

$$R(k, k) \leq \exp((U(1) - \Delta)k + o(k)).$$

A 512-bit Arb evaluation gives

$$\begin{aligned} e^{U(1)-\Delta} &< 3.7806852883796401133524 \\ &< 3.780685288379640114 < 3.780685290, \end{aligned}$$

where the first decimal is an outward-rounded upper endpoint. Its distance to the safe nine-decimal value is greater than $1.6203 \cdot 10^{-9}$. This proves both printed claims. $\square$

8. CERTIFIED COMPUTATION AND INDEPENDENT REPLAY

The computational proof follows three rules.

Exact data. All polynomial coefficients and parameters are parsed from decimal or rational strings, never from binary floating-point literals. Equality of the inner and outer correlation constants is checked in exact rational arithmetic.

Continuum coverage. Every compact domain is a union of closed rational intervals. Arb evaluates an inclusion extension on each cell. Analytic monotonicity or derivative signs treat the unbounded tails. A dense floating scan is not proof and is used only during search.

Independent implementations. The final exact-diagonal/outer-transfer package contains two author checkers, a non-importing replay, and an adaptive referee replay. The latter two reconstruct the functions and constants instead of importing author code. They use different subdivisions and endpoint reductions. These four final checkers use 512-bit precision. The logically upstream six-stage rate chain and its direct-region replay use 256-bit precision. Every one of these programs uses `python-flint==0.9.0` and therefore shares the same Arb containment kernel: implementation diversity is not an independent arithmetic kernel or a proof-assistant formalization.

The canonical rate loader rejects duplicate JSON keys, missing or unknown schema fields, non-string numeric data, and non-finite decimal strings. In `-require-paper-chain` mode it additionally checks the six filenames, their order, and their complete file hashes before interval evaluation. The canonical reproduction wrappers also refuse optimized Python mode, in which proof-critical `assert` statements would otherwise be disabled.

The current theorem files have the following SHA-256 identifiers.

Role	SHA-256 prefix
proof specification	e4ea58def6405936
inner checker	17db00d9374ce3ba
outer checker	e2035cbffefcb147
non-importing replay	fc379e3b861b6905
adaptive replay	3c30f72ddc24848f
independent proof review	a21402a5dad40490

This table, the six complete rate-certificate hashes in Appendix C, and `external_sources/manifest.json` form the upper-bound evidence index. The compact ratio, exact-diagonal, and outer-wedge cells are deterministic rational partitions generated by the tracked checkers; they are not separate untracked cell files. The top-level `artifacts/MANIFEST.tsv` has a different role: it lists only the large finite CNF/DRAT release payloads.

From the repository root, a lightweight replay of the exact-diagonal and outer-transfer certificates is

```
bash scripts/bootstrap.sh
bash scripts/reproduce_upper.sh
```

The first command creates a Python 3.11 environment with exact dependency versions. The second runs the four final transfer checkers; it deliberately does not replay the six-stage rate chain. The complete upper-bound replay, including all six rate certificates, both Ramsey-region orientations, the combinatorial audit, and all four final transfer checkers, is

```
.venv/bin/python reproduce.py full
```

The unified manuscript is compiled separately with `make paper`.

9. SCOPE, TRUST BOUNDARY AND FURTHER WORK

The theorem is asymptotic. The regularization and book interfaces contain eventual estimates, and this paper does not extract an effective threshold for k . Consequently the result has no direct implication for a specific finite Ramsey number.

The retained-spine parameters were found by bounded numerical search and then frozen. The proof certifies the frozen point; it does not prove global optimality. A better correlation inequality, a different off-diagonal rate, or a richer outer partition may improve the decimal further.

The external trust boundary consists of the explicitly stated Yang–Mao v1 interfaces, the repaired local book/rate theorem recorded in the appendices, and the containment semantics of Arb. Independent implementations reduce the risk of a transcription or dependency error but are not a proof-assistant formalization.

Finally, the numerical comparison in the introduction is made against published or source-stated bounds that are cited explicitly. This draft does not claim priority, novelty against all unpublished work, acceptance, or an unqualified world-best status.

Part 2. Controlled Gaussian residuals for fixed-ratio lower bounds

10. FIXED-RATIO LOWER BOUND: THEOREM AND PROOF ARCHITECTURE

The probabilistic argument of Erdős gives the classical exponential scale for diagonal and fixed-ratio Ramsey lower bounds [Erd47]. Ma, Shen and Xie recently obtained the first exponential improvement for $R(\ell, C\ell)$ at every fixed $C > 1$ [MSX25]. Hunter, Milojević and Sudakov gave a Gaussian

random-graph construction and stronger quantitative estimates [HMS25]; Lin and Niu sharpened the truncated Gaussian cumulant analysis [LN26]. Truncated Gaussian variance proxies and cumulants also have an independent probabilistic literature [BMA24].

This part works inside the explicit HMS Appendix-B induction. The new ingredient is not history conditioning by itself—that is already present in the cited constructions—but a deterministic nonexchangeable triangular mean ledger. A boundary derivative must normally match a projection weight exactly. We first show that this exact-merge requirement uniquely determines the array and leaves no scalar optimization freedom. We then retain the negative quadratic term from strong concavity and intentionally permit a small deterministic mismatch. Square completion pays this residual and creates a larger net gain.

10.1. Main theorem. For $C > 1$, let $p_C \in (0, 1/2)$ be the unique solution of

$$C = \frac{\log p_C}{\log(1 - p_C)}.$$

The parameters p, c, D, ω_0 , the source quantities $B_R(C)$ and $B_B(C)$, and the exact-CGF term $G_*(C)$ are defined explicitly in Section 11. The theorem imports the four version-pinned conditional probability, extraction, and blue-ledger items (S1), (S2), (S4), and (S5), stated in source interfaces 11.2 and 11.4. The weighted reverse-propagation rule (S3) is proved locally in Lemma 11.3, as are the one-step nonuniform estimate, first-moment conversion, and red–blue comparison.

Theorem 10.1 (Source-relative fixed-ratio bound). *Fix the absolute source constant K used in (L2.10). Under source interfaces 11.2 and 11.4, there exists $C_0 = C_0(K) > 1$ such that, for each fixed $C \geq C_0$,*

$$\liminf_{\ell \rightarrow \infty} \frac{1}{\ell} \log R(\ell, \lfloor C\ell \rfloor) \geq -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C), \quad (\text{L1.1})$$

where $\hat{H}_*(C) > 0$ and

$$\hat{H}_*(C) = \frac{1 + o(1)}{64 \log C} \quad (C \rightarrow \infty). \quad (\text{L1.2})$$

The public coarsening $B_R(C)/2 \geq 1/20$ may be substituted in (L1.1).

The quantifiers are important: K is fixed first, then C is fixed before $\ell \rightarrow \infty$, and the proof does not give a numerical $C_0(K)$. The source contains unspecified $O(\cdot)$ and $o(\cdot)$ constants, so an effective threshold cannot be extracted from the pinned text alone.

10.2. Contributions.

- (i) We prove rigidity of the exact-gradient triangular recursion.
- (ii) We construct an adapted deterministic residual and prove a uniform full-box Hessian bound.

- (iii) We compute the exact multiplicity of the residual and its square penalty.
- (iv) We insert a three-factor Hölder decomposition without counting the old exact-CGF gain twice.
- (v) We propagate the ledger through reverse induction and quantify the perfect-sequence extraction loss.
- (vi) We optimize a narrowly specified scalar residual class and obtain the leading coefficient $1/64$.

The result is stated as a source-relative strengthening. A direct claim of improvement over every printed final normalization in Lin–Niu would require both expressions to be derived from one common terminal probability formula; that comparison is not asserted here.

11. GAUSSIAN SOURCE LEDGER AND NOTATION

Fix $C > 1$. Let $p_C \in (0, 1/2)$ be the solution of

$$\log p_C = C \log(1 - p_C),$$

and, in the range of C used below, set

$$p = p_C + \frac{1}{C} < \frac{1}{2}, \quad \Phi(-c) = p, \quad a = \varphi(c), \quad D = \frac{4aC}{1-p}. \quad (\text{L2.1})$$

For the fixed- C limit $\ell \rightarrow \infty$, take

$$d = \lceil D^2 \ell^2 \rceil, \quad D_\ell = \frac{\sqrt{d}}{\ell} = D + O_C(\ell^{-2}). \quad (\text{L2.2})$$

The harmless replacement of D by $D_\ell \geq D$ in finite- ℓ feasibility inequalities will always be left implicit.

For a standard normal variable conditioned on $Z \leq -b$, define

$$m(b) = \frac{\varphi(b)}{\Phi(-b)}, \quad v(b) = 1 + b m(b) - m(b)^2, \quad Y_b = Z + m(b), \quad (\text{L2.3})$$

so that Y_b is centered and has variance $v(b)$. Its positive-direction cumulant generating function is

$$K_b(u) = \log \mathbb{E} e^{u Y_b} = \frac{u^2}{2} + u m(b) + \log \Phi(-b - u) - \log \Phi(-b), \quad u \geq 0, \quad (\text{L2.4})$$

and we write

$$\psi_b(u) = \begin{cases} K_b(u)/u^2, & u > 0, \\ v(b)/2, & u = 0. \end{cases} \quad (\text{L2.5})$$

We shall repeatedly use

$$K_b''(u) = v(b + u), \quad K_b(u) = \int_0^u (u - s) v(b + s) ds. \quad (\text{L2.6})$$

Lemma 11.1 (One-sided Gaussian calculus). *For every $b \in \mathbb{R}$, $m'(b) = m(b)(m(b) - b) > 0$, $m''(b) \geq 0$, and $v(b) = 1 - m'(b)$ is nonincreasing. Consequently, $0 < v(b) \leq 1$, $K_b''(u) \in [0, 1]$ for $u \geq 0$, and $\psi_b(u)$ is nonincreasing in both b and u .*

Proof. Differentiating $m = \varphi/\Phi(-\cdot)$ gives $m' = m(m - b)$. The classical inverse-Mills inequality $m(b) > b$ for $b > 0$ (and positivity of m for $b \leq 0$) gives $m' > 0$. Moreover $v = 1 - m'$. Monotonicity of the variance of an upper-truncated normal in its upper cutoff, applied at the cutoff $-b$, says that v is nonincreasing in b ; equivalently $m'' = -v' \geq 0$. The variance bounds give $0 < v \leq 1$. Formula (L2.6) then implies the assertions concerning K_b and ψ_b ; for example $\psi_b(u) = \int_0^1 (1 - t)v(b + ut) dt$. $\square$

Put

$$\begin{aligned} m_0 &= m(c) = \frac{a}{p}, & m'_0 &= m'(c), \\ \alpha_R &= 10\sqrt{\log(10/p)}, & \omega_0 &= \frac{\alpha_R^2}{D} = \frac{100 \log(10/p)}{D}, \end{aligned} \quad (\text{L2.7})$$

and

$$\rho = \frac{m_0 m'_0}{D}, \quad \theta = m_0^2, \quad A_0 = m_0^2 m'_0. \quad (\text{L2.8})$$

For fixed $w > \omega_0$, define the lower and upper cutoff extrema

$$\begin{aligned} m_w &= m(c - w), & M_w &= m(c + w), \\ j_-(w) &= m'(c - w), & j_+(w) &= m'(c + w). \end{aligned} \quad (\text{L2.9})$$

The distinction between m_w and M_w is essential: the former gives the history-uniform exact-CGF deficit, whereas the latter bounds the new linear tilts.

11.1. The frozen red and blue source ledgers. Fix one absolute source constant K that dominates the red $O(\sqrt{\log(1/p)}/D)$ errors in the pinned HMS induction, and put

$$\vartheta_C = 1 - K \frac{\sqrt{\log(1/p)}}{D} > 0, \quad (\text{L2.10})$$

$$B_R(C) = -\log \frac{p}{p_C} + \vartheta_C \frac{a^3}{3p^3 D}, \quad B_B(C) = \log \frac{1 - p_C}{1 - p} - \frac{4a^3 C}{3(1 - p)^3 D}. \quad (\text{L2.11})$$

The dependence of ϑ_C , $B_R(C)$, and the eventual threshold C_0 on this once-fixed K is suppressed in the notation; equivalently one may write $\vartheta_{C,K}$, $B_{R,K}(C)$, and $C_0(K)$. The literal red target envelope is

$$\mathcal{H}_C(r) = p^{(r)} \exp \left[-\vartheta_C \frac{a^3}{p^3 \sqrt{d}} \binom{r}{3} + \frac{a}{p} d^{-1/5} \binom{r}{2} + r \right]. \quad (\text{L2.12})$$

Thus $B_R/2$ and $CB_B/2$, respectively, are the exact limiting red and blue source bonuses relative to $-\frac{1}{2} \log p_C$. In particular, these symbols do not denote the conclusion of the present paper.

11.2. The pre-existing exact-CGF deficit. Set $t_0 = m_0\sqrt{d}$. For $1 \leq k \leq \ell$, set

$$P_{k,d}^0 = \frac{\sqrt{d}}{\sqrt{d} - 4m_0k}, \quad (\text{L2.13})$$

and, whenever the denominator is positive, define

$$\mathbf{d}_{k,w,d} = \frac{m_0^2 m_w^2}{d} k(k-1)^2 \left[1 - P_{k,d}^0 \psi_{c-w} \left(\frac{P_{k,d}^0 m_0 m_w (k-1)}{\sqrt{d}} \right) \right], \quad (\text{L2.14})$$

$$\mathcal{D}_{C,w,d}(r) = \sum_{k=1}^{r-1} \mathbf{d}_{k,w,d}. \quad (\text{L2.15})$$

For $0 \leq x \leq 1$, let

$$P_*(x) = \frac{D}{D - 4m_0x}, \quad u_*(x) = \frac{P_*(x)m_0m(c - \omega_0)x}{D}. \quad (\text{L2.16})$$

For all sufficiently large C , $D > 8m_0$ and $P_*(1)v(c - \omega_0) < 2$; hence (L2.13)–(L2.17) have positive denominators and nonnegative integrands throughout the range used below. The exact source-relative CGF term used in the theorem is

$$G_*(C) = \frac{m_0^2 m(c - \omega_0)^2}{D^2} \int_0^1 x^3 [1 - P_*(x) \psi_{c-\omega_0}(u_*(x))] dx. \quad (\text{L2.17})$$

Equivalently, if $G(C, w)$ is obtained from (L2.17) by replacing ω_0 with w , then

$$\lim_{\ell \rightarrow \infty} \ell^{-2} \mathcal{D}_{C,w,d}(\ell) = G(C, w), \quad G_*(C) = \lim_{w \downarrow \omega_0} G(C, w). \quad (\text{L2.18})$$

11.3. The controlled-residual constants. Set

$$a_- = \min\{A_0, \theta\}, \quad a_+ = \max\left\{\theta, \frac{A_0 + \rho\theta}{1 - \rho}\right\}, \quad (\text{L2.19})$$

$$\mu_w = j_-(w) - j_+(w)^2 \frac{m_0 + \theta/D}{D(1 - \rho)}, \quad (\text{L2.20})$$

and

$$\hat{L}_w(C) = \frac{4m_0 M_w^2}{D} + \frac{2M_w^2 a_+}{D^2} + \frac{32m_0^2 M_w^2 a_+}{a_- D^2}. \quad (\text{L2.21})$$

Finally put

$$\hat{B}_w(C) = m_0^2 - \frac{16}{3} - \hat{L}_w(C), \quad E_w(C) = \hat{B}_w(C)\theta - \frac{\theta^2}{2\mu_w}, \quad (\text{L2.22})$$

$$\hat{H}(C, w) = \frac{\hat{B}_w(C)A_0 + E_w(C)}{12D^2}, \quad \hat{H}_*(C) = \lim_{w \downarrow \omega_0} \hat{H}(C, w). \quad (\text{L2.23})$$

11.4. Probability space, exposure histories, and source events. For each $0 \leq r \leq \ell$, let $\mathbf{x}_1, \dots, \mathbf{x}_r$ be independent $N(0, d^{-1}I_d)$ vectors. An edge is blue when $\langle \mathbf{x}_i, \mathbf{x}_j \rangle \geq -c/\sqrt{d}$ and red otherwise. We use the Bartlett representation $\mathbf{y}_1, \dots, \mathbf{y}_r$: the entries $y_j(i)$, $i < j$, are independent $N(0, d^{-1})$, the diagonal entries are independent with $y_i(i) \sim \sqrt{\chi_{d-i+1}^2/d}$, and $y_j(i) = 0$ for $i > j$. This representation has the same joint Gram matrix as the original Gaussian vectors.

Let $M_i = (y_j(i))_{j=i+1}^r$ be the strict part of column i , and put

$$\mathcal{D}_r = \sigma(y_1(1), \dots, y_r(r)).$$

Define the exposure filtration

$$\mathcal{F}_s = \mathcal{D}_r \vee \sigma(M_1, \dots, M_s), \quad 0 \leq s \leq r. \quad (\text{L2.24})$$

All conditional assertions below are statements about regular conditional probabilities and hold almost surely. Write π_s for projection onto the first s Bartlett coordinates and put

$$h_{ij} = \langle \pi_{i-1}(\mathbf{y}_i), \pi_{i-1}(\mathbf{y}_j) \rangle, \quad 1 \leq i < j \leq r. \quad (\text{L2.25})$$

Thus h_{ij} is $\mathcal{F}_{i-1}$ -measurable. Let E_{ij} be the blue-edge event and set

$$I_s = \bigcap_{s < i < j \leq r} \overline{E}_{ij}, \quad \mathcal{C}_s = \bigcap_{s < i < j \leq r} E_{ij}. \quad (\text{L2.26})$$

For the source perfectness events put $\alpha_B = 10\sqrt{C \log(10/p)}$ and $\delta = \alpha_B d^{-1/4}$. The event B_r^R says that, for every $i \leq r$,

$$\|\mathbf{y}_i\| \in (1 - \delta, 1 + \delta), \quad \|\pi_{i-1}(\mathbf{y}_i)\| \leq \frac{\alpha_R \sqrt{\ell}}{\sqrt{d}}; \quad (\text{L2.27})$$

B_r^B is defined by replacing α_R with α_B . We use the probability notation

$$P_{R,r}^* = \mathbb{P}(I_0 \cap B_r^R), \quad P_{R,r} = \mathbb{P}(I_0), \quad P_{B,r}^* = \mathbb{P}(\mathcal{C}_0 \cap B_r^B), \quad P_{B,r} = \mathbb{P}(\mathcal{C}_0). \quad (\text{L2.28})$$

Empty intersections have probability one.

For $0 \leq s \leq r$, let

$$\kappa_{r,s}^R = \mathbb{P}(B_r^R \mid \mathcal{F}_s), \quad \mathfrak{H}_{r,s}^R = \{\kappa_{r,s}^R > 0\}. \quad (\text{L2.29})$$

We call an $\mathcal{F}_s$ -history *red-admissible* when it belongs to $\mathfrak{H}_{r,s}^R$. This definition is independent of a choice of coordinates for the conditional kernel. On the complementary histories, $\mathbb{P}(I_s \cap B_r^R \mid \mathcal{F}_s) = 0$ almost surely. Thus every conditional estimate below only needs to be checked on $\mathfrak{H}_{r,s}^R$.

At exposure i , set $X_j = y_j(i)$, $j > i$, and let $A_i^R = \bigcap_{j>i} \overline{E}_{ij}$. Conditional on $\mathcal{F}_{i-1}$, this is the coordinatewise event with exact upper cutoffs

$$X_j \leq -\frac{b_{ij}}{\sqrt{d}}, \quad b_{ij} = \frac{c + \sqrt{d} h_{ij}}{y_i(i)}. \quad (\text{L2.30})$$

The red-perfect extraction map Γ_R scans $\mathbf{x}_1, \dots, \mathbf{x}_\ell$ in order and retains an index when adjoining its vector preserves the two inequalities in (L2.27). For a specified $I \subseteq [\ell]$, define

$$p_I = \mathbb{P}(I_0 \cap \{\Gamma_R = I\}). \quad (\text{L2.31})$$

This fixes both the event and the retained-order relabeling used later.

For a deterministic nonnegative triangular array $T = (T_{jq})_{1 \leq j < q \leq r}$, define the weighted projection potential

$$\mathcal{P}_s^T = \sum_{s < j < q \leq r} T_{jq} \langle \pi_s(\mathbf{y}_j), \pi_s(\mathbf{y}_q) \rangle. \quad (\text{L2.32})$$

For $i \geq 1$, put $q_i = \mathbb{P}(A_i^R \mid \mathcal{F}_{i-1})$. Its normalized one-step indicator kernel is

$$\begin{aligned} \mathcal{K}_i^T &= \exp\left(\sum_{j>i} T_{ij} h_{ij}\right) \\ &\quad \times \mathbb{E}\left[\mathbf{1}_{A_i^R} \exp\left\{-\sum_{i<j<q\leq r} T_{jq} X_j X_q\right\} \middle| \mathcal{F}_{i-1}\right] \\ &= \exp\left(\sum_{j>i} T_{ij} h_{ij}\right) q_i \\ &\quad \times \mathbb{E}\left[\exp\left\{-\sum_{i<j<q\leq r} T_{jq} X_j X_q\right\} \middle| A_i^R, \mathcal{F}_{i-1}\right]. \end{aligned} \quad (\text{L2.33})$$

The first line is the definition. It remains meaningful on a history where $q_i = 0$; the second line is its usual conditional-event form on $\{q_i > 0\}$.

Source interface 11.2 (Version-pinned red source interface). Items (S1) and (S2) are imported from the content-addressed HMS v2 and Lin–Niu v2 sources identified in Appendix E.

- (S1) For every $r \leq \ell$ and $1 \leq i < r$, on every red-admissible $\mathcal{F}_{i-1}$ -history, the next-column coordinates in (L2.30), conditionally on A_i^R , are independent $N(0, d^{-1})$ variables truncated by the displayed upper cutoffs, and

$$|b_{ij} - c - \sqrt{d} h_{ij}| \leq d^{-1/5}, \quad |\sqrt{d} h_{ij}| \leq \omega_0. \quad (\text{L2.34})$$

Thus, for every fixed $w > \omega_0$, all cutoffs lie in $[c - w, c + w]$ once $\ell \geq \ell_0(C, w)$.

- (S2) Under the conditioning and history quantifiers in (S1), with $\zeta_j = \sqrt{d}(X_j - \mathbb{E}[X_j \mid A_i^R, \mathcal{F}_{i-1}])$, the variables $(\zeta_j)_{j>i}$ are conditionally independent, centered, and 1-subgaussian, and

$$\log \mathbb{E}[e^{x\zeta_j^2} \mid A_i^R, \mathcal{F}_{i-1}] \leq \frac{16}{3}x, \quad 0 \leq x \leq \frac{1}{8}. \quad (\text{L2.35})$$

For $k = r - i$, $Q_0 = d/(4t_0k)$, and centered ζ_j , the frozen uniform-quadratic factor obeys

$$\frac{1}{Q_0} \log \mathbb{E} \exp \left\{ -\frac{Q_0 t_0}{d} \sum_{i < j < q \leq r} \zeta_j \zeta_q \right\} \leq \frac{4t_0k}{d}. \quad (\text{L2.36})$$

The positive-direction CGF is exactly (L2.4). On this history define the source baseline tilt

$$u_j^0 = \frac{t_0}{d} \sum_{\substack{q > i \\ q \neq j}} m(b_{iq}), \quad j > i.$$

This is the quantity recalled in (L5.9). In the source pre-Cauchy allocation, the centered-linear proxy is precisely $\sum_{j > i} (u_j^0)^2$; replacing that proxy by the exact CGF produces the same-history deficit (L5.16). For the uniform array $T_{jq}^0 = t_0$, the version-pinned source scalar allocation and this exact-CGF replacement give, on every history in $\mathfrak{H}_{r,i-1}^R$,

$$\mathcal{K}_i^{T^0} \leq \frac{\mathcal{H}_C(k+1)}{\mathcal{H}_C(k)} e^{-d_{k,w,d}} \quad (k = r - i). \quad (\text{L2.36a})$$

This scalar inequality, rather than a final Ramsey bound, is the imported baseline. No nonuniform one-step estimate is imported.

No Ramsey-rate conclusion, red–blue comparison, or nonuniform residual term is included in (S1) or (S2).

Lemma 11.3 (Weighted reverse propagation (S3)). *For every target $r \leq \ell$, every deterministic nonnegative triangular T fixed before exposure, and every deterministic real sequence $\lambda_1, \dots, \lambda_{r-1}$, the following weighted reverse-induction rule holds. If, on every history in $\mathfrak{H}_{r,i-1}^R$, with $k = r - i$,*

$$\mathcal{K}_i^T \leq \frac{\mathcal{H}_C(k+1)}{\mathcal{H}_C(k)} e^{-\lambda_i}, \quad 1 \leq i < r, \quad (\text{L2.37})$$

then, for every $0 \leq s \leq r$,

$$\mathbb{P}(I_s \cap B_r^R \mid \mathcal{F}_s) \leq \mathcal{H}_C(r-s) \exp \left[-\mathcal{P}_s^T - \sum_{i=s+1}^{r-1} \lambda_i \right]. \quad (\text{L2.38})$$

On $(\mathfrak{H}_{r,s}^R)^c$, the left side of (L2.38) is zero, so the same inequality holds almost surely on the entire conditional state space.

Proof. Put

$$Z_s = \mathbb{P}(I_s \cap B_r^R \mid \mathcal{F}_s), \quad \Lambda_s = \sum_{t=s+1}^{r-1} \lambda_t.$$

We first justify the exponential conditional expectations. Given $\mathcal{F}_{i-1}$, write $u_j = -b_{ij}/\sqrt{d}$, so that $A_i^R = \{X_j \leq u_j : j > i\}$. These cutoffs are finite

almost surely and

$$q_i = \prod_{j>i} \Phi(-b_{ij}) > 0$$

almost surely. For a vector x in this truncated region, put $\bar{u}_j = (u_j)_+$ and $x_j^- = (-x_j)_+$. Since $T \geq 0$,

$$\sum_{i<j<q\leq r} T_{jq} x_j x_q \geq - \sum_{i<j<q\leq r} T_{jq} (\bar{u}_j x_q^- + \bar{u}_q x_j^-).$$

Consequently the integrand in (L2.33) is bounded by an exponential of a nonnegative linear combination of the x_j^- . Gaussian linear exponential moments are finite, even after the coordinatewise upper truncation. Thus $\mathcal{K}_i^T$ and $\mathbb{E}[e^{-\mathcal{P}_i^T} \mid A_i^R, \mathcal{F}_{i-1}]$ are finite almost surely. The primary indicator form in (L2.33) also supplies an unambiguous version on any exceptional history with $q_i = 0$.

Since $I_{i-1} = A_i^R \cap I_i$ and $A_i^R \in \mathcal{F}_i$, the law of total probability, first in its indicator form, is

$$Z_{i-1} = \mathbb{E}[\mathbf{1}_{A_i^R} Z_i \mid \mathcal{F}_{i-1}] = q_i \mathbb{E}[Z_i \mid A_i^R, \mathcal{F}_{i-1}], \quad (\text{L2.39})$$

where the final form is only needed when $q_i > 0$. The exact projection identities are

$$\begin{aligned} \mathcal{P}_i^T &= \sum_{i<j<q\leq r} T_{jq} \langle \pi_{i-1} \mathbf{y}_j, \pi_{i-1} \mathbf{y}_q \rangle + \sum_{i<j<q\leq r} T_{jq} X_j X_q, \\ \mathcal{P}_{i-1}^T &= \sum_{i<j<q\leq r} T_{jq} \langle \pi_{i-1} \mathbf{y}_j, \pi_{i-1} \mathbf{y}_q \rangle + \sum_{j>i} T_{ij} h_{ij}. \end{aligned} \quad (\text{L2.40})$$

Denote the first sum on the right of each identity by U_i , the second sum in the first identity by V_i , and the second sum in the second identity by W_i . Thus $\mathcal{P}_i^T = U_i + V_i$, $\mathcal{P}_{i-1}^T = U_i + W_i$, and (L2.33) gives the exact conjugation identity

$$\mathbb{E}[\mathbf{1}_{A_i^R} e^{-\mathcal{P}_i^T} \mid \mathcal{F}_{i-1}] = e^{-\mathcal{P}_{i-1}^T} \mathcal{K}_i^T.$$

There are two terminal cases. At $s = r$, we have $I_r = \Omega$, $\mathcal{P}_r^T = 0$, $\Lambda_r = 0$, and $Z_r \leq 1 = \mathcal{H}_C(0)$. At $s = r - 1$, we again have an empty future edge set, $\mathcal{P}_{r-1}^T = 0$, and $\Lambda_{r-1} = 0$, so $Z_{r-1} \leq 1 \leq e = \mathcal{H}_C(1)$. The latter is the load-bearing base case because the one-step premise is indexed only by $i < r$.

Assume now that (L2.38) holds at level $s = i$, where $1 \leq i < r$, and put $k = r - i$. On $\mathfrak{H}_{r,i-1}^R$, conditional monotonicity, (L2.39), and the conjugation identity give

$$\begin{aligned} Z_{i-1} &\leq \mathcal{H}_C(k) e^{-\Lambda_i} \mathbb{E}[\mathbf{1}_{A_i^R} e^{-\mathcal{P}_i^T} \mid \mathcal{F}_{i-1}] \\ &= \mathcal{H}_C(k) e^{-\mathcal{P}_{i-1}^T - \Lambda_i} \mathcal{K}_i^T \\ &\leq \mathcal{H}_C(k+1) e^{-\mathcal{P}_{i-1}^T - \lambda_i - \Lambda_i}. \end{aligned}$$

Since $\Lambda_{i-1} = \lambda_i + \Lambda_i$, this is (L2.38) at level $i - 1$. On $(\mathfrak{H}_{r,i-1}^R)^c$, the same claim is automatic because $Z_{i-1} = 0$. Reverse induction from $s = r - 1$, together with the separately checked endpoint $s = r$, completes the proof. $\square$

Sections 13–15 compare the nonuniform kernel with the scalar baseline (L2.36a) and prove (L2.37) for the array used here.

Source interface 11.4 (Version-pinned extraction and blue source interfaces). Items (S4) and (S5) are imported from the content-addressed HMS v2 source identified in Appendix E.

- (S4) If red-perfect extraction deletes u vertices from a target of size ℓ , then, for each specified retained set I of size $\ell - u$,

$$p_I \leq P_{R,\ell-u}^* \left(\frac{p}{10} \right)^{10\ell u}, \quad (\text{L2.41})$$

for every $0 \leq u \leq \ell$, with $P_{R,0}^* = \mathcal{H}_C(0) = 1$. The probability $P_{R,\ell-u}^*$ uses the retained-order relabeling defined after (L2.31).

- (S5) For each fixed C , there is a real sequence $\epsilon_{C,\ell} \rightarrow 0$ such that, with $q_\ell = \lfloor C\ell \rfloor$, the unchanged blue induction and its perfect-sequence extraction give

$$P_{B,q_\ell} \leq (1-p)^{\binom{q_\ell}{2}} \exp \left[\frac{4a^3}{(1-p)^3 \sqrt{d}} \binom{q_\ell}{3} + \epsilon_{C,\ell} \ell^2 \right]. \quad (\text{L2.42})$$

No Ramsey-rate conclusion, no red–blue comparison, and no new residual term is included in (S1), (S2), (S4), or (S5). The scalar source baseline is (L2.36a); the nonuniform one-step estimate is proved locally, and its propagation is the local Lemma 11.3, not an imported source hypothesis.

The constant K in (L2.10) is not effective in the pinned source. Consequently all sufficiently-large- C statements below are existential.

12. RIGIDITY OF EXACT GRADIENT MERGE

For a target ordered set $[r]$, consider deterministic triangular weights W_{ij} and the connection-plus-mean potential

$$F_i(b) = \sum_{j>i} \log \Phi(-b_j) - \frac{1}{d} \sum_{i<j<q} W_{jq} m(b_j) m(b_q). \quad (\text{L3.1})$$

At the central cutoff,

$$\partial_j F_i(c\mathbf{1}) = -m_0 - \frac{m_0 m'_0}{d} \sum_{\substack{q>i \\ q \neq j}} W_{jq}. \quad (\text{L3.2})$$

Proposition 12.1 (Unique exact-merge array). *Normalize the coefficient of $\sum_{j>i} \log \Phi(-b_j)$ in (L3.1) to be one. If the linear term in (L3.2), multiplied by $b_j - c = \sqrt{d} h_{ij}$, must equal $-W_{ij} h_{ij}$ at every boundary, then the weights are uniquely determined by*

$$W_{ij} = m_0 \sqrt{d} + \frac{m_0 m'_0}{\sqrt{d}} \sum_{\substack{q>i \\ q \neq j}} W_{jq}. \quad (\text{L3.3})$$

There is no further normalized scalar degree of freedom.

Proof. The coefficient comparison gives (L3.3). Process the first endpoint in decreasing order. Every weight on the right side has already been defined, so the system is triangular and has a unique solution. $\square$

Remark 12.2 (Global presentation scale). Multiplying the entire inequality by $\kappa > 0$ is a presentation equivalence, not a new normalized array. After dividing the connection term back to coefficient one, the normalized array still satisfies (L3.3). If one instead calls $W^{\text{eff}} = \kappa W$ the effective coefficient without renormalizing, then its affine recursion has base term $\kappa m_0 \sqrt{d}$, not $m_0 \sqrt{d}$. Thus a normalized solution and a κ -scaled effective coefficient are never identified. No assertion is made about edge-dependent rescalings.

The feedback operator in (L3.3) has row norm at most $\rho = m_0 m'_0 / D = o(1)$. Its accumulated increment is

$$2A_0 \binom{r}{4} [1 + O(\rho)]. \quad (\text{L3.4})$$

With the source centered coefficient $m_0^2[1 + o(1)]$, the exact-merge ledger contributes the previously reviewed term $(1 + o(1))/(96 \log C)$. Exact feedback evaluation can change only the next order.

13. CONTROLLED RESIDUAL AND STRONG CONCAVITY

Recall $t_0 = m_0 \sqrt{d}$ from Section 11, and set

$$\gamma_d = \frac{m_0 m'_0}{\sqrt{d}}, \quad e_{ij} = \theta(i - 1). \quad (\text{L4.1})$$

In decreasing order of the first endpoint define

$$T_{ij} = t_0 + \gamma_d \sum_{\substack{q > i \\ q \neq j}} T_{jq} + e_{ij}. \quad (\text{L4.2})$$

The array is deterministic and adapted: when column i is exposed, every future weight in (L4.2) has already been fixed.

With F_i as in (L3.1), now using T , we obtain the exact residual identity

$$\partial_j F_i(c\mathbf{1}) = -\frac{T_{ij} - e_{ij}}{\sqrt{d}}. \quad (\text{L4.3})$$

Thus the mismatch is explicit and deterministic.

Lemma 13.1 (Uniform Hessian bound). *For all sufficiently large fixed C , every $b \in [c - w, c + w]^{V_i}$ satisfies*

$$-\nabla^2 F_i(b) \succeq \mu_w I, \quad (\text{L4.4})$$

where $\mu_w > 0$ is defined in (L2.20).

Proof. The recursion gives

$$T_{\max} \leq \frac{t_0 + \theta r}{1 - \rho}, \quad \frac{1}{d} \max_j \sum_{q \neq j} T_{jq} \leq \frac{m_0 + \theta/D}{D(1 - \rho)}.$$

The diagonal of $-\nabla^2 F_i$ is at least $j_-(w)$. The absolute off-diagonal row sum is at most the second term subtracted in (L2.20), multiplied by $j_+(w)^2$. Gershgorin's theorem proves (L4.4). $\square$

Write

$$\delta_j = b_j - c = \sqrt{d} h_{ij} + \varepsilon_{ij}, \quad |\varepsilon_{ij}| \leq d^{-1/5}.$$

Strong concavity, (L4.3), and

$$\frac{e_{ij}\delta_j}{\sqrt{d}} - \frac{\mu_w \delta_j^2}{2} \leq \frac{e_{ij}^2}{2\mu_w d} \quad (\text{L4.5})$$

give

$$F_i(b) \leq F_i(c\mathbf{1}) - \sum_{j>i} T_{ij} h_{ij} + \sum_{j>i} \frac{e_{ij}^2}{2\mu_w d} + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (\text{L4.6})$$

The full boundary coefficient merges into the projection potential, while the failure of exact gradient merge is paid by the square term.

14. DETERMINISTIC MASS AND THE CENTERED-MOMENT LEDGER

Write $\Delta_{ij} = T_{ij} - t_0$. The recursion implies

$$A_0(r-j-1) + \theta(j-1) \leq \Delta_{jq} \leq \frac{A_0 + \rho\theta}{1-\rho}(r-j-1) + \theta(j-1). \quad (\text{L5.1})$$

Consequently

$$a_-(r-2) \leq \Delta_{jq} \leq a_+(r-2). \quad (\text{L5.2})$$

For $V_i = \{i+1, \dots, r\}$, of size $k = r-i$, put

$$S_i = \sum_{\{j,q\} \subset V_i} \Delta_{jq}, \quad R_i = \max_{j \in V_i} \sum_{q \neq j} \Delta_{jq}.$$

Then

$$S_i \geq a_-(r-2) \binom{k}{2}, \quad R_i \leq a_+(r-2)(k-1), \quad \frac{R_i}{S_i} \leq \frac{2a_+}{a_-k}. \quad (\text{L5.3})$$

Lemma 14.1 (Exact accumulated multiplicities). *Let*

$$K_r = \sum_{i=1}^{r-1} (r-i)(i-1)^2 = \binom{r}{3} + 2 \binom{r}{4}. \quad (\text{L5.4})$$

Then

$$\sum_{i=1}^{r-1} S_i \geq 2A_0 \binom{r}{4} + \theta K_r, \quad \sum_{1 \leq i < j \leq r} e_{ij}^2 = \theta^2 K_r. \quad (\text{L5.5})$$

Proof. Sum the two terms of the lower bound in (L5.1) with their exposure multiplicities. The future-degree term counts every ordered quadruple twice, giving $2\binom{r}{4}$; the residual term is exactly the sum in (L5.4). The second identity follows directly from $e_{ij} = \theta(i-1)$. $\square$

14.1. Exact conditional factorization. Fix a red-admissible $\mathcal{F}_{i-1}$ -history for target r at exposure step i , and condition further on A_i^R . For $j \in V_i$, let X_j be the coordinate in (L2.30), write $b_j = b_{ij}$, put $m_j = m(b_j)$, and standardize its centered part by

$$\zeta_j = \sqrt{d}(X_j - \mathbb{E}X_j) = \sqrt{d}X_j + m_j. \quad (\text{L5.6})$$

By (S2), the ζ_j are conditionally independent, centered and 1-subgaussian. Expanding the weighted quadratic form gives the exact identity

$$\begin{aligned} \log \mathbb{E} \exp \left\{ - \sum_{\{j,q\} \subset V_i} T_{jq} X_j X_q \right\} \\ = -\frac{1}{d} \sum_{\{j,q\} \subset V_i} T_{jq} m_j m_q + \Lambda_i, \end{aligned} \quad (\text{L5.7})$$

$$\Lambda_i = \log \mathbb{E} \exp \left\{ \begin{aligned} & \sum_{j \in V_i} (u_j^0 + \delta u_j) \zeta_j \\ & - \frac{t_0}{d} \sum_{j < q} \zeta_j \zeta_q - \frac{1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \end{aligned} \right\}. \quad (\text{L5.8})$$

where

$$u_j^0 = \frac{t_0}{d} \sum_{q \neq j} m_q = \frac{m_0}{\sqrt{d}} \sum_{q \neq j} m_q, \quad \delta u_j = \frac{1}{d} \sum_{q \neq j} \Delta_{jq} m_q. \quad (\text{L5.9})$$

All tilts in (L5.9) are nonnegative.

14.2. Three-factor Hölder and the square-MGF cost. For $k \geq 2$, choose

$$Q_0 = \frac{d}{4t_0k}, \quad Q_1 = \frac{d}{4R_i}, \quad P^{-1} = 1 - Q_0^{-1} - Q_1^{-1}. \quad (\text{L5.10})$$

If $R_i = 0$, the increment factor is absent and we set $Q_1 = \infty$. For all sufficiently large C ,

$$Q_0^{-1} + Q_1^{-1} \leq \frac{4m_0}{D} + \frac{4a_+}{D^2} < \frac{1}{2}, \quad (\text{L5.11})$$

so $Q_0, Q_1 > 1$ and $1 < P \leq 2$. Hölder's inequality applied to the three random factors in (L5.8) yields

$$\begin{aligned} \Lambda_i &\leq \frac{1}{P} \sum_j K_{b_j}(P(u_j^0 + \delta u_j)) \\ &\quad + \frac{1}{Q_0} \log \mathbb{E} \exp \left\{ -\frac{Q_0 t_0}{d} \sum_{j < q} \zeta_j \zeta_q \right\} \\ &\quad + \frac{1}{Q_1} \log \mathbb{E} \exp \left\{ -\frac{Q_1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \right\}. \end{aligned} \quad (\text{L5.12})$$

The middle line is the unchanged HMS uniform-quadratic factor and costs at most $4t_0k/d$.

For the last line, write $R_j = \sum_{q \neq j} \Delta_{jq}$. Nonnegativity of Δ gives

$$-\sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \leq \frac{1}{2} \sum_j R_j \zeta_j^2. \quad (\text{L5.13})$$

Because $Q_1 R_j / (2d) \leq 1/8$, the pinned square-MGF estimate

$$\log \mathbb{E} e^{x \zeta_j^2} \leq \frac{16}{3} x, \quad 0 \leq x \leq \frac{1}{8}, \quad (\text{L5.14})$$

and conditional independence imply

$$\begin{aligned} \frac{1}{Q_1} \log \mathbb{E} \exp \left\{ -\frac{Q_1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \right\} &\leq \frac{1}{Q_1} \sum_j \frac{16}{3} \frac{Q_1 R_j}{2d} \\ &= \frac{16}{3} \frac{S_i}{d}, \end{aligned} \quad (\text{L5.15})$$

where $\sum_j R_j = 2S_i$. This derives the constant $16/3$.

14.3. The same-history CGF bridge. Let $P^0 = P_{k,d}^0$, so that $(P^0)^{-1} = 1 - Q_0^{-1}$. The old optimized uniform ledger on this same conditional history has deficit

$$\mathfrak{d}_i(b) = \sum_{j \in V_i} \left[(u_j^0)^2 - \frac{1}{P^0} K_{b_j}(P^0 u_j^0) \right]. \quad (\text{L5.16})$$

Equations (L2.6), the cutoff window, and $\sum_{q \neq j} m_q \geq (k-1)m_w$ give the deterministic lower bound

$$\mathfrak{d}_i(b) \geq \mathfrak{d}_{k,w,d}. \quad (\text{L5.17})$$

Indeed, after writing $K_b(P^0 u)/P^0 = P^0 u^2 \psi_b(P^0 u)$, this is exactly the monotonicity comparison used in (L2.14). Thus the term whose accumulated limit is $G_*(C)$ is identified before any new factor is introduced.

It remains to compare the new linear factor with the second term in (L5.16). Since $K_b'' \in [0, 1]$, for $u, \delta \geq 0$ and $1 < P \leq 2$,

$$\frac{K_b(P(u + \delta)) - K_b(Pu)}{P} \leq 2u\delta + \delta^2. \quad (\text{L5.18})$$

Moreover, differentiating $K_b(Pu)/P$ and using (L2.6) gives

$$0 \leq \frac{\partial}{\partial P} \frac{K_b(Pu)}{P} \leq u^2. \quad (\text{L5.19})$$

As

$$P - P^0 = PP^0 \left(\frac{1}{P^0} - \frac{1}{P} \right) = \frac{PP^0}{Q_1} \leq \frac{16R_i}{d}, \quad (\text{L5.20})$$

we obtain

$$\begin{aligned} &\frac{1}{P} \sum_j K_{b_j}(P(u_j^0 + \delta u_j)) - \frac{1}{P^0} \sum_j K_{b_j}(P^0 u_j^0) \\ &\leq 2 \sum_j u_j^0 \delta u_j + \sum_j (\delta u_j)^2 + \frac{16R_i}{d} \sum_j (u_j^0)^2. \end{aligned} \quad (\text{L5.21})$$

Each term is now bounded without probabilistic input. From (L5.3), (L5.9), $m_j \leq M_w$, and $k/\sqrt{d} \leq 1/D$,

$$2 \sum_j u_j^0 \delta u_j \leq \frac{4m_0 M_w^2 S_i}{D} \frac{1}{d}, \quad (\text{L5.22})$$

$$\sum_j (\delta u_j)^2 \leq \frac{2M_w^2 a_+ S_i}{D^2} \frac{1}{d}, \quad (\text{L5.23})$$

$$\frac{16R_i}{d} \sum_j (u_j^0)^2 \leq \frac{32m_0^2 M_w^2 a_+ S_i}{a_- D^2} \frac{1}{d}. \quad (\text{L5.24})$$

For completeness, (L5.22) uses $\sum_j \delta u_j \leq 2M_w S_i/d$; (L5.23) uses $\max_j \delta u_j \leq M_w R_i/d$; and (L5.24) uses

$$\sum_j (u_j^0)^2 \leq \frac{2m_0^2 M_w^2 S_i}{a_-} \frac{1}{d}, \quad (\text{L5.25})$$

which follows from the first inequality in (L5.3) and $r-2 \geq k-1$. The sum of (L5.22)–(L5.24) is precisely $\hat{L}_w(C) S_i/d$.

Finally, at the central cutoff the deterministic Δ -part of (L5.7) is $-m_0^2 S_i/d$. Combining it with (L5.15) and (L5.21)–(L5.24) leaves

$$-\hat{B}_w(C) \frac{S_i}{d}. \quad (\text{L5.26})$$

Equations (L5.16)–(L5.17) charge the old exact-CGF deficit once; (L5.26) charges only the new increment factor and the change $P^0 \rightarrow P$; and (L4.6) pays the boundary residual. Hence the three ledger entries are disjoint on the same conditional history.

15. REVERSE INDUCTION AND PERFECT-SEQUENCE EXTRACTION

By (L7.7)–(L7.11), after increasing the existential threshold $C_0(K)$ and restricting w to a sufficiently small right-neighborhood of ω_0 , we have $\hat{B}_w(C) > 0$, $E_w(C) > 0$, and $\mu_w > 0$. We fix this range throughout the section; in particular, every monotonicity assertion below has its sign justified before it is used.

Define the new accumulated deterministic deficit

$$\mathcal{N}_{C,w,d}(r) = \frac{1}{d} \left[2\hat{B}_w A_0 \binom{r}{4} + E_w K_r \right]. \quad (\text{L6.1})$$

At an exposure step with $k = r-i$ future vertices, the same-history calculation of Section 14, followed by the pathwise boundary estimate (L4.6), proves the one-step kernel bound (L2.37) with

$$\lambda_i = \mathbf{d}_{k,w,d} + \hat{B}_w(C) \frac{S_i}{d} - \sum_{j>i} \frac{e_{ij}^2}{2\mu_w d} - \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (\text{L6.2})$$

Every coefficient is deterministic and fixed before column i is sampled. Equations (L2.33) and (L2.40) retain every future coefficient T_{jq} , while (L4.6) supplies each new boundary edge ij with its full coefficient T_{ij} . Thus Lemma 11.3 applies without an exchangeability assumption.

Apply (L2.38) at $s = 0$, average over $\mathcal{F}_0$, and then use Lemma 14.1. Since $\mathcal{P}_0^T = 0$, this gives

$$P_{R,r}^* \leq \mathcal{H}_C(r) \exp[-\mathcal{D}_{C,w,d}(r) - \mathcal{N}_{C,w,d}(r) + J_{r,d}], \quad (\text{L6.3})$$

where

$$0 \leq J_{r,d} \leq \frac{m_0 + \theta/D}{1 - \rho} d^{-1/5} \binom{r}{2} = o_C(r^2). \quad (\text{L6.4})$$

Here $J_{r,d}$ denotes the displayed deterministic upper envelope, so $J_{q,d} \leq J_{\ell,d}$ for $q \leq \ell$. The base cases $r = 0, 1, 2$ follow from the empty-event convention and the same one-step identity; no asymptotic estimate is needed there.

15.1. Uniform restoration after deleting vertices. Let extraction retain $q = \ell - u$ vertices. Relabel them in retained order and rebuild the deterministic array for target size q . Since $q/\sqrt{d} \leq \ell/\sqrt{d}$, all row-sum, Hessian and Hölder feasibility inequalities only improve.

First, the bracket in (L2.14) lies in $[0, 1]$, and $m_w \leq m_0$. Consequently

$$\begin{aligned} 0 &\leq \mathcal{D}_{C,w,d}(\ell) - \mathcal{D}_{C,w,d}(\ell - u) \\ &\leq \frac{m_0^4}{d} \sum_{k=\ell-u}^{\ell-1} k(k-1)^2 \leq \frac{m_0^4}{D^2} \ell u. \end{aligned} \quad (\text{L6.5})$$

For sufficiently large C , $m_0^4/D^2 = (1 + o(1))/(8 \log C) < 1$, so the right side is at most ℓu .

Next,

$$\binom{\ell}{4} - \binom{\ell-u}{4} \leq u \binom{\ell}{3}, \quad \binom{\ell}{3} - \binom{\ell-u}{3} \leq u \binom{\ell}{2}. \quad (\text{L6.6})$$

Using $K_r = \binom{r}{3} + 2\binom{r}{4}$, (L6.6) yields

$$\begin{aligned} 0 &\leq \mathcal{N}_{C,w,d}(\ell) - \mathcal{N}_{C,w,d}(\ell - u) \\ &\leq \frac{1}{d} \left[2\hat{B}_w A_0 u \binom{\ell}{3} + E_w \left(u \binom{\ell}{2} + 2u \binom{\ell}{3} \right) \right] \\ &\leq \left[\frac{\hat{B}_w A_0 + E_w}{3D^2} + \frac{E_w}{2D^2 \ell} \right] \ell u. \end{aligned} \quad (\text{L6.7})$$

The last bracket is $O(1/\log C) + O_C(\ell^{-1}) < 1$ for sufficiently large C and then sufficiently large ℓ , so (L6.7) also costs at most ℓu .

Finally, the ratio between the source envelopes at sizes $\ell - u$ and ℓ costs at most

$$[\log(1/p) + 1] \ell u. \quad (\text{L6.8})$$

Indeed, $\binom{\ell}{2} - \binom{\ell-u}{2} \leq \ell u$ pays the missing p -factors, and $\binom{\ell}{3} - \binom{\ell-u}{3} \leq \ell^2 u/2$, together with $\vartheta_C a^3/(p^3 D) \leq 2$ for all sufficiently large C , pays the cubic term. The positive $d^{-1/5} \binom{r}{2}$ and r terms in $\mathcal{H}_C(r)$, as well as (L6.4), are monotone in r and require no restoration charge.

15.2. The binomial subset sum. Apply the pinned extraction inequality (L2.41), then (L6.3) at $q = \ell - u$, and finally (L6.5)–(L6.8). For every specified retained set I and every $1 \leq u \leq \ell$,

$$\begin{aligned} p_I &\leq P_{R,\ell-u}^* \left(\frac{p}{10} \right)^{10\ell u} \\ &\leq \mathcal{H}_C(\ell) \exp[-\mathcal{D}_{C,w,d}(\ell) - \mathcal{N}_{C,w,d}(\ell) + J_{\ell,d}] e^{-\eta_C \ell u}, \end{aligned} \quad (\text{L6.9})$$

where the explicit residual exponent

$$\eta_C = 10 \log(10/p) - \log(1/p) - 3 > 0 \quad (\text{L6.10})$$

pays, respectively, the source-envelope, old-CGF, and new-residual restoration charges. Summing over all possible deleted sets is now an exact binomial calculation:

$$\sum_{u=1}^{\ell} \binom{\ell}{u} e^{-\eta_C \ell u} = (1 + e^{-\eta_C \ell})^{\ell} - 1 = o(1). \quad (\text{L6.11})$$

Including the $u = 0$ perfect term proves the unconditional red probability bound

$$P_{R,\ell} \leq \mathcal{H}_C(\ell) \exp[-\mathcal{D}_{C,w,d}(\ell) - \mathcal{N}_{C,w,d}(\ell) + J_{\ell,d}] (1 + o(1)). \quad (\text{L6.12})$$

This proves extraction of the complete new exponent, rather than merely asserting that a source failure factor is large enough.

16. FIRST MOMENTS, RED-BLUE CROSSING AND ASYMPTOTICS

16.1. The two explicit first-moment thresholds. Fix C and $w > \omega_0$, and let $\ell \rightarrow \infty$. From (L2.12), (L2.18), (L6.1), and (L6.12),

$$\begin{aligned} \limsup_{\ell \rightarrow \infty} \frac{1}{\ell^2} \log P_{R,\ell} &\leq \frac{1}{2} \log p - \vartheta_C \frac{a^3}{6p^3 D} \\ &\quad - G(C, w) - \hat{H}(C, w). \end{aligned} \quad (\text{L7.1})$$

The discretization term vanishes because $J_{\ell,d} = o_C(\ell^2)$. Let $n = \lfloor e^{\sigma \ell} \rfloor$. Since $\binom{n}{\ell} \leq n^{\ell}$, the expected number of red K_{ℓ} 's tends to zero whenever

$$\begin{aligned} \sigma &< -\frac{1}{2} \log p + \vartheta_C \frac{a^3}{6p^3 D} + G(C, w) + \hat{H}(C, w) \\ &= -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G(C, w) + \hat{H}(C, w). \end{aligned} \quad (\text{L7.2})$$

This proves the red threshold directly from the probability ledger.

For $q_\ell = \lfloor C\ell \rfloor$, (L2.42) and $q_\ell/\ell = C + O(\ell^{-1})$ give

$$\limsup_{\ell \rightarrow \infty} \frac{1}{\ell^2} \log P_{B,q_\ell} \leq \frac{C^2}{2} \log(1-p) + \frac{2a^3 C^3}{3(1-p)^3 D}. \quad (\text{L7.3})$$

The bound $\binom{n}{q_\ell} \leq n^{q_\ell}$ therefore makes the expected number of blue K_{q_ℓ} 's tend to zero whenever

$$\sigma < -\frac{C}{2} \log(1-p) - \frac{2a^3 C^2}{3(1-p)^3 D} = -\frac{1}{2} \log p_C + \frac{CB_B(C)}{2}. \quad (\text{L7.4})$$

No blue improvement is used. A union bound over red and blue vertex sets therefore produces a coloring with neither forbidden clique for every σ strictly below the minimum of the right sides of (L7.2) and (L7.4).

16.2. The crossing is separated by a constant. The pinned Gaussian asymptotics, all in the order $C \rightarrow \infty$, are

$$\begin{aligned} \frac{p}{p_C} &= 1 + O((\log C)^{-1}), & \frac{a^3}{p^3 D} &= \frac{1}{2} + o(1), & \vartheta_C &= 1 - o(1), \\ B_R(C) &= \frac{1}{6} + o(1), & CB_B(C) &= 1 + o(1). \end{aligned} \quad (\text{L7.5})$$

The exact integral (L2.17) also gives

$$G_*(C) = \frac{1 + o(1)}{32 \log C} = o(1). \quad (\text{L7.6})$$

Indeed $m(c - \omega_0)/m_0 \rightarrow 1$, $P_*(x) \rightarrow 1$ uniformly, and the bracket in (L2.17) tends uniformly to one, while $m_0^4/D^2 = (1 + o(1))/(8 \log C)$.

The controlled-residual constants satisfy

$$\begin{aligned} \rho &= o(1), & m'_0 &\rightarrow 1, & m_0 m''(c) &= o(1), \\ M_w/m_0 &\rightarrow 1, & j_\pm(w) &\rightarrow 1, & \mu_w &\rightarrow 1, \end{aligned} \quad (\text{L7.7})$$

where the right limit $w \downarrow \omega_0$ is taken for each fixed C before $C \rightarrow \infty$, and

$$\hat{L}_w = o(m_0^2), \quad \hat{B}_w = m_0^2[1 + o(1)]. \quad (\text{L7.8})$$

Thus every positivity and feasibility condition used above holds for all sufficiently large fixed C , uniformly for w in a sufficiently small right-neighborhood of ω_0 .

Equations (L7.5)–(L7.8) show

$$\frac{CB_B(C)}{2} - \left[\frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C) \right] = \frac{5}{12} + o(1) > 0. \quad (\text{L7.9})$$

This positive limiting gap, rather than the mere fact that the added terms are $o(1)$, proves that red remains the bottleneck. Taking the supremum of (L7.2) over $w > \omega_0$, using continuity as $w \downarrow \omega_0$, and then using (L7.9) proves Theorem 10.1.

16.3. The new leading coefficient. Finally,

$$m_0^2 = (2 + o(1)) \log C, \quad D^2 = (32 + o(1))(\log C)^3, \quad (\text{L7.10})$$

and hence

$$\hat{B}_w A_0 = (4 + o(1))(\log C)^2, \quad E_w = (2 + o(1))(\log C)^2. \quad (\text{L7.11})$$

Substitution into (L2.23) gives

$$\hat{H}_*(C) = \frac{(6 + o(1))(\log C)^2}{12(32 + o(1))(\log C)^3} = \frac{1 + o(1)}{64 \log C}. \quad (\text{L7.12})$$

The exact-merge predecessor is $H_*(C) = (1 + o(1))/(96 \log C)$, so the new term is eventually strictly larger. Since $B_R(C)/2 \rightarrow 1/12$, the public coarsening $B_R(C)/2 \geq 1/20$ also follows after increasing the existential threshold.

16.4. A precisely delimited scalar method cap.

Definition 16.1 (Admissible scalar residual family). Fix constants $M_0, M_{\text{ext}} > 0$. An (M_0, M_{ext}) -admissible family $\mathcal{E}$ assigns, for every sufficiently large C , every w in a right-neighborhood of ω_0 , and every $0 \leq r \leq \ell$, deterministic data

$$(e^{C,w,\ell,r}, T^{C,w,\ell,r}, \beta_{C,w,\ell,r}, \mu_{C,w,\ell,r}).$$

The same constants M_0, M_{ext} apply to every member of the family and to every C, w, ℓ, r . At $r = 0, 1$, the arrays are empty and we set $\beta = 0, \mu = 1$, and $\mathcal{G}_{\mathcal{E}} = 0$.

(A1) The increments are fixed before exposure and

$$0 \leq e_{ij} \leq M_0 m_0^2 r \quad (1 \leq i < j \leq r). \quad (\text{L7.13})$$

(A2) The array is the unique solution of the generalized recursion

$$T_{ij} = t_0 + \gamma_d \sum_{\substack{q>i \\ q \neq j}} T_{jq} + e_{ij}, \quad \Delta_{ij} = T_{ij} - t_0 \geq 0. \quad (\text{L7.14})$$

Thus neither the recursion nor the admissible class refers to the particular choice $e_{ij} = \theta(i-1)$.

(A3) Put

$$S_i = \sum_{\{j,q\} \subset V_i} \Delta_{jq}, \quad R_i = \max_{j \in V_i} \sum_{q \neq j} \Delta_{jq}, \quad k = |V_i|.$$

The three-factor split is feasible at every step:

$$\frac{4t_0 k}{d} + \frac{4R_i}{d} \leq \frac{1}{2}. \quad (\text{L7.15})$$

For $r \geq 2$, writing F_i^T for (L3.1) with coefficient array T , the two scalar coefficients obey the exact constraints

$$0 < \mu_{C,w,\ell,r} \leq \inf_{1 \leq i < r} \inf_{b \in [c-w, c+w]^{V_i}} \lambda_{\min}(-\nabla^2 F_i^T(b)), \quad 0 \leq \beta_{C,w,\ell,r} \leq m_0^2. \quad (\text{L7.16})$$

With these coefficients, define

$$\lambda_i^\varepsilon = \mathbf{d}_{r-i,w,d} + \beta_{C,w,\ell,r} \frac{S_i}{d} - \sum_{j>i} \frac{e_{ij}^2}{2\mu_{C,w,\ell,r}d} - \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (\text{L7.17})$$

The family must certify, on every history in $\mathfrak{H}_{r,i-1}^R$, the one-step inequality

$$\mathcal{K}_i^T \leq \frac{\mathcal{H}_C(r-i+1)}{\mathcal{H}_C(r-i)} e^{-\lambda_i^\varepsilon}. \quad (\text{L7.18})$$

Thus this scalar class credits exactly the displayed $\beta S_i/d$ term, pays the displayed scalar square penalty, and discards every favorable term not listed in (L7.17).

- (A4) Relative to the weighted propagation lemma (S3), the family's credited deterministic contribution is the normalized functional

$$\mathcal{G}_\mathcal{E}(C, w, \ell, r) = \frac{1}{d} \left[\beta_{C,w,\ell,r} \sum_{i=1}^{r-1} S_i - \frac{1}{2\mu_{C,w,\ell,r}} \sum_{1 \leq i < j \leq r} e_{ij}^2 \right]. \quad (\text{L7.19})$$

This definition excludes adaptive weights, matrix-valued completion, and unlisted source credits.

- (A5) After deleting u vertices, relabeling in retained order, and rebuilding the family member, with $x_+ = \max\{x, 0\}$,

$$0 \leq [\mathcal{G}_\mathcal{E}(C, w, \ell, \ell) - \mathcal{G}_\mathcal{E}(C, w, \ell, \ell - u)]_+ \leq \frac{M_{\text{ext}}}{\log C} \ell u \quad (\text{L7.20})$$

for every $0 \leq u \leq \ell$. Thus extraction is a quantified condition, not a reference to the optimizer-specific estimate.

Define

$$\begin{aligned} \text{Lead}_C(\mathcal{E}) &= \limsup_{w \downarrow \omega_0} \limsup_{\ell \rightarrow \infty} \frac{\mathcal{G}_\mathcal{E}(C, w, \ell, \ell)}{\ell^2}, \\ \mathfrak{M}_{M_0, M_{\text{ext}}}(C) &= \sup_{\mathcal{E}} \text{Lead}_C(\mathcal{E}), \end{aligned} \quad (\text{L7.21})$$

where the supremum is over (M_0, M_{ext}) -admissible families. The right-neighborhood in the first limsup may depend on C , but not on ℓ or r .

Proposition 16.2 (Cap within Definition 16.1). *For every fixed M_0, M_{ext} , the leading functional in (L7.21) satisfies*

$$\limsup_{C \rightarrow \infty} 64 \log C \mathfrak{M}_{M_0, M_{\text{ext}}}(C) \leq 1. \quad (\text{L7.22})$$

There are fixed absolute constants M_0^*, M_{ext}^* for which the family $e_{ij} = \theta(i-1)$, $\beta = \hat{B}_w(C)$, and $\mu = \mu_w$ is admissible and satisfies

$$\lim_{C \rightarrow \infty} 64 \log C \text{Lead}_C(\mathcal{E}^*) = 1. \quad (\text{L7.23})$$

Thus $1/64$ is both the cap and an attained leading coefficient within the precisely delimited scalar class, for fixed admissibility constants large enough to contain $\mathcal{E}^*$.

Proof. Fix an admissible family and abbreviate its two scalar coefficients by β and μ . The exact constraint (L7.16) gives

$$\beta \leq m_0^2. \quad (\text{L7.24})$$

At $b = c\mathbf{1}$, the row condition (L7.15) and the diagonal formula (L-A.A.1) give, for every $j \in V_i$,

$$\mu \leq \lambda_{\min}(-\nabla^2 F_i^T(c\mathbf{1})) \leq (-\nabla^2 F_i^T(c\mathbf{1}))_{jj} \leq m'_0 + \frac{1}{8}m_0 m''(c) = 1 + o(1). \quad (\text{L7.25})$$

Here $t_0 k + R_i \leq d/8$, $m'_0 \rightarrow 1$, and $m_0 m''(c) = o(1)$. Hence every valid common scalar completion constant satisfies, uniformly over the class,

$$\mu \leq 1 + o(1). \quad (\text{L7.26})$$

Let $\mathcal{L}$ be the feedback operator in (L7.14). Its row norm is at most $\gamma_{dr} \leq m_0 m'_0 / D = \rho = o(1)$. Writing

$$g_{ij} = A_0(r - i - 1),$$

the recursion for Δ is $\Delta = g + \mathcal{L}\Delta + e$. Therefore

$$\Delta = (I - \mathcal{L})^{-1}g + e + O(\rho m_0^2 r) \quad (\text{L7.27})$$

uniformly on every edge, with an implicit constant depending only on M_0 , by (L7.13). The exact exposure multiplicity is

$$\sum_{i=1}^{r-1} S_i = \sum_{1 \leq j < q \leq r} (j-1) \Delta_{jq}. \quad (\text{L7.28})$$

The first term of (L7.27) contributes

$$2A_0 \binom{r}{4} [1 + O(\rho)] \quad (\text{L7.29})$$

to (L7.28). The last term changes the numerator in (L7.19) by at most $O_{M_0}(\rho m_0^4 r^4) = o(m_0^4 r^4)$.

For the direct residual term, scalar square completion is pointwise:

$$\beta(j-1)e_{jq} - \frac{e_{jq}^2}{2\mu} \leq \frac{\mu\beta^2}{2}(j-1)^2. \quad (\text{L7.30})$$

Summing (L7.30) gives $\mu\beta^2 K_r/2$. Set $r = \ell$, divide (L7.19) by ℓ^2 , and use $\binom{\ell}{4}/\ell^4 \rightarrow 1/24$, $K_\ell/\ell^4 \rightarrow 1/12$, and $d/\ell^2 \rightarrow D^2$. Equations (L7.24)–(L7.30) yield

$$\text{Lead}_C(\mathcal{E}) \leq \frac{\beta A_0}{12D^2} + \frac{\mu\beta^2}{24D^2} + o\left(\frac{m_0^4}{D^2}\right) \leq \frac{m_0^4}{8D^2} [1 + o(1)] = \frac{1 + o(1)}{64 \log C}. \quad (\text{L7.31})$$

The error is uniform for fixed M_0, M_{ext} , so taking the supremum and then the limsup proves (L7.22).

For $e_{ij} = \theta(i-1)$, equations (L5.5), (L7.7), and (L7.8) make (L7.30) asymptotically tight. Equations (L5.11), (L6.2), and (L4.4) verify (L7.15)–(L7.18), and (L7.13) holds with $M_0^* = 1$. To verify (L7.20), write $\Delta^{(r)}$ for the

array rebuilt at target r , and put $r_- = \ell - u$. Subtracting the two positive recursions (L7.14), and using $\|\mathcal{L}\|_\infty \leq \rho < 1/2$, gives absolute constants K_1, K_2 such that

$$\max_{1 \leq j < s \leq r_-} \left| \Delta_{js}^{(\ell)} - \Delta_{js}^{(r_-)} \right| \leq K_1 m_0^2 u, \quad \max_{j < s \leq \ell} \Delta_{js}^{(\ell)} \leq K_2 m_0^2 \ell. \quad (\text{L7.32})$$

Indeed, the second bound is (L5.1). If U denotes the first maximum, subtraction of (L7.14) gives $U \leq A_0 u + \rho U + \gamma_d u K_2 m_0^2 \ell$. Now $A_0 = O(m_0^2)$, $\gamma_d \ell \leq \rho$, and $\rho < 1/2$ give the first bound. The multiplicity identity (L7.28) and (L5.4) now give

$$\left| \sum_i S_i^{(\ell)} - \sum_i S_i^{(r_-)} \right| \leq K_3 m_0^2 \ell^3 u, \quad |K_\ell - K_{r_-}| \leq K_4 \ell^3 u \quad (\text{L7.33})$$

for absolute K_3, K_4 . Since $\beta \leq m_0^2$, $\mu_w \rightarrow 1$, and $m_0^4/D^2 = O(1/\log C)$, (L7.33) proves (L7.20) for some absolute M_{ext}^* . Finally, the functional is exactly (L2.23) up to $o(1/\log C)$, and (L7.12) gives the asserted limit (L7.23). $\square$

The proposition does not quantify over outcome-adaptive weights, matrix-valued square completion, higher-order potentials, a sharper source quadratic MGF, or a different extraction invariant.

17. REPRODUCIBILITY AND CLAIM BOUNDARY

The frozen pre-canonical route note, arithmetic checker, and its independent source-level review have SHA-256 prefixes **2234a769**, **616040b9** and **ef6ddbce**, respectively. Their historical status labels do not override this manuscript's canonical theorem statement or proof-status ledger. The checker replays the recursion, multiplicity identities, Hessian samples, exact-CGF increments, finite margins at four large C -values and convergence to the leading constant. It is an diagnostic arithmetic witness, not a substitute for the reverse-induction proof.

From the repository root run

```
bash scripts/bootstrap.sh
bash scripts/reproduce_lower.sh
```

to reproduce the bridge, exact-CGF predecessor, nonexchangeable ledger and controlled-residual arithmetic calculations with pinned dependencies. A passing arithmetic replay is not labelled a theorem proof.

The theorem is source-relative, fixed- C , asymptotic in ℓ , and non-effective in $C_0(K)$. It supplies no new blue-side construction and no finite Ramsey number. The current manuscript does not claim a common- normalization strict improvement over the final printed Lin–Niu theorem, global optimality, priority, publication, or world-best status.

Logical proof status. As a standalone theorem without source interfaces 11.2 and 11.4, (L1.1) is PROVABLE AFTER WEAKENING / EXTRA ASSUMPTION, not an unconditional claim. Under the explicit source items (S1), (S2), (S4), and (S5), the weighted one-step comparison, local propagation lemma (S3), residual contribution, extraction algebra, and crossing argument are PROVABLE AS STATED. In particular, the manuscript does not replace the imported source interfaces or the local propagation proof by numerical evidence.

Part 3. A proof-carrying finite Ramsey barrier

18. FINITE RAMSEY BARRIER: PROBLEM AND PROOF ARCHITECTURE

Write $R(3, k)$ for the least n such that every graph on n vertices contains either a triangle or an independent set of order k . The current dynamic survey records

$$100 \leq R(3, 18) \leq 120;$$

see Radziszowski’s continuously maintained table [Rad26]. The public construction of Nagda, Raghavan, and Thakurta supplies a triangle-free graph on 99 vertices with independence number 17, thereby certifying the lower endpoint [NRT26]. Extending that graph by one vertex produces a natural, but logically delicate, local search around the next possible certificate.

This part studies one labelled 100-vertex extension H . The graph has no independent 18-set but has a unique triangle. It is therefore a near miss, not a Ramsey graph. We ask a deliberately local question: how many *original* edges must be deleted before the triangle can be removed while arbitrary original nonedges are allowed to change into edges? This asymmetric edit model makes additions completely free; consequently, a lower bound in this model is stronger than a lower bound obtained by fixing the nonedges or by restricting the total symmetric-difference distance.

The principal result is a proof-carrying lower bound of seven on that local deletion radius. The argument has four independently inspectable layers.

- (i) A direct graph check establishes the seed identity, its unique triangle, and the absence of an independent 18-set.
- (ii) The unique triangle gives an exhaustive three-way branch split. A previous proof-carrying package excludes total deletion budget at most five, reducing the new layer to exact budget six.
- (iii) Each exact-six branch is translated into a finite CNF *relaxation* of the graph problem. All structural clauses are reconstructed from the seed; the independent-set clauses are validated as universally sound.
- (iv) CaDiCaL emits a DRAT refutation for each branch, and a separately pinned `drat-trim` executable verifies every refutation. Solver status strings are not used as proof evidence.

This layering is important. An unsatisfiable CNF proves the graph claim only after formula soundness and branch completeness have been established.

A solver’s report of unsatisfiability proves the CNF claim only after an independently checkable proof trace has been accepted. We record both implications explicitly.

The computation is also informative about what remains open. We first tested the next deletion layer, exact budget seven, with one bounded solver call in each branch. Every call reached its wall-time limit before producing a separated model or a proof. A follow-up branch-1 analysis proves the necessary degree cap $\deg(v) \leq 17$ and thereby excludes 87.5813...% of its raw six-edge residual deletion sets. A deletion-first pilot encoded that lift and initially encountered two apparent exact independent-set separator stalls. An order-aware reconstruction showed that both stalls were vertex-order artefacts: reverse ordering returned the requested witness batches in milliseconds. Seven repaired gates then spent their bounded walls in the master after fast separation, but again produced neither a verified model nor a checked global proof. A subsequent proof-carrying generalized-core route produced four stronger, checked singleton no-goods for branch 1: every satisfying assignment of the common relaxation must preserve each of four named seed edges. Their CNFs were rebuilt byte for byte without the production encoder and their DRAT traces were replayed by separately built checker binaries on two architectures. A checked satisfying model shows that the common relaxation is nonempty, although that model itself contains an independent 18-set. The singleton clauses exclude an additional 1307748575589 residual supports beyond the degree cap, about 2.431110% of its survivors. This is a rigorous family-level pruning result, but it leaves branch 1 and the overall exact-seven problem open. A subsequent exact projection found 396 forced original-nonedge assignments but no additional pure deletion-support clause. We then proved, using the checked budget-six obstruction, that every feasible branch-1 repair has a maximal-triangle-free representative. Combining a full all-pair encoding of that normal form with the complete 530525-mask union gives a fixed CNF with 639290 variables and 1972360 clauses. Independent reconstruction was byte-exact, but its single 300-second bounded call timed out with no model or proof. We record this and every other unresolved endpoint as UNKNOWN; none is evidence either for or against the existence of a seven-deletion repair.

Claim boundary. The theorem concerns one fixed, labelled seed and one one-sided edit metric. It neither constructs a 100-vertex $(3, 18)$ -Ramsey graph nor rules out such graphs elsewhere in the search space. Thus it does not improve either endpoint of the globally tabulated interval for $R(3, 18)$. The value of the result is instead methodological and local: it gives a complete, proof-carrying obstruction around a particularly close seed, with all negative conclusions separated from incomplete search.

19. THE FROZEN SEED AND THE PRECISE STATEMENT

All vertices are labelled by

$$V = \{0, 1, \dots, 99\}.$$

The input graph H is the adjacency matrix in the artifact

`r3_18_n100_nearmiss.txt`.

Its SHA-256 identity is

`e6527601f72ebb2d3aed11adcdd2fffb0c6aab5e326d81d439c10ee468844e85e`.

Content addressing is part of the mathematical statement: replacing the matrix, even by an isomorphic relabelling, changes the frozen input unless the new identity is separately certified.

Lemma 19.1 (Seed facts). *The graph H has 827 edges, satisfies $\alpha(H) < 18$, and has exactly one triangle, namely*

$$T = \{97, 98, 99\}.$$

Its first 99 vertices induce a graph G_0 with 809 edges, no triangle, and $\alpha(G_0) = 17$.

Computational proof. The matrix parser checks that the input is a symmetric 100×100 binary matrix with zero diagonal. Exhaustive enumeration of the $\binom{100}{3} = 161700$ triples finds only T . A bitset maximum-clique search on the complement returns no clique of order 18, which is equivalent to $\alpha(H) < 18$. The 99×99 principal block agrees entry-for-entry with the separately frozen matrix G_0 , whose SHA-256 is

`3e9d16e29111f3a2f25ae3b992235804c2612de2f6ada5570901d17ceedfca45`.

The same verifier finds no independent 18-set in G_0 and returns an independent 17-set. It also finds no triangle in G_0 . The new vertex has neighbourhood

$$N_H(99) = \{33, 34, \dots, 48, 97, 98\},$$

and the only edge induced by this neighbourhood is $(97, 98)$, providing a second direct check of uniqueness. The negative clique routine was compared with literal subset enumeration on 8640 small random graph/target pairs; all decisions and returned witnesses agreed. Reproduction commands are given in Section 24 and Appendix G. $\square$

Evidence grade and logical role. The assertion $\alpha(H) < 18$ is an exact deterministic trusted-code result: the frozen matrix and inspectable bitset search reproduce it, and exhaustive small-instance comparisons test the pruning invariant. It is not backed by a DRAT/LRAT trace or a proof-assistant certificate. This assertion describes why H is a near miss, but it is not a logical premise of Theorem 19.3. The deletion-barrier proof uses the fixed edge set and seed triangle of H , while the independence-number hypothesis is imposed on the putative final graph F . Thus the implication proved below does not infer anything from $\alpha(H) < 18$.

The graph G_0 reproduces the already tabulated lower bound $R(3, 18) \geq 100$. The graph H , by contrast, contains a triangle and therefore does not certify $R(3, 18) \geq 101$.

Definition 19.2 (Free-addition deletion distance). For a graph F on V , define

$$d_H(F) = |E(H) \setminus E(F)|.$$

Only missing input edges are charged. Every pair in $\binom{V}{2} \setminus E(H)$ may independently be added to F at zero cost. Let

$$\mathcal{F} = \{F \text{ on } V : F \text{ is triangle-free and } \alpha(F) < 18\}$$

and define

$$\rho(H) = \inf\{d_H(F) : F \in \mathcal{F}\},$$

with the convention $\inf \emptyset = \infty$.

Theorem 19.3 (Fixed-seed deletion barrier). *For the content-addressed graph H above,*

$$\rho(H) \geq 7.$$

Equivalently, no $F \in \mathcal{F}$ satisfies $d_H(F) \leq 6$, even when arbitrary subsets of the 4123 original nonedges are added.

The convention in the definition matters: Theorem 19.3 is compatible both with $\rho(H) = 7$ and with a larger or infinite radius. We do not determine which occurs.

20. REDUCTION TO SIX PROOF-CARRYING BRANCH FORMULAS

Let

$$f_0 = (97, 98), \quad f_1 = (97, 99), \quad f_2 = (98, 99)$$

be the three edges of the unique seed triangle. We first state the checked dependency at the preceding deletion layer.

Proposition 20.1 (Budget-five dependency). *There is no $F \in \mathcal{F}$ with $d_H(F) \leq 5$.*

Proof-carrying computation. Every triangle-free F omits at least one f_i . In branch i , the unit $\neg x_{f_i}$ is imposed and a sequential counter permits at most four further deletions among the other 826 input edges. All triangle clauses and a finite bank of universally valid independent-18 hitting clauses are imposed. The resulting three CNFs have respectively 169382, 184967, and 177799 clauses. All three formulas were reconstructed clause-by-clause from H , and all three DRAT refutations were replayed with a freshly compiled checker from the pinned source revision. The full identities are in Table 7; the formula-soundness argument is the same as Proposition 21.2. Therefore each branch is impossible, and the three branches cover every candidate of total deletion budget at most five. $\square$

By Proposition 20.1, a counterexample to Theorem 19.3 would have exactly six deleted input edges. Fix any one absent edge f_i of the seed triangle. The other five deletions are counted among $E(H) \setminus \{f_i\}$. Notice that the other two triangle edges are not fixed present; if either is also deleted, it simply consumes one of these five residual deletions. Thus the branches cover all candidates and may overlap.

For each $i \in \{0, 1, 2\}$, let Φ_i be the formula defined precisely in Section 21. It contains the fixed negative unit for f_i , an exact-five counter on the residual seed edges, all triangle clauses, and a branch-specific bank of independent-set hitting clauses.

Lemma 20.2 (Exact-six branch cover). *If $F \in \mathcal{F}$ and $d_H(F) = 6$, then the edge assignment of F extends to a satisfying assignment of at least one of Φ_0, Φ_1, Φ_2 .*

Proof. Because F is triangle-free, at least one edge f_i of T is absent. Choose such an i . The chosen unit clause holds. Exactly five members of $E(H) \setminus \{f_i\}$ are absent, so the primary edge assignment extends to the auxiliary variables of the exact-five counter. Triangle-freeness satisfies every triangle clause. Finally, $\alpha(F) < 18$ implies that each installed 18-set contains a present pair, so every installed hitting clause holds. Hence F 's primary assignment extends to a model of Φ_i . $\square$

Proposition 20.3 (Certified branch refutations). *Each formula Φ_0, Φ_1, Φ_2 is unsatisfiable.*

Proof certificate. The three frozen DIMACS files have the identities and dimensions in Table 2. The corresponding DRAT traces were accepted by the pinned `drat-trim` checker, each reporting the exact status line `s VERIFIED`. Before replay, independent semantic checkers reconstructed the entire triangle prefix, cardinality block, fixed unit, and every 153-literal hitting clause; they also checked the compressed and raw identities. Details are given in Section 22 and Appendix G. $\square$

Proof of Theorem 19.3. Suppose $F \in \mathcal{F}$ and $d_H(F) \leq 6$. Proposition 20.1 implies $d_H(F) = 6$. By Lemma 20.2, F 's edge assignment extends to a model of at least one Φ_i . This contradicts Proposition 20.3. Therefore no such F exists and $\rho(H) \geq 7$. $\square$

21. CNF ENCODING AND ITS SOUNDNESS

21.1. Primary variables and structural clauses. For every unordered pair $e = \{u, v\} \in \binom{V}{2}$, a primary variable x_e records whether $e \in E(F)$. The 4950 primary variables are numbered in lexicographic pair order. Crucially, variables exist for all original nonedges; those variables are absent from the deletion counter and are not fixed false.

Triangle-freeness is encoded without separation: for every $a < b < c$, the formula contains

$$\neg x_{ab} \vee \neg x_{ac} \vee \neg x_{bc}.$$

There are exactly $\binom{100}{3} = 161700$ such clauses.

For every installed set $S \in \binom{V}{18}$, the formula contains the positive hitting clause

$$C_S = \bigvee_{e \in \binom{S}{2}} x_e. \quad (\text{F4.1})$$

It has $\binom{18}{2} = 153$ distinct primary literals. Every graph with independence number below 18 satisfies C_S . We install only a finite selected bank $\mathcal{B}_i$, not all $\binom{100}{18}$ clauses. Consequently, Φ_i is a relaxation of the exact Ramsey branch. This is the useful direction: unsatisfiability of the relaxation implies unsatisfiability of the exact branch.

21.2. Exact deletion counter. In branch i , let

$$L_i = (\neg x_e : e \in E(H) \setminus \{f_i\})$$

be the ordered list obtained by restricting the lexicographic pair order to the seed edges other than f_i . The list has 826 literals. A PySAT sequential-counter equality encoding implements

$$\sum_{\ell \in L_i} [\ell \text{ is true}] = 5. \quad (\text{F4.2})$$

We use the standard sequential counter of Sinz [Sin05]. The concrete generator is exposed by PySAT [IMaMS18]; its semantics is not used as an unchecked black box.

For completeness, define the irredundant sequential schema explicitly. Given signed literals $z_0, \dots, z_{n-1}$ and $1 \leq q \leq n-2$, introduce $y_{r,j}$ for $0 \leq r < q$ and $0 \leq j < n-q$, and let $\text{SC}_{\leq q}(z)$ contain

$$\begin{aligned} \neg z_j \vee y_{0,j} & \quad (0 \leq j < n-q), \\ \neg y_{r,j} \vee y_{r,j+1} & \quad (0 \leq r < q, 0 \leq j < n-q-1), \\ \neg z_{j+r+1} \vee \neg y_{r,j} \vee y_{r+1,j} & \quad (0 \leq r < q-1, 0 \leq j < n-q), \\ \neg z_{j+q} \vee \neg y_{q-1,j} & \quad (0 \leq j < n-q). \end{aligned}$$

Lemma 21.1 (Sequential-counter projection semantics). *For every assignment of the primary variables,*

$$\exists y \text{ SC}_{\leq q}(z, y) \iff \sum_{j=0}^{n-1} [z_j \text{ is true}] \leq q. \quad (\text{F4.3})$$

Consequently, for $n = 826$,

$$\text{SC}_{\leq 821}(\neg z) \wedge \text{SC}_{\leq 5}(z)$$

has an auxiliary extension if and only if exactly five members of z are true. The budget-five block $\text{SC}_{\leq 4}(z)$ has an auxiliary extension if and only if at most four members are true.

Proof. Suppose first that at most q input literals are true. Assign $y_{r,j} = 1$ exactly when at least $r + 1$ of $z_0, \dots, z_{j+r}$ are true. The first clause family records the first threshold crossing. The second propagates an attained threshold to a longer prefix. In the third family, truth of both z_{j+r+1} and $y_{r,j}$ raises the prefix count from at least $r + 1$ to at least $r + 2$. The last family cannot be violated, because such a violation would exhibit $q + 1$ true inputs. This gives the required auxiliary extension.

Conversely, induction on r shows that the clauses force $y_{r,j} = 1$ whenever at least $r + 1$ of $z_0, \dots, z_{j+r}$ are true. For $r = 0$, use the first clause at the position of a true input and then the propagation clauses. For the induction step, let p be the position of the $(r + 1)$ -st true input and put $j' = p - r$. The induction hypothesis forces $y_{r-1,j'}$; the third family, together with z_p , forces $y_{r,j'}$, and propagation reaches $y_{r,j}$. If $q + 1$ inputs were true, let p be the position of the $(q + 1)$ -st and put $j = p - q$. The preceding argument forces $y_{q-1,j}$, while the last family forbids its conjunction with z_p , a contradiction. This proves (F4.3). Applying (F4.3) to z and to its 826 complements gives respectively the upper and lower sides of the equality claim; the at-most-four claim is the direct case $q = 4$. $\square$

In the exact-five DIMACS block, the complement counter at bound 821 comes first and has 4105 auxiliary variables and 7394 clauses; the positive counter at bound 5 has another 4105 auxiliaries and 9026 clauses. Thus the equality block has 8210 auxiliary variables and 16420 clauses, ending at variable 13160. The checker `routes/finite/check_r3_18_seqcounter.py` does not import or call PySAT's cardinality generator. It independently generates the four clause families, compares all six actual counter blocks clause-for-clause in their canonical literal order, checks their digests and variable ranges, and constructs the prefix-threshold auxiliary assignment from the proof above. Exhaustive small-instance tests enumerate both primary and auxiliary assignments and verify both directions of the projected at-most and equality semantics.

Thus

$$\Phi_i = \left(\bigwedge_{a < b < c} (\neg x_{ab} \vee \neg x_{ac} \vee \neg x_{bc}) \right) \wedge \text{Card}_{=5}(L_i) \wedge \neg x_{f_i} \wedge \bigwedge_{S \in \mathcal{B}_i} C_S. \quad (\text{F4.4})$$

TABLE 2. Exact-six branch dimensions. All branches have 4950 primary variables, 8210 counter auxiliaries, 161700 triangle clauses, 16420 counter clauses, and one fixed unit.

branch	fixed absent edge	hitting clauses	variables	total clauses
0	(97, 98)	251771	13160	429892
1	(97, 99)	63943	13160	242064
2	(98, 99)	5422	13160	183543

TABLE 3. Audited construction of the branch-0 universal cut union.

source	source masks	new masks	cumulative
branch-0 discovery checkpoint	183133	183133	183133
branch-0 earlier bank	141149	4096	187229
branch-1 bank	63943	52515	239744
branch-2 bank	5422	3931	243675
budget-five branch 0	287	286	243961
budget-five branch 1	15872	4262	248223
budget-five branch 2	8704	3548	251771

Proposition 21.2 (CNF soundness). *For each i , every graph $F \in \mathcal{F}$ with $d_H(F) = 6$ and $f_i \notin E(F)$ induces a primary assignment that extends to a model of Φ_i .*

Proof. The triangle block holds by triangle-freeness. The fixed unit holds by the branch premise. Since f_i is one of exactly six missing input edges, exactly five literals in L_i are true; Lemma 21.1 supplies an auxiliary assignment. Each $S \in \mathcal{B}_i$ contains a present pair because $\alpha(F) < 18$, so every C_S holds. No condition was imposed on an original nonedge except through triangle and hitting clauses, hence arbitrary additions remain represented. $\square$

21.3. Sound reuse of the branch-0 cut union. Branch 0 uses 251771 distinct 18-sets deduplicated from seven checked sources. The ordered mask digest is

f10690b826b86eb03567a2ffaaffb553801fae32af20cfe4337118bddf4e41afa.

Only vertex-set masks were transferred. No learned clause, branch assumption, or solver conclusion was reused. Since every clause C_S in (F4.1) follows solely from $\alpha(F) < 18$, it is valid in every branch. The source counts and cumulative deduplication totals appear in Table 3.

22. PROOF PRODUCTION, CHECKING, AND TRUST

22.1. From finite formulas to checked refutations. The final proof-producing runs used CaDiCaL [BFF⁺24] at source commit

c60730422e758ef1cebe7aeddff2dda31c996bf04.

An exit code 20 and the text UNSATISFIABLE were treated only as requests to check the emitted proofs. The final mathematical evidence is the successful replay of the DRAT traces [WHH14, Heu16] by the official `drat-trim` source at commit

2e3b2dc0ecf938adbbd779d42877b6ed69d9a985.

The principal recorded executable has SHA-256

31d9e8e04bc76a65f0a18ea212ac44af4cf7bc398629495cec093ced772c50f4.

TABLE 4. Exact-six proof artifacts and replay scale. Hashes are abbreviated here and given in full in Table 8.

branch	raw DRAT bytes	proof lines	compressed bytes	replay
0	3493847713	6649939	656144450	verified
1	2137671968	5590150	433552903	verified
2	759008359	2837771	152784369	verified

All three checks terminated with exit code zero and reported an exact `s VERIFIED` status line.

For branch 0, a fresh replay in a temporary directory took 273.448 seconds inside the checker and reported 135152646 resolution steps in the retained core. Branch 1 reported 209897525 resolution steps in its independently checked run. These timings are descriptive, not logical premises. The source commit and successful replay are the portable evidence; the executable hash records the historical build and is not required to match a fresh binary compiled on another platform.

22.2. Semantic reconstruction. The proof checker establishes unsatisfiability of the bytes in each DIMACS file. Separate semantic checkers establish that those bytes are the intended relaxations. For every branch they independently:

- (1) parse the frozen seed and regenerate the lexicographic pair-variable map;
- (2) compare all 161700 triangle clauses in order;
- (3) regenerate the production PySAT counter and, in a separate checker that does not import `CardEnc`, independently generate the sequential-counter schema and compare its variables and clauses in order;
- (4) check the branch unit and the exact clause partition;
- (5) reconstruct every hitting clause as all 153 pair variables of one 18-vertex mask;
- (6) verify the compressed and raw sizes and SHA-256 identities; and
- (7) verify the budget-five dependency and the three-edge cover.

Branch 0 additionally reparses all seven cut sources and reproduces the deduplicated union. Branches 1 and 2 likewise reconstruct their complete stored cut banks. An earlier interrupted branch-1 trace fails gzip integrity, remains quarantined, and is excluded from every manifest and conclusion.

22.3. Trust boundary. The claim is proof-carrying, not formally verified in a proof assistant. Its earliest machine dependencies are:

- the bitset independent-set verifier used for the descriptive seed fact $\alpha(H) < 18$, which is not a premise of the deletion-barrier implication;
- the `drat-trim` proof kernel, together with its compiler and execution hardware; and

- the independently inspectable scripts that map matrices, sequential schemas, and masks to DIMACS.

The audit narrows this boundary through independent matrix parsing, exhaustive triangle enumeration, small-instance cross-checks of the clique routine, exhaustive small-instance projection tests for the independent counter schema, constructive auxiliary assignments, complete clause reconstruction, content hashes, gzip integrity tests, and proof replay with a freshly compiled checker. PySAT remains production provenance, but Lemma 21.1 and the CardEnc-free comparison mean that its counter semantics is no longer assumed merely because PySAT emitted the bytes. A future LRAT conversion and formally verified LRAT checker could reduce the trusted kernel further, but is not required for the conventional proof-carrying SAT standard used here.

23. THE UNRESOLVED EXACT-SEVEN LAYER

The theorem implies only $\rho(H) \geq 7$. To distinguish this lower bound from an exact value, we report bounded experiments at deletion budget seven.

In branch i , the fixed unit $\neg x_{f_i}$ accounts for one deletion and an equality counter requires exactly six of the other 826 input edges to be absent. All 4123 original nonedges remain free. The universal bank of 251771 hitting clauses from branch 0 is valid in every branch and was preloaded. Every initial formula therefore had 4950 primary variables, 9840 counter auxiliaries, 161700 triangle clauses, 19680 counter clauses, one fixed unit, and 251771 hitting clauses, for a total of 433152 clauses.

Computational observation 23.1 (Bounded endpoints, not a theorem). One solver call was launched per branch with a parent-enforced 300-second discovery wall. Branch 0 used CaDiCaL, branch 1 Glucose 4.2, and branch 2 MapleChrono. Each remained inside its first SAT call until the wall expired. No branch produced a model that reached independent-set separation, and no branch produced a checked refutation. The endpoint of all three branches is therefore UNKNOWN.

The zero recorded separation iterations means that no SAT model returned before interruption; it does not mean the formula is unsatisfiable. The recoverable state consists of the completely reconstructed initial formula and universal cut bank. Internal CDCL state was not serialized. Exact hashes of the frozen result and checkpoint records are listed in Appendix H.

Proposition 23.2 (A degree-cap lift for branch 1). *Let F be triangle-free with $\alpha(F) < 18$. Then $\deg_F(v) \leq 17$ for every vertex v . In branch 1, where (97, 99) is fixed absent, every feasible set of six residual seed-edge deletions contains an edge incident with vertex 98. Consequently this condition alone excludes*

$$\frac{\binom{808}{6}}{\binom{826}{6}} = 0.875813014052041 \dots$$

of the raw residual deletion sets.

Proof. The open neighbourhood of any vertex in a triangle-free graph is independent, so its size is at most 17. After deleting the fixed branch edge (97, 99), vertex 98 still has seed degree 18. If D denotes the six residual deletions and A the selected seed-nonedge additions, then

$$\deg_F(98) = 18 - |D \cap \delta(98)| + |A \cap \delta(98)| \leq 17.$$

Thus D must meet the 18 residual seed edges incident with vertex 98. There are 808 residual edges not incident with 98, so precisely $\binom{808}{6}$ of the $\binom{826}{6}$ raw six-edge sets are excluded. $\square$

For the rest of this section put

$$E_1 = E(H) \setminus \{(97, 99)\}, \quad \Delta_{98} = \{e \in E_1 : 98 \in e\}.$$

Thus $|E_1| = 826$, $|\Delta_{98}| = 18$, and a branch-1 repair has a residual deletion support $D \in \binom{E_1}{6}$.

Lemma 23.3 (Soundness of the branch-1 common relaxation). *Let Ψ_1 be the CNF consisting, in order, of all triangle clauses, an exact-six counter on the literals $\{\neg x_e : e \in E_1\}$, the unit $\neg x_{97,99}$, one degree-at-most-17 counter for every vertex, and the ordered bank of 251771 universal independent-18 hitting clauses. Every feasible branch-1 exact-seven repair extends to a satisfying assignment of Ψ_1 .*

Proof. The primary variables x_{uv} record the final graph. Triangle-freeness satisfies every triangle clause and the branch choice satisfies the fixed unit. Exactly six edges of E_1 are absent, so the primary assignment has an extension through the exact counter. The degree-cap proposition gives an extension through every degree counter. Finally, for each stored 18-set S , the clause

$$\bigvee_{\{u,v\} \in \binom{S}{2}} x_{uv}$$

is necessary when $\alpha(F) < 18$. Hence every clause family is satisfied. The bank need not contain every 18-set, so Ψ_1 is a relaxation rather than an exact encoding. $\square$

Proposition 23.4 (Four proof-checked singleton entailments). *Define*

$$a = (11, 62), \quad b = (18, 61), \quad c = (18, 64), \quad d = (18, 69).$$

All four edges lie in $E_1 \setminus \Delta_{98}$. The frozen relaxation satisfies

$$\Psi_1 \models x_a \wedge x_b \wedge x_c \wedge x_d.$$

Thus none of these four seed edges can belong to the residual deletion support of a feasible branch-1 repair. No minimality assertion is made.

Proof. For each $e \in \{a, b, c, d\}$, a standalone CaDiCaL run refuted

$$\Psi_1 \wedge \neg x_e,$$

and the pinned `drat-trim` checker accepted each emitted proof. These four checked refutations are equivalent to the displayed entailment by Ψ_1 . They are not asserted to be equivalent to the target-graph statement; the necessary condition for target graphs follows only in the forward direction from Lemma 23.3.

Each DIMACS instance has 154190 variables and 718453 clauses. A PySAT-free checker that imports neither the production formula builder nor the exporter reconstructed all four compressed CNFs byte for byte, including the candidate-unit polarity. A separately compiled checker binary on a second architecture replayed all four proofs to the exact status `s VERIFIED`. The formula, proof, checker, and artifact identities are recorded in Appendix H. $\square$

Computational observation 23.5 (The common relaxation is nonempty). A complete CaDiCaL model for Ψ_1 was checked against an independent byte-for-byte reconstruction of all 718452 clauses. The model sets all four displayed edge variables to true, as the proposition requires. It also contains the independent 18-set

$$\{99, 96, 97, 95, 92, 90, 87, 91, 85, 93, 84, 73, 70, 88, 82, 81, 78, 83\}.$$

Hence the singleton entailments are not consequences of an empty common relaxation, but this particular model is not an exact-seven Ramsey repair. The satisfiability of Ψ_1 resolves neither existence nor nonexistence for the target problem because the universal bank is incomplete.

Corollary 23.6 (Support exclusion beyond the degree cap). *Every feasible branch-1 residual support D meets Δ_{98} and avoids $\{a, b, c, d\}$. The singleton no-goods exclude an additional*

$$1307748575589$$

supports beyond those already excluded by the degree cap. Together the two filters exclude

$$380670699850897$$

of the $\binom{826}{6} = 433155188594590$ raw supports and leave 52484488743693 outside these two necessary-condition filters.

Proof. The support restrictions follow from the degree-cap proposition, Lemma 23.3, and Proposition 23.4. Avoiding the four distinguished edges leaves 822 eligible residual edges. Because those four edges avoid Δ_{98} , the supports that both avoid them and fail the degree test are precisely the six-subsets of the other 804 nonincident edges. Hence the number surviving both filters is

$$\binom{822}{6} - \binom{804}{6} = 52484488743693.$$

Subtracting this from the degree-cap survivor count gives the additional 1307748575589 exclusions. Equivalently, the singleton clauses remove about 2.431110% of the degree-cap survivors, and the combined filters exclude

about 87.883214% of the raw support universe. These are support-index counts, not a wall-time speedup or a count of graphs. $\square$

The four retention units also permit a stricter audit of what follows from the local triangle and degree systems before any new SAT call. Propagating those systems to the original nonedges gives exactly 396 forced-absent addition variables: 388 come from pairs having more than six common neighbours in the fixed branch base, two are forced by the retained-star configuration, and six arise only from the coupled exact-six, $\delta(98)$, and degree constraints. Substitution of all 396 units into the finite I_{18} bank leaves no clause involving deletion variables alone. Thus this closure is useful preprocessing, but it supplies no new deletion-support no-good beyond the filters proved above.

There is, however, a clean normal form that applies to the target problem rather than merely to a finite bank.

Proposition 23.7 (Maximal-triangle-free normal form). *A feasible branch-1 exact-seven repair exists if and only if one exists that is maximal triangle-free. Equivalently, one may require every absent pair to have a common neighbour.*

Proof. Only the forward direction needs proof. Starting from a feasible repair F , repeatedly add an absent pair $uv \notin E(H)$ whenever u and v have no common neighbour. Such an addition creates no triangle, cannot increase the independence number, and does not change $E(H) \setminus E(F)$. The finite process terminates with the same exact seven-edge deletion support and with every still-absent original nonedge triangle-blocked.

It remains to consider an absent pair $e \in E(H)$. If e were safe to add, doing so would again preserve triangle-freeness and could not increase the independence number, while reducing the one-sided deletion count from seven to six. This contradicts the proof-checked theorem $\rho(H) \geq 7$. The argument also applies when $e = (97, 99)$: after adding that fixed branch edge, at least one of the other two edges of the seed's unique triangle remains absent, so the resulting graph is covered by the global three-branch budget-six exclusion. Hence every nonedge in the final graph has a common neighbour, which is precisely maximal triangle-freeness. The reverse implication is immediate. $\square$

This is an existence normal form, not a claim that every raw repair is already maximal. It is the fixed-support analogue of the standard use of maximal triangle-free graphs in computational Ramsey searches [BGSP12]. For a scoped pair uv , an exact existential CNF encoding introduces variables $y_{uv,w}$ and uses

$$x_{uv} \vee \bigvee_{w \notin \{u,v\}} y_{uv,w}, \quad \neg y_{uv,w} \vee x_{uw}, \quad \neg y_{uv,w} \vee x_{vw}.$$

Reverse implications are unnecessary: the y -variables only select one common-neighbour witness when uv is absent. Scoping this construction to the 4123 original nonedges uses 404054 auxiliaries and 812231 clauses;

the full maximal-triangle-free encoding uses 485100 auxiliaries and 975150 clauses. A complete three-vertex truth-table audit checks the existential semantics.

The two checked relaxation models illustrate the claim boundary. The common model has one unsafe original nonedge, (37, 86). Adding it would create a degree-18 vertex, so the resulting graph is not asserted to remain a model of the degree-capped incomplete-bank relaxation; the same pair instead recovers the independent-18 domination witness recorded below. The A+ model has exactly three unsafe original nonedges, (16, 64), (32, 98), (40, 89). Adding all three gives a maximal triangle-free primary graph with the same seven deletions and maximum degree 17, but it still contains the already known independent set $\{81, 82, \dots, 97, 99\}$. Maximality is therefore complementary to, not a replacement for, independent-set separation.

Nor can a branch-1 conclusion be transferred by relabelling. A deterministic degree-initialized one-dimensional Weisfeiler–Leman refinement has 3, 15, and 100 color classes at initialization, round one, and round two. Every automorphism preserves every refinement color, so the singleton round forces the automorphism group of the labelled seed to be trivial. The other two triangle-edge branches therefore require independent treatment.

For completeness, the domination diagnostic used above is the following elementary consequence of the target conditions. Put

$$Z_v = \{u \neq v : uv \notin E(F), N_F(u) \cap N_F(v) = \emptyset\}.$$

Then $N_F(v) \cup I$ is independent for every independent $I \subseteq F[Z_v]$, and hence $\alpha(F[Z_v]) \leq 17 - \deg_F(v)$. In particular, degree 17 forces $Z_v = \emptyset$, while degree 16 forces Z_v to be a clique of order at most two. On the common model this exposes the new mask 2000000404414192982822020; it has zero overlap with the universal, historical, A+, and fixed-base mask families. On the A+ model it yields no new mask, consistently with all three unsafe pairs having degree 16 at both endpoints. These are checked separators for relaxation models, not new theorems about existence.

We encoded the same degree cap at both levels of a deletion-first branch-1 pilot. The master selects six residual deletions and addition-eligibility variables, separates conditional independent-set clauses, and passes a fixed deletion set to an exact add-only subproblem. An incomplete master separation call may fall back to that exact subproblem. If the fixed-deletion subproblem is itself incomplete, the run stops without learning a deletion no-good. A completed fixed-deletion UNSAT may be used as solver-trusted search telemetry, but it is not promoted without a checked proof trace.

Computational observation 23.8 (Deletion-first endpoints, still not a theorem). A six-configuration bounded matrix learned between zero and 2816 new conditional independent-set cuts per run. One MapleChrono configuration reached two fixed-deletion UNSAT endpoints, but emitted no proof for them; the same run then stopped at the independent-set oracle limit. Three

predeclared positive-signal extensions raised that oracle limit from two million to twenty million nodes. Two stopped at exactly 20,000,001 recursive nodes inside a fixed-deletion subproblem after learning 10,780 and 13,362 new cuts, respectively; the third learned 12,800 cuts but reached its 300-second global wall before entering a subproblem. No run returned a verified candidate or a checked refutation. Every endpoint therefore remains UNKNOWN.

The two apparent fixed-deletion separator stalls were then reconstructed under an order-aware implementation. The historical ascending recursion exhausted its two-million-node allowance on both supports. Reversing the vertex order, which leaves the graph property unchanged, instead returned 512 validated independent 18-sets in 11,261 and 10,057 recursive nodes, respectively, taking about 0.006 seconds in each case. Thus the old twenty-million-node endpoints were an ordering artefact on these supports, not evidence of a combinatorial obstruction.

We consequently ran seven further bounded gates with a reverse-first, bidirectional exact oracle. On every master model for which separation was invoked, a batch of either 512 or 4096 validated cuts was returned in 0.005–0.075 seconds. Nevertheless, each gate reached its global wall after zero to five master models; none entered a fixed-deletion subproblem, returned a verified candidate, emitted a proof, or learned a new fixed-deletion no-good. Under these particular checkpoints and limits, every run reached its global wall while still in the master. This remains bounded, configuration-specific telemetry rather than a claim of intrinsic hardness. The command lines, stable implementation hashes, all sixteen deletion-first endpoints, and the stop rule are recorded in Appendix H.

We also tested a genuinely different deletion-only projection. It retains the exact residual counter, local wedge implications for addition eligibility, the complete 235504-member fixed-base I_{18} covering family, and the vertex-98 hit constraint, while forgetting the full edge assignment. Its natural exact-six control has 13159 variables and 264757 clauses. A pinned CaDiCaL call reached its predeclared 300-second wall without a status line. The incomplete proof prefix was hashed and deleted, and the exact-seven instance was not run under the predeclared stop rule. This endpoint is UNKNOWN; more importantly, the route failed its strength gate on an already known-unsatisfiable control and therefore supplied no reason to spend a second long call.

Finally, a no-repeat A+ bank was generated from a checked common-relaxation model after excluding the universal, historical, and exhaustive fixed-base families. Appending its 4096 new I_{18} clauses and the four proved retention units produced a formula with 154190 variables and 722552 clauses. The single authorized solver call returned a complete satisfying model after 2532.362 seconds. An independent checker evaluated every clause, recovered a triangle-free 823-edge primary graph, and found the independent 18-set

$$\{81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 99\}.$$

That set was absent from the installed universal and A+ banks but already belonged to both the historical and exhaustive fixed-base families. Thus A+ is a rigorously audited negative endpoint and, specifically, shows why a fresh prefix is inferior to installing the complete deduplicated bank. It does not provide a repair or a branch refutation.

We therefore reconstructed that stronger bank explicitly. The universal, historical, and exhaustive fixed-base families contain 251771, 64591, and 235504 masks, respectively; their deduplicated union has 526429 members. The 4096 A+ masks are disjoint from all three, giving 530525 distinct necessary I_{18} clauses. Because the common formula already contains the universal family, the strengthened formula appends 278754 new hitting clauses and the four retention units. Its deterministic DIMACS image has 154190 variables, 997210 clauses, 390604816 bytes, and SHA-256

4f7e8f5b724a657888c7814d2c25cac41c283c3db356fb8f1a15f4c7322c375d.

An independent reconstruction agrees byte for byte. This is strictly stronger than A+: it installs the two families that were previously used only as no-repeat filters and directly blocks the displayed A+ witness. It also differs from the deletion-first master with all fixed-base cuts, because it retains the full edge formula and adds 43614 history-only masks plus the 4096 A+ masks. Formula construction alone is not a SAT or UNSAT result.

We finally combined this complete mask bank with the full all-pair maximal-triangle-free encoding of Proposition 23.7. The resulting direct primary-graph CNF has 639290 variables, 1972360 clauses, 408370088 bytes, and SHA-256

09e1784c3f43c4901dc6f6b4749fc5a74b025b08f74be58133edd1ed1096ebdb.

A separate reconstruction agreed byte for byte, both before and after a command-line symlink-validation repair that did not change the formula. Exactly one externally bounded Sirius call used CaDiCaL 3.0.1 at source commit c60730422e758ef1cebe7aeddff2dda31c996bf04, a 300-second wall, non-binary DRAT, and a 10-GiB proof cap. The wrapper exited 124, and the result file contained exactly c UNKNOWN. The run recorded 13347 conflicts, 105834 decisions, and 1216037481 propagations; these are resource telemetry only. Its incomplete 922038097-byte proof prefix was hashed and deleted without replay. Consequently the run produced no model, certificate, learned cut, branch closure, or global Ramsey implication.

Accordingly, none of the following is established: a repair with exactly seven deletions; nonexistence of such a repair; the exact value of $\rho(H)$; or a 100-vertex (3, 18)-Ramsey graph. A machine-readable Boolean field whose default value is false is likewise not evidence of nonexistence unless the companion establishment flag and a checked proof support it.

24. REPRODUCIBILITY AND ARTIFACT PUBLICATION

24.1. Two verification tiers. The public source repository [Xin26] separates lightweight tracked checks from large proof replay. Its accompanying evidence release is designated at <https://github.com/jackal092927/ramsey-bounds-reproducible/releases/tag/evidence-2026-08-30> and is valid only if it contains exactly the 26 assets listed in `artifacts/MANIFEST.tsv`. Publication, immutability, credential-free download, and final-tag replay are release gates recorded separately; they are not premises of the graph theorem. Tier 1: source and semantics. Python 3.11.15 and the locked environment reproduce graph verification, unit tests, the PySAT-independent sequential-counter schemas and their frozen block digests, and exact-seven frozen-state checks. From the repository root, the lightweight entry point is

```
bash scripts/bootstrap.sh
.venv/bin/python reproduce.py quick
```

The retained maximal-union timeout endpoint is audited separately, without recreating or promoting its deleted proof prefix, by

```
.venv/bin/python \
  routes/finite/check_r3_18_budget7_branch1_maximal_union_probe.py
```

That checker requires exit 124, the exact `c UNKNOWN` result, the fixed formula and solver hashes, and the absence of any retained file named as a proof certificate. The proof-critical Python packages are pinned to `python-flint==0.9.0`, `mpmath==1.4.1`, `python-sat==1.9.dev13`, and `six==1.17.0`. The finite route uses PySAT; the other entries are repository-wide proof dependencies.

Tier 2: large proof replay. Once published, the release is downloaded without GitHub credentials by

```
bash scripts/download_release_artifacts.sh
```

The helper first requires the complete release asset-name set to equal the manifest exactly and then checks the content before decompression:

```
.venv/bin/python scripts/verify_artifacts.py \
  --directory /path/to/release-assets --exact
```

The exact mode also rejects unmanifested files, so a stale or mixed download directory cannot masquerade as the immutable release payload. The manifest includes audited Linux x86-64 and macOS arm64 `drat-trim` binaries, their source-commit marker, and the upstream license. The strict branch-1 replay accepts exactly either audited binary hash; `scripts/build_drat_trim.sh` remains a non-normative manual comparison helper. The one integrated command

```
.venv/bin/python reproduce.py finite-heavy \
  --artifact-dir /path/to/release-assets \
  --drat-trim /absolute/path/to/audited-manifest-binary
```

materializes a temporary overlay containing tracked semantic metadata and the 26 release assets. It compares the independent sequential schemas with all six theorem counter blocks, reconstructs the six theorem formulas and

four singleton formulas, replays all ten proofs, checks the complete common relaxation model, and audits the exact-seven state. A proof is accepted only when the checker exits zero and prints an exact `s VERIFIED` status line.

24.2. Why large proofs are release assets. The six exact-six compressed files already occupy about 1.26 GB, and the four singleton compressed proofs add several more gigabytes. Adding these or the smaller budget-five proofs to ordinary Git history would make cloning needlessly expensive. The repository therefore tracks code, matrices, cut banks, semantic records, tests, the paper, and a content-addressed manifest. CNF and DRAT payloads are designated for distribution as immutable GitHub Release assets, keyed by SHA-256 and byte count. No DOI-bearing archival mirror is claimed. This design keeps ordinary clones reviewable while preserving bit-for-bit replay.

The release is complete only if it includes the three budget-five proof pairs used by Proposition 20.1, the three exact-six proof pairs used by Proposition 20.3, the four strongest singleton proof pairs, the common CNF/model pair, and the audited checker toolchain files. Their identities are listed in Tables 7, 8 and 12 and in the strict manifest. Dominated pair-core, incomplete, corrupt, and exploratory traces are explicitly excluded.

24.3. Reproducibility contract. A reproduction should report five logically distinct outcomes:

- (1) the seed and artifact hashes match;
- (2) semantic reconstruction passes;
- (3) each required DRAT replay is verified;
- (4) the common-relaxation SAT model is checked and its independent 18-set is recovered; and
- (5) the exact-seven endpoint remains UNKNOWN, unless a new, separately reviewed certificate supersedes it.

Passing only a solver run, only a hash check, or only unit tests does not by itself reproduce the full theorem.

25. SCOPE, LIMITATIONS, AND CONCLUSION

We established a complete local statement: every triangle-free graph with independence number below 18 on the labelled vertex set of H deletes at least seven of H 's 827 edges, even if additions among all original nonedges are free. The proof is carried by six checked SAT refutations: three for budget at most five and three for the exact-six layer.

The conclusion does not transfer automatically to an isomorphism class, to a different extension neighbourhood, or to a symmetric edit metric. It does not rule out a remote 100-vertex construction, and it does not improve the global bound on $R(3, 18)$. The exact-seven layer remains open. These are mathematical scope restrictions, not merely caveats about running time.

The main methodological lesson is that a computational near miss can support a rigorous negative result without being oversold as a Ramsey-number improvement. A content-addressed seed, a proof of case coverage, independently reconstructed CNF semantics, and replayable UNSAT proofs together establish a precise local barrier. Keeping bounded UNKNOWN runs outside the theorem makes the remaining target equally precise: either exhibit a seven-deletion repair and verify it directly, or produce proof-carrying refutations of all exact-seven branches.

Part 4. Quantum implicit-majority Ramsey search

26. QUANTUM CONSTRUCTIVE RAMSEY SEARCH: THEOREM AND CONTRIBUTION

The elementary Erdős–Szekeres proof of the diagonal Ramsey theorem is constructive [ES35]: choose a pivot, retain its larger colour neighbourhood, and iterate. On a graph with M vertices this uses $O(M)$ edge queries, because every retained set is represented explicitly. This part shows that the same proof architecture has a substantially cheaper quantum implementation when the nested candidate sets are instead represented by their membership predicates.

We work in the promised valid-graph adjacency-oracle model. One coherent query returns the colour of a requested unordered vertex pair. The output is a set of vertices inducing one colour. The main theorem is the following.

Theorem 26.1 (Implicit-majority quantum Ramsey search). *Let $K \geq 2$, $N \geq 4^{K-1}$, and $0 < \eta < 1/2$. Given coherent edge-oracle access to a simple undirected graph on N vertices, there is a quantum algorithm which, with probability at least $1 - \eta$, outputs a K -clique or a K -vertex independent set using*

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

edge queries in the worst case. The algorithm verifies every returned witness and may output $\perp$ on the exceptional event.

Corollary 26.2. *For a promised simple graph on $N = 2^n$ vertices, one can find a clique or independent set of order $\lfloor n/2 \rfloor + 1$ using*

$$O\left(\sqrt{N} \log N \log \frac{\log N}{\eta}\right)$$

edge queries.

The first idea is to use capped unknown-solution quantum search to sample from a set specified only by the $O(K)$ pivot constraints accumulated so far. There are two ways to choose the next colour class. The cleaner, estimation-free method samples one uniform vertex and follows the colour of its edge to the current pivot. This chooses a class with probability proportional to its size. A new survival lemma shows that this size-biased recursion

completes with probability greater than $1/2$ from 4^{K-1} vertices, giving an $O(2^K K^2 \log K \log(1/\eta))$ -query algorithm.

The sharper method uses a scale-aware concentration budget. Sampling a deep candidate set costs exponentially more than sampling an early one, so we estimate early majorities more accurately and permit larger errors later. The sum of all errors is kept below a fixed constant, whereas the work is balanced across the recursion levels. A dyadic schedule removes two avoidable factors of K from the uniform-accuracy analysis and gives Theorem 26.1. The size-biased argument extends directly to any fixed number of edge colours; we record the multicolour form as Corollary K.1.

Contributions. This part makes four precise contributions.

- It gives a worst-case $O(2^K K \log(K/\eta))$ quantum-query algorithm for constructive diagonal Ramsey search, or $\tilde{O}(\sqrt{N})$ at the succinct Ramsey scale.
- It introduces two ways to navigate the implicit recursion: a size-biased rule with a short survival proof, and a sharper scale-aware majority estimator whose accuracy budget is matched to the cost of each level.
- It extends the size-biased mechanism to a fixed number of colours and places the result against a randomized classical lower bound of $\Omega(N^{1-1/\sqrt{2}})$ in the same parameterization.
- It gives an explicit parameter audit of the closest published quantum-query lower-bound claim, exposing a conflict that any comparison in this model must resolve.

The result is algorithmic rather than extremal. It does not change any numerical Ramsey bound or construct an avoidance colouring. It gives a square-root query improvement, up to polylogarithmic factors, over the standard $O(N)$ constructive recursion, but no matching lower bound against every randomized classical algorithm is presently known.

There is also a nontrivial literature issue. Jain, Li, Robere, and Xun print an $N^{1-o(1)}$ quantum-query lower bound for the $N = 2^n$, $K = n/2$ default Ramsey problem in their formal definition [JLRX24]. Their hard instances are valid graph-hash products, while Theorem 26.1 gives an $O(\sqrt{N} \text{polylog } N)$ upper bound on promised valid graphs. Direct substitution into the Ramsey reduction and the cited Liu–Zhandry theorem gives exponent at most $1/24$, not $1 - o(1)$ [LZ19]. We state this incompatibility precisely but do not claim an author-confirmed correction.

The rest of the paper defines the oracle model and capped sampler, proves the recurrence and query bound, derives a randomized classical lower bound from the random-Painter theorem, and records the novelty and nonclaim boundaries.

27. ORACLE MODEL AND CAPPED UNIFORM SEARCH

Let $G: \binom{[N]}{2} \rightarrow \{0, 1\}$ be a promised simple undirected graph. The coherent edge oracle acts as

$$O_G|u, v, b\rangle = |u, v, b \oplus G(u, v)\rangle.$$

Only calls to O_G are counted in the query bound. Reversible operations on $O(\log N)$ -bit labels are charged in the gate-complexity statement.

The promised formulation is the clean graph-theoretic core, but the usual locally verifiable TFNP relation also follows at the same asymptotic cost. Given an arbitrary Boolean circuit $C(u, v)$, define the canonical simple graph

$$A(u, v) = \begin{cases} C(\min\{u, v\}, \max\{u, v\}), & u \neq v, \\ 0, & u = v. \end{cases}$$

One A -query uses one C -query. Run Theorem 26.1 on A and let W be the returned homogeneous set. Query the $O(K^2)$ diagonal and both ordered entries induced by W . If a self-loop or asymmetry is found, return it as the invalid-encoding certificate. Otherwise C agrees with A on W , so W is a valid clique or independent-set witness. This gives the following model-alignment corollary.

Corollary 27.1 (Locally verifiable arbitrary encoding). *The standard Ramsey TFNP relation that accepts either a local invalidity certificate or a locally verified homogeneous set has the same $O(2^K K \log(K/\eta))$ quantum-query upper bound.*

We use the following standard consequence of the unknown-solution search of Boyer, Brassard, Hoyer, and Tapp [BBHT98]. A detailed formulation is included in Appendix I because bounded termination and conditional uniformity are both needed here.

Lemma 27.2 (Capped uniform marked-item search). *Let $T \subseteq [M]$ have density at least $\lambda > 0$, and suppose its membership predicate costs q edge queries. There is a quantum block which uses at most $O(q/\sqrt{\lambda})$ edge queries, succeeds with probability at least an absolute constant $p_0 > 0$, and, conditional on verified success, returns an exactly uniform element of T . The same predetermined cap makes the block terminate if the density promise is false or T is empty.*

Moreover, m independent uniform samples can be collected, except with probability δ , using a predetermined $O(m + \log(1/\delta))$ blocks whenever the density promise holds.

The batch statement is what avoids paying a separate $\log(1/\delta)$ factor for each of the many samples. Every block starts from the uniform state, uses fresh internal randomness, measures, and verifies exact membership. Conditioned on its success indicator, the returned identity is uniform; identities from distinct successful blocks are independent.

For comparison, the exact classical majority recursion queries all edges from each pivot to its current explicit set. Starting with M vertices, its geometric sum is less than $2M$. Our quantum algorithm uses the same combinatorial invariant but accesses each set implicitly.

28. TWO IMPLICIT-SET RECURSIONS

28.1. An estimation-free size-biased recursion. The following algorithm isolates the simplest new mechanism. Set $M = 4^{K-1}$ and $r = 2K - 3$, restrict to the first M vertices, let $S_0 = [M]$, and fix any $v_0 \in S_0$. At step $i = 0, \dots, r - 1$, define $T_i = S_i \setminus \{v_i\}$. Use the permutation-symmetric repeated capped search of Lemma 27.2, with the universal density lower bound $1/M$, to sample an exactly uniform $x_i \in T_i$, or abort after a fixed cap. Query

$$c_i = G(v_i, x_i),$$

and set

$$S_{i+1} = \{x \in T_i : G(v_i, x) = c_i\}, \quad v_{i+1} = x_i.$$

After the last step put $w = v_r$. Among the r labels c_i , choose a colour appearing at least $K - 1$ times and return the corresponding pivots together with w , after direct verification.

The set S_i is never materialized. Its membership predicate consists of the exact edge constraints from the earlier pivots and the exclusions of those pivots. It therefore costs $O(i + 1)$ edge queries. Repeating the capped block $O(\log(2r))$ times makes the conditional failure probability of each nonempty-set sample $O(1/r)$. The structural probability that all r sets remain nonempty is controlled in Lemma 29.1.

28.2. Scale-aware approximate majorities. Fix $M = 4^{K-1}$ input vertices and set

$$r = 2K - 3, \quad E = \frac{1}{16}.$$

For $0 \leq i < r$, define

$$d_i = \left\lceil \frac{r - 1 - i}{6} \right\rceil, \quad z_i = 2^{-d_i}, \quad Z = \sum_{j=0}^{r-1} z_j,$$

and allocate

$$\varepsilon_i = E \frac{z_i}{Z}, \quad a_i = \frac{1}{2} - \varepsilon_i. \quad (1)$$

Thus $\sum_i \varepsilon_i = E$. The error allowance doubles once every six levels as the search sets become more expensive.

Before round i , retain classical pivots $v_0, \dots, v_{i-1}$ and colours $c_0, \dots, c_{i-1}$. They define the exact implicit set

$$S_i = \{u \in [M] \setminus \{v_0, \dots, v_{i-1}\} : G(v_j, u) = c_j \text{ for every } j < i\}. \quad (2)$$

Its reversible membership predicate uses $O(i)$ edge queries.

At round i , use Lemma 27.2 to obtain a verified pivot $v_i \in S_i$. Then collect, with replacement,

$$m_i = \left\lceil \frac{1}{2\varepsilon_i^2} \log \frac{4r}{\eta} \right\rceil \quad (3)$$

independent uniform samples from $S_i \setminus \{v_i\}$. Query their edges to v_i and write $\hat{p}_i$ for the observed fraction of colour 1. Set $c_i = 1$ if $\hat{p}_i \geq 1/2$ and $c_i = 0$ otherwise, and append this exact constraint to Equation (2) to obtain S_{i+1} .

After r rounds, use one final capped search to obtain $w \in S_r$. Select a colour occurring at least $K - 1$ times among $c_0, \dots, c_{r-1}$, take any $K - 1$ pivots carrying it, and add w . Before returning, query all $\binom{K}{2}$ induced edges and accept only an exactly homogeneous witness.

Every search uses a predetermined density bound proved in the next section. If an earlier random estimate was bad and a later set violates that density promise, the fixed cap still terminates and the algorithm returns $\perp$. There is no repeat-until-success loop that can hang on an empty set.

29. CORRECTNESS AND SURVIVAL OF THE NESTED SETS

We begin with the estimation-free algorithm. For integers $d, s \geq 0$, let $P_d(s)$ be the infimum, over every graph, every candidate set of size s , and every distinguished pivot in it, of the probability that the ideal size-biased recursion completes d further steps. Put $P_0(s) = 1$ for $s \geq 1$, $P_d(0) = 0$, and $P_d(1) = 0$ for $d \geq 1$.

Lemma 29.1 (Size-biased survival). *For every $d \geq 0$ and $s \geq 1$,*

$$P_d(s) \geq \frac{(s - 2^d + 1)_+}{s}, \quad (y)_+ = \max\{y, 0\}.$$

In particular, for $M = 4^{K-1}$ and $r = 2K - 3$, the ideal recursion completes with probability at least $1/2 + 1/M$.

Proof. The case $d = 0$ is immediate. For $d \geq 1$ and $s \geq 2$, let the two colour classes of the $s - 1$ non-pivot vertices have sizes a and b . A uniform sample selects them with probabilities $a/(s - 1)$ and $b/(s - 1)$, so induction gives

$$P_d(s) \geq \frac{aP_{d-1}(a) + bP_{d-1}(b)}{s - 1} \geq \frac{(a - 2^{d-1} + 1)_+ + (b - 2^{d-1} + 1)_+}{s - 1}.$$

Since $(x)_+ + (y)_+ \geq (x + y)_+$ and $a + b = s - 1$, the numerator is at least $(s - 2^d + 1)_+$. Replacing the denominator $s - 1$ by s only weakens the bound. The definitions cover $s \leq 1$. Finally $M = 2^{r+1}$, so the displayed bound is $(M - 2^r + 1)/M = 1/2 + 1/M$. $\square$

Exact conditional uniformity of each successful capped search lets us couple an implemented run to this ideal recursion. Give each of its r searches failure probability at most $1/(100r)$. A union bound reduces the completion probability by at most $1/100$, so one fixed-cap run succeeds with probability

at least 0.49. Independent repetition and final exact verification amplify this to $1 - \eta$ using $O(\log(1/\eta))$ runs. On every completed run, nesting gives $G(v_i, v_j) = G(v_i, w) = c_i$ for $j > i$; the pigeonhole argument used in Lemma 29.4 therefore proves exact homogeneity.

We now prove the sharper scale-aware algorithm.

We first isolate the adaptive concentration event.

Lemma 29.2 (Conditional near-majorities). *Except with probability at most $\eta/2$, every level reached after an accurate prefix satisfies*

$$|\hat{p}_i - p_i| \leq \varepsilon_i,$$

where p_i is the true colour-1 fraction from v_i to $S_i \setminus \{v_i\}$.

Proof. Condition on the complete classical history $\mathcal{H}_i$ before level i 's samples. The graph, S_i , and v_i are then fixed. Conditional on verified batch success, Lemma 27.2 gives independent uniform samples from the fixed set. Hoeffding's inequality and Equation (3) give

$$\Pr(|\hat{p}_i - p_i| > \varepsilon_i \mid \mathcal{H}_i) \leq 2e^{-2m_i\varepsilon_i^2} \leq \frac{\eta}{2r}.$$

Apply this inequality to the first inaccurate level following an accurate prefix. The tower property removes the conditioning, and a union bound over the r possible levels proves the claim. No independence between levels is used. $\square$

There are r pivot searches, r sampling batches, and one final search. Allocate failure probability at most $\eta/(4r + 2)$ to each one by the capped amplification in Lemma 27.2. Apply these conditional bounds only after a good prefix, just as in the preceding proof. Their union has probability at most $\eta/2$. Together with Lemma 29.2, every search and estimate needed below is therefore good except with probability at most η .

Let $s_i = |S_i|$ and

$$A_0 = 1, \quad A_i = \prod_{j=0}^{i-1} a_j.$$

Lemma 29.3 (Variable recurrence and density). *On every prefix satisfying Lemma 29.2,*

$$s_i > MA_i - 1, \quad A_i \geq \frac{7}{8} 2^{-i}.$$

Consequently S_r is nonempty, and for every sampling level $i < r$,

$$\frac{|S_i \setminus \{v_i\}|}{M} > \frac{3}{8} 2^{-i}.$$

The final set also satisfies $|S_r|/M \geq 2^{-(r+1)} \geq (3/8)2^{-r}$.

Proof. An accurate empirical majority retains at least fraction $a_i = 1/2 - \varepsilon_i$ of the non-pivot vertices, so

$$s_{i+1} \geq a_i(s_i - 1). \tag{4}$$

Since $\sum_i \varepsilon_i = 1/16$ and $\prod_j (1 - x_j) \geq 1 - \sum_j x_j$ for $x_j \in [0, 1]$,

$$A_i = 2^{-i} \prod_{j < i} (1 - 2\varepsilon_j) \geq 2^{-i} \left(1 - 2 \sum_{j < i} \varepsilon_j \right) \geq \frac{7}{8} 2^{-i}.$$

We prove $s_i > MA_i - 1$ by induction. It holds at $i = 0$. If it holds at i , then Equation (4) and $a_i < 1/2$ give

$$s_{i+1} > a_i(MA_i - 2) = MA_{i+1} - 2a_i > MA_{i+1} - 1.$$

Because $M2^{-r} = 2$, the final lower bound is $s_r > (7/8)2 - 1 = 3/4$, and the integer s_r is positive. If $i < r$, then $M2^{-i} \geq 4$ and

$$s_i - 1 > MA_i - 2 \geq \frac{7}{8}M2^{-i} - 2 \geq \frac{3}{8}M2^{-i}.$$

Finally $s_r \geq 1$ and $1/M = 2^{-(r+1)}$, proving the last assertion. $\square$

Lemma 29.4 (Exact homogeneous output). *Whenever the algorithm reaches its final verified output, that output is a K -clique or a K -vertex independent set.*

Proof. For $i < j$, the definition of S_{i+1} gives $G(v_i, v_j) = c_i$. It also gives $G(v_i, w) = c_i$. Among $r = 2K - 3$ colours, one occurs at least $\lceil r/2 \rceil = K - 1$ times. The corresponding pivots and w therefore induce exactly that colour. The final direct check makes the assertion unconditional on earlier search or estimation errors. $\square$

30. QUERY COMPLEXITY AND THE CLASSICAL LANDSCAPE

For the size-biased recursion, a capped sample from a nonempty subset of the M -element universe costs $O(\sqrt{M})$ membership tests and $O(\log(2r))$ repetitions give failure probability $O(1/r)$. The predicate at level i costs $O(i + 1)$ edge queries, hence one fixed-cap run costs

$$O\left(\sqrt{M} \log(2r) \sum_{i=0}^{r-1} (i + 1)\right) = O(K^2 2^K \log K).$$

The repetition in the preceding section proves the following preliminary bound in worst-case, rather than merely expected, query complexity.

Proposition 30.1 (Size-biased quantum recursion). *Under the hypotheses of Theorem 26.1, the estimation-free algorithm returns a verified homogeneous K -set with probability at least $1 - \eta$ using*

$$O\left(K^2 2^K \log K \log \frac{1}{\eta}\right)$$

edge queries.

We next analyze the sharper scale-aware recursion.

By Lemma 29.3, a capped search block at level i uses $O(2^{i/2})$ membership tests. Computing and uncomputing membership and checking the sampled edge costs $O(i+1)$ edge queries. Put

$$b_i = (i+1)2^{i/2}.$$

The batch cap uses $O(m_i + \log(r/\eta)) = O(m_i)$ blocks. Hence its total sampling cost is

$$O\left(\log \frac{r}{\eta} \sum_{i=0}^{r-1} b_i \varepsilon_i^{-2}\right). \quad (5)$$

There are at most six indices at each value of d_i , so

$$Z = \sum_i 2^{-d_i} < 6 \sum_{d \geq 0} 2^{-d} = 12.$$

Using Equation (1),

$$\sum_{i=0}^{r-1} b_i \varepsilon_i^{-2} = O\left(\sum_{i=0}^{r-1} (i+1)2^{i/2}4^{d_i}\right).$$

Write $h = r - 1 - i$. Since $4^{\lceil h/6 \rceil} \leq 4 \cdot 2^{h/3}$, the last display is bounded by

$$O\left(2^{r/2} \sum_{h=0}^{r-1} (r-h)2^{-h/6}\right) = O(r2^{r/2}).$$

The high-confidence pivot searches, final search, and K^2 verification fit the same bound. Since $r = 2K - 3$, this proves Theorem 26.1.

The gate complexity is

$$2^K \text{poly}\left(K, \log N, \log \frac{1}{\eta}\right)$$

plus the gate cost of one supplied edge-oracle evaluation. No QRAM is assumed. The $O(K)$ measured pivot labels and colours are classical controls for reversible comparisons, conjunctions, and uncomputation.

The known classical lower bound remains below the quantum upper bound. The following direct consequence of the random-Painter analysis of Conlon–Fox–Grinshpun–He [CFGH20] records the best bound we obtain from the current literature; details appear in Appendix J.

Proposition 30.2 (Randomized classical lower bound). *For $K \geq 8$ and $M = 4^{K-1}$, every randomized edge-query algorithm that finds a homogeneous K -set with worst-case probability at least $2/3$ uses*

$$\Omega\left(2^{(2-\sqrt{2})K}\right) = \Omega\left(M^{1-1/\sqrt{2}}\right)$$

queries.

The exponent $1 - 1/\sqrt{2} \approx 0.292893$ does not reach the quantum upper exponent $1/2$, but it strengthens the earlier $M^{1/4}$ black-box lower bound recorded by Komargodski–Naor–Yogev in this parameterization [KNY17]. Therefore no quantum-versus-best-randomized-classical separation is claimed. A randomized lower bound $2^{(1+\delta)K}$ for any fixed $\delta > 0$ would, through the online Ramsey connection, improve the long-standing exponential lower bound for diagonal Ramsey numbers. The missing separation is tied to a major combinatorial barrier rather than a routine query lemma.

The size-biased mechanism also extends to any fixed number of edge colours. The statement and its one-line recurrence proof are given in Appendix K.1; this extension is not needed for the sharper binary theorem.

31. PRIOR ART AND COMPARISON WITH A REPORTED LOWER BOUND

Previous quantum papers under the phrase “Ramsey algorithm” solve different problems. Gaitan and Clark encode the existence of avoidance colourings in an adiabatic Hamiltonian [GC12], and Bian et al. implement small instances experimentally [BCM⁺13]; Wang searches the space of $2^{\binom{V}{2}}$ labelled graphs when testing a Ramsey threshold [Wan15]; and Pion and Mniszewski study QUBO reduction and quantum annealing for bound discovery [PM25]. Their square roots refer to the number of graph colourings, not the number of vertices of one given graph. Dörn’s clique and independent-set query algorithms are closer in oracle model but do not exploit the Ramsey totality promise or the majority recursion [Dör05]. Learning-graph algorithms for fixed subgraph containment [LMS12] and quantum backtracking [Mon18] provide powerful generic primitives, but neither by itself supplies the implicit-set survival invariant used here. To our knowledge, no earlier work uses implicit-set quantum sampling to implement the constructive Erdős–Szekeres proof; this priority statement is limited to the exact oracle model and sources compared here.

The closest exact-model source is Jain–Li–Robere–Xun [JLRX24]. Their formal definitions use $N = 2^n$ vertices and default target $K = n/2$, and their query-complexity paragraph states an $N^{1-o(1)}$ quantum lower bound via a graph-hash reduction and the Liu–Zhandry multicollision theorem [LZ19]. The hard instances are symmetric valid graphs, while Corollary 27.1 also aligns our upper bound with the locally verifiable arbitrary-circuit relation.

For collision range size B and fixed multiplicity t , the reduction has $M \asymp B^{4t}$, whereas Liu–Zhandry’s exponent in B is $\alpha_t = (2^{t-1} - 1)/(2^t - 1)$. The inherited exponent is therefore

$$\frac{\alpha_t}{4t} = \frac{2^{t-1} - 1}{4t(2^t - 1)} \leq \frac{1}{24}, \quad t \geq 2,$$

with equality at $t = 2$. Thus these two ingredients do not yield $M^{1-o(1)}$.

The upper theorem and the printed near-linear lower cannot both hold in the same model. We have found no public correction. We leave clarification

of the discrepancy as an open issue and make no claim about the intended parameterization.

32. EVIDENCE BOUNDARY, OPEN PROBLEMS, AND CONCLUSION

The audit in Appendix K.2 checks the recurrence, error schedule, split-tree bound, exponent translation, and small sampler instances. These are falsification checks, not speedup evidence; the theorem is the analytic argument in Sections 27 to 30.

Small proof-of-concept experiments. Appendix K.2 reports 5,000 seeded runs at $K = 3, N = 16$, an explicit 16-state Grover sampler check, and exact small split-tree programs. They check sampler symmetry and end-to-end combinatorial logic, but are not a full coherent simulation or evidence of wall-clock or hardware speedup.

AI-assisted research process. OpenAI Codex assisted brainstorming, proof and query-analysis checks, literature comparison, executable checks, and editing; ChatGPT Pro assisted brainstorming and adversarial review. Outputs were treated as untrusted: the claims rest on the proofs, primary sources, and reproducible checks, and the human authors retain full responsibility.

Three open problems delimit the contribution. First, resolve the discrepancy with the printed Jain–Li–Robere–Xun quantum lower bound. Second, determine whether a randomized classical algorithm can match the $\tilde{O}(\sqrt{M})$ upper bound; Proposition 30.2 leaves a large exponent gap. Third, determine whether the implicit-set method can be adapted to off-diagonal or hypergraph Ramsey total search, and whether the multicolour bound in Corollary K.1 can be sharpened by a scale-aware estimator.

In summary, the constructive Ramsey recursion can be implemented without materializing its exponentially shrinking candidate sets. Capped uniform search and a scale-aware error schedule give a $\tilde{O}(\sqrt{N})$ query algorithm at the succinct Ramsey scale. This does not imply a new numerical Ramsey bound, a physical speedup, or a quantum-versus-all-classical separation.

33. UNIFIED REPRODUCIBILITY AND PLANNED PUBLIC ARTIFACT RELEASE

The public source repository at <https://github.com/jackal092927/ramsey-bounds-reproducible> exposes one canonical paper and a tiered verification entry point. The evidence package is designated for a future release at <https://github.com/jackal092927/ramsey-bounds-reproducible/releases/tag/evidence-2026-08-30> and is valid only if it contains exactly the 26 content-addressed assets listed in `artifacts/MANIFEST.tsv`. At the date of this draft, the final tag and immutable 26-asset Release have not been published; credential-free asset verification and final-tag replay therefore remain pending gates outside the mathematical argument.

The 30 August ChatGPT Pro second-opinion review of the upper, lower, and finite parts has completed. Every recoverable concrete objection was checked against those canonical sources and either repaired or explicitly

dispositioned. A separate public-only quantum packet was sent on 31 August; its review was still running at the date of this draft, so no conclusion from it is counted. AI-assisted author-side review is not external human peer review and does not change any theorem’s stated trust boundary. A clean checkout with Python 3.11.15, exactly `uv 0.10.2`, and the locked dependencies is prepared by

```
bash scripts/bootstrap.sh
```

The bootstrap fails closed when that `uv` version is absent and has no unhashed package-manager fallback. The following tiers intentionally make different guarantees.

Tier	Coverage	Logical scope
quick	Four current upper checkers, the current lower arithmetic replay, finite unit/seed checks, tracked graph certificates, and the exact quantum-recurrence diagnostics.	Fast integrity check; it does not replay the large finite DRAT proofs.
full	quick , the complete long upper interval chain and the full lower checker suite.	Replays the computer-assisted upper layer and the lower arithmetic witnesses.
finite-heavy	Hash verification, semantic CNF reconstruction, ten DRAT replays, and the complete common-model audit with an audited checker.	Replays the proof-carrying finite theorem and the four branch-1 singleton consequences while preserving exact seven as UNKNOWN.
quantum	Exact rational recurrence checks, exhaustive small binary/ternary split-tree dynamic programs, and deterministic ideal-sampler diagnostics.	Checks theorem formulas and invariants; it is not a physical quantum simulation or empirical speedup claim.
paper	Deterministic materialization and compilation of this single manuscript.	Checks source integration and typesetting, not the mathematical certificates by itself.
sources	Networked downloads of the three pinned arXiv members and five upstream graph matrices.	Verifies source identities; it is separate from the offline theorem replay.

The fast and full tiers are invoked as

```
.venv/bin/python reproduce.py quick
.venv/bin/python reproduce.py full
```

```
.venv/bin/python reproduce.py sources
make verify-quantum
```

Large compressed CNF/DRAT pairs are designated for distribution as immutable, manifest-verified Release assets rather than ordinary Git objects. Once the Release is published, download and authenticate every entry of `artifacts/MANIFEST.tsv` without GitHub credentials by

```
bash scripts/download_release_artifacts.sh
```

The manifest supplies audited Linux x86-64 and macOS arm64 checker binaries. Executable mode is set only after hash verification. A source build is available only for non-normative manual comparison:

```
bash scripts/build_drat_trim.sh /absolute/new/checker-directory
```

The strict branch-1 gate accepts exactly either audited manifest binary hash. The proof-carrying tier is

```
.venv/bin/python reproduce.py finite-heavy \
  --artifact-dir artifacts/downloads \
  --drat-trim /absolute/path/to/audited-manifest-binary
```

The archival unified PDF and the standalone anonymous quantum PDF are generated by `make paper`; `make papers` is retained as an alias. The directories `papers/upper`, `papers/lower`, and `papers/finite` are historical source components. The directory `papers/quantum` is both the shared source component for this part and the standalone ITCs submission source.

The build fixes `SOURCE_DATE_EPOCH`. Two clean builds made with the recorded pdf $\text{\TeX}$ 1.40.24 ($\text{\TeX}$ Live 2022) and Latexmk 4.77 toolchain were byte-identical. Byte identity across different $\text{\TeX}$ distributions, font packages, or engines is not claimed.

The download helper first requires the release asset-name set to equal the manifest names exactly. The release manifest then authenticates compressed artifacts before decompression. The finite-heavy verifier then reconstructs the structural clauses and independent-set hitting clauses, checks the exact branch counters, and replays the proof traces. This ordering prevents a valid proof of an unrelated or malformed CNF from being mistaken for evidence about the graph theorem.

34. CONCLUSIONS, NON-IMPLICATIONS, AND OPEN PROBLEMS

The four Parts give complementary information about Ramsey theory without pretending that their techniques or normalizations coincide. The diagonal upper result isolates an exact-diagonal correlation gain inside a retained-spine transfer. The fixed-ratio lower result shows how a controlled deterministic residual can be paid by full-box strong concavity. The finite result turns a near miss into a precise, proof-carrying local obstruction. The quantum result turns the elementary constructive recursion into a sublinear-query total-search algorithm, with both an estimation-free size-biased proof and a sharper scale-aware implementation.

Four restrictions remain part of the conclusions. First, the upper theorem is source-relative and asymptotic, with no effective finite- k threshold. Second, the lower theorem is conditional on the pinned items (S1), (S2), (S4), and (S5); weighted reverse propagation (S3) is proved locally. After the source constant K is fixed, it fixes sufficiently large C before taking $\ell \rightarrow \infty$, then sends $w \downarrow \omega_0$ before studying $C \rightarrow \infty$. It has no effective $C_0(K)$ and makes no cross-normalization priority claim. Third, the finite theorem does not imply $R(3, 18) \geq 101$. Its four proof-checked branch-1 singleton consequences are non-vacuous and materially strengthen the necessary support filter, but they do not close branch 1. A maximal-triangle-free normal form and complete four-bank mask union give a cleaner frozen branch-1 formula, but its only bounded endpoint is UNKNOWN; the exact-seven layer remains open. Fourth, the quantum theorem changes query complexity rather than a Ramsey number. Its randomized classical lower bound does not match the quantum upper bound, and the printed Jain–Li–Robere–Xun lower-bound remark remains in parameter conflict pending author clarification.

The most direct next steps are correspondingly separate: strengthen the correlation or outer partition in Part I; seek adaptive or higher-order residual potentials beyond the scalar class of Part II; either exhibit a seven-deletion repair or produce checked exact-seven refutations in Part III; and resolve the model/literature conflict and randomized-classical gap in Part IV. Any future improvement should retain the same separation between discovery, human-readable implication, independently replayable certificate, and open computational state.

Completion of the earlier three-part ChatGPT Pro review does not close the separate public-artifact gate, and the quantum-specific Pro review has not yet returned a verdict. The final tag, immutable Release, and credential-free final-tag replay remain pending at the date of this draft.

APPENDIX A. EXACT CERTIFICATE DATA

The proof specification is the immutable file `routes/upper/EXACT_DIAGONAL_NEXT_CANDIDATE.md`. Its SHA-256 is

`e4ea58def640593690a7545e111c3b38f1bfcf8a5735fe7985600481e0bf36d4`.

The exact constants are

$$\begin{aligned} u_0 &= 1235783/500000, & \sigma_0 &= 10^{-8}, & D_0 &= 88053/10^9, \\ \beta &= 330867/500000, & \varepsilon &= 6893/10^7, \\ C &= 12366348252219/1250000000000. \end{aligned}$$

The compact ratio proof uses 65,536 cells on $[u_0, 2.9]$; the exact diagonal uses 131,072 cells on $[0, 20]$. Both tails are proved by explicit monotone envelopes. The outer checker verifies the full ordered wedge and the final decimal rounding.

The repository integrity checker validates all six frozen proof files before running any expensive enclosure. In particular, a successful numerical run

against a modified proof specification is not accepted as a reproduction of this theorem.

A.1. Pinned source interface. The imported source member is `main.tex` from Yang–Mao v1, arXiv:2608.01962v1. Its SHA-256 is

155b7104ec5b6935a576ae9f2b161976a966b0b46bd2b69153c0934ca688da2a.

The exact mapping for the conclusions actually imported in source interface 4.2 is:

source lines	imported conclusion
318–332	preliminary colour spines, common reservoir, reservoir-size lower bound, and minimum colour degrees
1146–1565	parameterized monochromatic-book theorem and proof with the exact ρ_β, Π, Ξ constants
1569–1618	compatibility of preliminary spines with the final book

For source lineage, but not as additional imported assumptions, the analytic antecedents are:

source lines	locally used lineage
370–418	positive tensor-moment lemma, reproved in the specialized form Lemma 6.5
426–791, 793–983	generic root-filter, separator, and expectation–tail argument; the stronger exact-diagonal specialization is proved in Section 6
989–1009	definition and parameter domain of $\mathcal{G}_2^{(3)}(\beta, C)$, restated in (U2.2a)
1017–1142	density-increment lemma internal to the pinned source proof of the imported book theorem

The source archive is downloaded on demand and checked against this digest; it is not silently replaced by a later arXiv version. The manuscript proves the stronger local correlation input (U2.7); only the regularization, parameterized-book, and compatibility conclusions in the first table are imported.

A.2. Outer-wedge certificate data. All outer parameters are exact terminating rationals. Substitution in (U2.3)–(U2.4) gives outward-rounded enclosures

$$q \in [0.7559369286692751849342481567047, 0.7559369286692751849342481567048],$$

$$\Xi \in [940.3014597301789400852772355749, 940.3014597301789400852772355750].$$

The proof-critical lower or upper endpoints are:

quantity	certified endpoint
$\sup_{(0,1]} U''$	< -0.3602795262399473
degree slack	$> 1.5056762000000000 \cdot 10^{-9}$
red boundary derivative	$< -3.6952685632128072 \cdot 10^{-4}$
red page margin	$> 5.1006707822905727 \cdot 10^{-9}$
blue diagonal derivative	$< -1.8476200284472833 \cdot 10^{-4}$
blue page margin	$> 3.4825658438202073 \cdot 10^{-6}$
reservoir margin	$> 6.0468779601096049 \cdot 10^{-6}$
base upper endpoint	$< 3.7806852883796401133524$
safe-decimal rounding margin	$> 1.6203598866476283 \cdot 10^{-9}$

The primary outer checker has SHA-256

`e2035cbffefcb147141fcee4831cac2af085f26d9183091ee20d90acc89ac87b,`

the non-importing replay has SHA-256

`fc379e3b861b69054aadaa80ebc3c791cb8358f22c9b0aa01070c56aa131c26c,`

and the adaptive referee replay has SHA-256

`3c30f72ddc24848f2b72c278f35b5d8bd6296ae8a40b045d30c5b7e087bd7150.`

They use different subdivision and endpoint-reduction structures, but all three trust python-flint/Arb containment and the same exact mathematical specification; they are not described as independent formal proofs.

APPENDIX B. STANDALONE PROOF OF THE GNNW BOOK LEMMA

Date: 2026-08-12

Source being repaired: Gupta–Ndiaye–Norin–Wei, arXiv:2407.19026v1, Lemmas 7–11 and Theorem 12.

B.1. Claim boundary. This note proves the combinatorial input used by the certified diagonal Ramsey descent without invoking GNNW Lemma 11 or Theorem 12 as black boxes. It retains their definitions of a candidate and of the rate region, and it uses only elementary Ramsey recursion, convexity, averaging, and induction.

Let a **candidate** be a pair of nonempty disjoint vertex sets (X, Y) in a red–blue complete graph. Its red cross-density is

$$d(X, Y) = \frac{e_R(X, Y)}{|X||Y|}.$$

It is (k, ℓ, t) -good if $X \cup Y$ contains a red K_k , or X contains a blue K_t , or Y contains a blue K_ℓ .

Let $\mathcal{R}$ be the closure of the set of pairs $(x, y) \in (0, 1)^2$ for which, for all sufficiently large $k + \ell$,

$$R(k, \ell) \leq x^{-k} y^{-\ell},$$

and let $\mathcal{R}_*$ be its interior.

B.1.1. *Theorem A (book induction).* Suppose $0 < \mu_0, x_0, y_0, p < 1$,

$$x_0 < p^{1/(1-\mu_0)}(1 - \mu_0), \quad (x_0, y_0) \in \mathcal{R}_*. \quad (\text{U-B.A.1})$$

There is L_0 such that, for all positive integers k, ℓ, t with $\ell \geq L_0$, every candidate of density at least p and satisfying

$$|X||Y| \geq x_0^{-k} y_0^{-\ell} \mu_0^{-t} \quad (\text{U-B.A.2})$$

is (k, ℓ, t) -good.

B.1.2. *Corollary B (balanced-book bound).* Suppose $0 < \mu, x, y, p < 1$,

$$x < p^{1/(1-\mu)}(1 - \mu), \quad (x, y) \in \mathcal{R}_*. \quad (\text{U-B.B.1})$$

For all sufficiently large ℓ , every red–blue coloring of K_N of red density at least p , where

$$N \geq x^{-k/2}(\mu y)^{-\ell/2}, \quad (\text{U-B.B.2})$$

contains a red K_k or a blue K_ℓ .

The statements hold for all $p, \mu \in (0, 1)$; no relation $p > \mu$ is needed.

B.2. Four elementary inputs.

B.2.1. *Lemma 1 (an interior point gives its own eventual bound).* If $(u, v) \in \mathcal{R}_*$, then, for all sufficiently large $a + b$,

$$R(a, b) \leq u^{-a} v^{-b}. \quad (\text{U-B.1.1})$$

Proof. Choose $\eta > 0$ so that $(u + 2\eta, v + 2\eta) \in \mathcal{R}$. Since $\mathcal{R}$ is the closure of the eventual-bound set, there is a point (u', v') in that set with $u' > u$ and $v' > v$. Its defining bound is stronger than (U-B.1.1). $\square$

B.2.2. *Lemma 2 (density convexity).* For every candidate,

$$\sum_{v \in X} d(X, N_R(v) \cap Y) |N_R(v) \cap Y| \geq e_R(X, Y) d(X, Y), \quad (\text{U-B.2.1})$$

where an empty-neighborhood summand is interpreted as zero.

Proof. Put $d_y = |N_R(y) \cap X|$. Multiplying the left side by $|X|$ gives $\sum_{y \in Y} d_y^2$. Cauchy–Schwarz bounds this below by $|Y|^{-1} (\sum_y d_y)^2 = e_R(X, Y)^2 / |Y|$, which is $|X|$ times the right side. $\square$

B.2.3. Lemma 3 (blue-book extraction). Let $0 < \nu < 1$, and let b, k, m be positive integers with $m \geq 5\nu^{-1}b^2$ and $b \leq m$. Suppose $|X| \geq 5m^2$ and at least $R(k, m)$ vertices $v \in X$ have blue degree at least $\nu|X|$ inside X . Then X contains a red K_k , or it contains disjoint sets S, T such that S is a blue clique, every edge from S to T is blue, and

$$|S| = b, \quad |T| \geq \frac{\nu^b}{2}|X|. \quad (\text{U-B.3.1})$$

Proof. The high-blue-degree vertices contain a red K_k or a blue K_m . In the latter case call its vertex set U . Put $d_z = |N_B(z) \cap U|$ for $z \in X \setminus U$ and let D be their average. Counting ordered blue pairs from U to $X \setminus U$ gives

$$D \geq \frac{m(\nu|X| - m)}{|X| - m} = \nu m - \frac{(1 - \nu)m^2}{|X| - m} > \nu m - \frac{1}{4}, \quad (\text{U-B.3.2})$$

where $|X| \geq 5m^2$ and $m \geq 1$ were used in the last inequality. Let $q = \lfloor D \rfloor$. The discrete convexity of $j \mapsto \binom{j}{b}$ on nonnegative integers (with value zero for $j < b$) shows that a uniform b -subset S of U satisfies

$$\mathbb{E}|N_B(S) \cap (X \setminus U)| = \binom{m}{b}^{-1} \sum_{z \in X \setminus U} \binom{d_z}{b} \geq \binom{m}{b}^{-1} \binom{q}{b} |X \setminus U|. \quad (\text{U-B.3.3})$$

Since $q > D - 1 > \nu m - 5/4$ and $m \geq 5\nu^{-1}b^2$, every $0 \leq i < b$ satisfies

$$\frac{q - i}{m - i} > \nu - \frac{(1 - \nu)i + 5/4}{m - i} \geq \nu \left(1 - \frac{b + 1/4}{5b^2}\right).$$

For the second inequality, the fraction being subtracted increases with i , and at $i = b - 1$ it is at most the claimed value because

$$\frac{(1 - \nu)(b - 1) + 5/4}{5b^2/\nu - (b - 1)} \leq \frac{\nu(b + 1/4)}{5b^2};$$

after cross-multiplication, the difference between the right numerator and the left numerator is $\nu(b - 1)(1 - (b + 1/4)/(5b^2)) \geq 0$.

(If $q < b$, the last strict lower bound would be positive at $i = q$, an impossibility, so the binomial ratio is well-defined.) Hence

$$\frac{\binom{q}{b}}{\binom{m}{b}} \geq \nu^b \left(1 - \frac{b + 1/4}{5b^2}\right)^b \geq \frac{3}{4}\nu^b. \quad (\text{U-B.3.4})$$

The last elementary bound is weakest at $b = 1$, where its left factor is $3/4$; for $b \geq 2$, Bernoulli's inequality gives at least $1 - (b + 1/4)/(5b) > 3/4$. Finally $|X \setminus U| \geq (4/5)|X|$, so (U-B.3.3)–(U-B.3.4) exceed $3\nu^b|X|/5 > \nu^b|X|/2$. Some S therefore has a common blue page T satisfying (U-B.3.1). $\square$

B.2.4. *Lemma 4 (the limit needed to choose the moment).* For every $p, \mu \in (0, 1)$,

$$\lim_{r \rightarrow \infty} (p^{1/r} - \mu)^r (1 - \mu)^{1-r} = p^{1/(1-\mu)} (1 - \mu), \quad (\text{U-B.4.1})$$

where r runs through sufficiently large positive integers, for which the base is positive.

Proof. Since $p^{1/r} \rightarrow 1 > \mu$, positivity is eventual, and

$$r \log \left(1 + \frac{p^{1/r} - 1}{1 - \mu} \right) \rightarrow \frac{\log p}{1 - \mu}.$$

Exponentiation and the remaining factor $1 - \mu$ prove (U-B.4.1). $\square$

B.3. **Proof of Theorem A.** The strict inequality in (U-B.A.1) implies $x_0 + \mu_0 < 1$. By Lemma 4 choose an integer $r > 1$ such that

$$x_0 < (p^{1/r} - \mu_0)^r (1 - \mu_0)^{1-r}. \quad (\text{U-B.5.1})$$

After fixing r , choose $\varepsilon > 0$ sufficiently small that

$$\begin{aligned} p &\geq 2\varepsilon, & (1 + \varepsilon)(\mu_0 + \varepsilon) &\leq 1, & x_0 + \mu_0 + 2\varepsilon &< 1, \\ & & (x_0 + 2\varepsilon, y_0 + 2\varepsilon) &\in \mathcal{R}_*, \\ \mu_0 + 2\varepsilon &\leq \frac{\mu_0 + \varepsilon}{(\mu_0 + \varepsilon) + (x_0 + \varepsilon)}, & (p - \varepsilon)^{1/r} &> \mu_0 + 3\varepsilon, \\ x_0 &\leq ((p - \varepsilon)^{1/r} - \mu_0 - 3\varepsilon)^r (1 - \mu_0 - \varepsilon)^{1-r} - \varepsilon. \end{aligned} \quad (\text{U-B.5.2})$$

All requirements follow by openness, continuity, (U-B.5.1), and the strict inequality $x_0 + \mu_0 < 1$. Set

$$x = x_0 + \varepsilon, \quad y = y_0 + \varepsilon, \quad \mu = \mu_0 + \varepsilon, \quad \delta_n = \frac{\varepsilon}{n}. \quad (\text{U-B.5.3})$$

We prove the following stronger assertion by induction on $n = k + t$, with $\ell \geq L_0$ fixed. If

$$d(X, Y) \geq p - \delta_n \quad (\text{U-B.5.4})$$

and

$$(d(X, Y) + \delta_n - p)^r |X||Y| \geq x^{-k} y^{-\ell} \mu^{-t}, \quad (\text{U-B.5.5})$$

then (X, Y) is (k, ℓ, t) -good. The cases $k = 1$ or $t = 1$ are immediate.

First note that (U-B.A.2) and $d(X, Y) \geq p$ imply (U-B.5.5): because $x_0 \leq x/(1 + \varepsilon)$, $y_0 \leq y/(1 + \varepsilon)$, and $\mu_0 \leq \mu/(1 + \varepsilon)$, it is enough that

$$\frac{\varepsilon^r}{n^r} (1 + \varepsilon)^{n+\ell} \geq 1. \quad (\text{U-B.5.6})$$

A single sufficiently large L_0 makes (U-B.5.6) true for every $n \geq 2$.

B.3.1. *Regularization and the size of X .* Delete from X any vertex having fewer than $(p - \delta_n)|Y|$ red neighbors in Y . Both the nonnegative excess

$$e_R(X, Y) - (p - \delta_n)|X||Y|$$

and its normalization by $|X||Y|$ do not decrease, so the left side of (U-B.5.5) does not decrease. The positive right side prevents deletion of all vertices. We may consequently assume

$$d(X', Y) \geq p - \delta_n \quad \text{for every nonempty } X' \subseteq X. \quad (\text{U-B.5.7})$$

Lemma 1 and the interior condition in (U-B.5.2) imply, for sufficiently large L_0 , that if

$$|Y| \geq (x + \varepsilon)^{-k}(y + \varepsilon)^{-\ell},$$

then Y already contains a red K_k or a blue K_ℓ . Otherwise, using $d + \delta_n - p \leq 1$ in (U-B.5.5),

$$|X| \geq \left(\frac{x + \varepsilon}{x}\right)^k \left(\frac{y + \varepsilon}{y}\right)^\ell \mu^{-t} \geq (1 + \varepsilon)^{n+\ell}. \quad (\text{U-B.5.8})$$

B.3.2. *The big-blue branch.* Put

$$b = \left\lceil \frac{2r \log n - r \log \varepsilon + \log 2}{\log(1 + \varepsilon)} \right\rceil, \quad m = \lceil 5\mu^{-1}b^2 \rceil, \quad w = R(k, m). \quad (\text{U-B.5.9})$$

Here $m = O_{r,\varepsilon}(\log^2 n)$, $m \geq b$, and $w = R(k, m) \leq \binom{k+m-2}{m-1} \leq (k+m)^m \leq \exp(O_{r,\varepsilon}(\log^3 n))$. Since (U-B.5.8) is exponential in $n + \ell$, increasing L_0 once makes, uniformly for every n ,

$$|X| \geq 5m^2, \quad w \leq \frac{\varepsilon(p - \varepsilon)}{n^3}|X|. \quad (\text{U-B.5.10})$$

Let

$$W = \{v \in X : |N_B(v) \cap X| \geq (\mu + \varepsilon)|X|\}.$$

If $|W| \geq w$, Lemma 3, with $\nu = \mu + \varepsilon$, produces a red K_k or a blue book with base S of size b and page T of size at least $(\mu + \varepsilon)^b|X|/2$. If $b \geq t$, its base already contains a blue K_t . Suppose $b < t$. From (U-B.5.7), $d(T, Y) \geq p - \delta_n$, and hence

$$d(T, Y) + \delta_{n-b} - p \geq \delta_{n-1} - \delta_n \geq \frac{\varepsilon}{n^2}.$$

Using (U-B.5.5), $d + \delta_n - p \leq 1$, and the definition of b gives

$$\begin{aligned}
& (d(T, Y) + \delta_{n-b} - p)^r |T| |Y| \\
& \geq \frac{1}{2} \left(\frac{\varepsilon}{n^2} \right)^r (\mu + \varepsilon)^b |X| |Y| \\
& \geq \frac{1}{2} \left(\frac{\varepsilon}{n^2} \right)^r \left(\frac{\mu + \varepsilon}{\mu} \right)^b x^{-k} y^{-\ell} \mu^{-t+b} \\
& \geq x^{-k} y^{-\ell} \mu^{-t+b}.
\end{aligned} \tag{U-B.5.11}$$

Induction applied to (T, Y) with parameter $t - b$ finishes this branch: a blue K_{t-b} in T joins the blue base S .

We may therefore assume

$$|W| < w. \tag{U-B.5.12}$$

B.3.3. The red-step/density-increment branch. Lemma 2 and (U-B.5.10)–(U-B.5.12) show that the contribution from W is at most

$$w|Y| \leq \frac{\varepsilon}{n^3} e_R(X, Y).$$

Consequently some $v \in X \setminus W$ satisfies

$$d(X, N_R(v) \cap Y) \geq d(X, Y) - \frac{\varepsilon}{n^3}. \tag{U-B.5.13}$$

Write

$$X_R = N_R(v) \cap X, \quad X_B = N_B(v) \cap X, \quad Y' = N_R(v) \cap Y, \quad p' = p - \delta_{n-1},$$

and

$$\alpha = d(X, Y') - p', \quad \alpha_R = d(X_R, Y') - p', \quad \alpha_B = d(X_B, Y') - p'.$$

An empty X_R or X_B is assigned zero weighted contribution and its induction branch is skipped; its corresponding α_R or α_B need not be defined. By (U-B.5.7), $|Y'| \geq (p - \varepsilon)|Y|$. Also, (U-B.5.4) and (U-B.5.13) give

$$\alpha \geq \delta_{n-1} - \delta_n - \frac{\varepsilon}{n^3} \geq \frac{\varepsilon}{n^2} > 0. \tag{U-B.5.14}$$

Since all edges from v to Y' are red,

$$\frac{\alpha_R}{\alpha} \frac{|X_R|}{|X|} + \frac{\alpha_B}{\alpha} \frac{|X_B|}{|X|} + \frac{1}{\alpha|X|} \geq 1. \tag{U-B.5.15}$$

If $\alpha_R \geq 0$ and

$$\alpha_R^r |X_R| \geq \frac{x \alpha^r |X|}{p - \varepsilon}, \tag{U-B.5.16}$$

then, using $|Y'| \geq (p - \varepsilon)|Y|$, (U-B.5.13), and

$$d(X, Y') - p' \geq d(X, Y) + \delta_n - p,$$

condition (U-B.5.16) and (U-B.5.5) show that (X_R, Y') satisfies the inductive moment inequality for $(k-1, \ell, t)$. Its goodness implies that of (X, Y) because v is red to $X_R \cup Y'$. The analogous condition, including its necessary sign hypothesis, is

$$\alpha_B \geq 0, \quad \alpha_B^r |X_B| \geq \frac{\mu \alpha^r |X|}{p - \varepsilon} \quad (\text{U-B.5.17})$$

gives the inductive branch $(k, \ell, t-1)$, since v is blue to X_B .

Assume both branches fail. Set

$$a = |X_R|/|X|, \quad c = |X_B|/|X|, \quad q = 1 - 1/r.$$

Using the strict failures of (U-B.5.16)–(U-B.5.17) in (U-B.5.15) gives

$$x^{1/r} a^q + \mu^{1/r} c^q + \frac{(p - \varepsilon)^{1/r}}{\alpha |X|} > (p - \varepsilon)^{1/r}. \quad (\text{U-B.5.18})$$

Now $a + c = 1 - |X|^{-1} < 1$ and $c \leq \mu + \varepsilon$. The function

$$f(z) = x^{1/r}(1 - z)^q + \mu^{1/r} z^q$$

is increasing on $0 \leq z \leq \mu/(\mu + x)$. The special smallness condition in (U-B.5.2) is exactly

$$\mu + \varepsilon \leq \frac{\mu}{\mu + x}.$$

Therefore

$$x^{1/r} a^q + \mu^{1/r} c^q \leq f(c) \leq f(\mu + \varepsilon) \leq x^{1/r}(1 - \mu)^q + \mu + \varepsilon. \quad (\text{U-B.5.19})$$

Equations (U-B.5.8) and (U-B.5.14), after one final uniform increase of L_0 , give

$$\frac{(p - \varepsilon)^{1/r}}{\alpha |X|} \leq \frac{n^2}{\varepsilon(1 + \varepsilon)^{n+\ell}} \leq \varepsilon. \quad (\text{U-B.5.20})$$

Substituting (U-B.5.19)–(U-B.5.20) into the strict inequality (U-B.5.18) yields

$$x^{1/r}(1 - \mu)^{1-1/r} + \mu + 2\varepsilon > (p - \varepsilon)^{1/r}.$$

The positivity condition in (U-B.5.2) allows the rearrangement

$$x > ((p - \varepsilon)^{1/r} - \mu - 2\varepsilon)^r (1 - \mu)^{1-r}. \quad (\text{U-B.5.21})$$

But $x = x_0 + \varepsilon$ and $\mu = \mu_0 + \varepsilon$, so (U-B.5.21) strictly contradicts the final inequality of (U-B.5.2). The induction, and hence Theorem A, is complete.

□

B.4. Proof of Corollary B. By openness choose $y_0 > y$ with $(x, y_0) \in \mathcal{R}_*$. Take a uniformly random equitable bipartition of the N vertices. Each edge has the same probability of crossing, so some partition (X, Y) has cross-red density at least the global red density p .

For all sufficiently large ℓ ,

$$(y_0/y)^{\ell/2} \geq 3.$$

Thus (U-B.B.2) makes both equitable parts at least $x^{-k/2}(\mu y_0)^{-\ell/2}$, and hence

$$|X||Y| \geq x^{-k} y_0^{-\ell} \mu^{-\ell}.$$

Apply Theorem A with $t = \ell$. A red K_k in $X \cup Y$, a blue K_ℓ in X , or a blue K_ℓ in Y gives the desired conclusion. $\square$

B.5. Source defects neutralized by this proof. The standalone proof explicitly repairs all of the following source-level issues rather than silently importing them:

1. Lemma 10 does not require $p > \mu$.
2. The moment order r is chosen before ε .
3. The northeast rate point is chosen in $\mathcal{R}_*$, and Lemma 1 supplies the exact eventual bound used later.
4. The printed comparison for y_0 uses x in place of y .
5. The case $b \geq t$ and empty X_R, X_B are treated separately.
6. The normalized sizes satisfy $a + c < 1$, not the dimensionally false printed inequality.
7. The concavity calculation supplies an upper bound in (U-B.5.19), and the strictness in (U-B.5.18) is preserved to obtain a genuine contradiction.

The only non-symbolic dependency left in the diagonal descent is the stated semantics of the Arb interval implementation used to certify its numerical inequalities. A post-repair adversarial replay of this standalone proof and a generated-copy fidelity audit have both passed. Independent external human review and a DOI-bearing archival deposit remain pending.

APPENDIX C. SIXTH CHAINED UPPER-BOUND DESCENT: BOUNDED SEARCH LOG

C.1. Certificate semantics and correctness. We first state the theorem-to-JSON implication used by all six links. For a finite exact-decimal coefficient tuple $c = (c_1, \dots, c_d)$, write

$$F_c(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda + e^{-\lambda} \sum_{j=1}^d c_j \lambda^j. \quad (\text{U-C.C.1})$$

A file with schema `corrected-two-sided-ramsey-v1` contains exact tuples $c^{\text{prior}}, c^{\text{target}}$, an exact splice λ_* , and a finite ordered list of records

$(\lambda_i^-, \lambda_i^+, M_i, Y_i)$. Acceptance means that outward-rounded interval evaluation establishes all of the following obligations.

- (i) The records form an exact partition of $[\lambda_*, 1]$, with $0 < M_i, Y_i < 1$, and the prior rate is strictly concave on $(0, 1]$.
- (ii) On every record and every λ in its closed interval, put

$$X = (1 - M_i)(1 - e^{-F'_{\text{c}^{\text{target}}(\lambda)}})^{1/(1-M_i)}, \quad a = -\log X, \quad b = -\log Y_i.$$

The verifier proves $F_{\text{c}^{\text{target}}} > 0$, $F'_{\text{c}^{\text{target}}} > 0$, and both prior-region inequalities

$$a + \mu b > F_{\text{c}^{\text{prior}}}(\mu), \quad b + \mu a > F_{\text{c}^{\text{prior}}}(\mu) \quad (0 < \mu \leq 1). \quad (\text{U-C.C.2})$$

Strict concavity reduces the two suprema in μ to their endpoint and unique stationary-point candidates; every candidate is enclosed.

- (iii) On the same complete record interval the main descent slack satisfies

$$F_{\text{c}^{\text{target}}}(\lambda) + \frac{1}{2}\{\log X + \lambda \log M_i + \lambda \log Y_i\} > 0. \quad (\text{U-C.C.3})$$

- (iv) On the open interval $(0, \lambda_*]$, the verifier takes $M = \lambda e^{-\lambda}$, defines X by the same formula and $Y = 1 - X$, and proves positivity of F, F' and a strictly positive lower bound for the left side of (U-C.C.3) divided by λ . Bounds $q_- \lambda \leq e^{-F'} \leq q_+ \lambda$ cancel the two $\lambda \log \lambda$ singularities, so this is an analytic tail proof rather than extrapolation from λ_* .
- (v) At the first link, the prior is exactly the rate in Lemma C.2. At every later link, c^{prior} agrees entry by entry, as exact decimal data, with the preceding accepted target tuple.

We next close the theorem-to-certificate bridge, including the two steps that are easiest to lose in a numerical presentation: the density dichotomy and the induction in the smaller clique order.

Lemma C.1 (Finite-net rate descent). *Let $G : [0, 1] \rightarrow \mathbb{R}$ be continuous, twice continuously differentiable on $(0, 1]$, and satisfy $G(0) = 0$, $G, G' > 0$. Suppose $G(\lambda) = H(\lambda) + O(\lambda)$ at zero, where $H(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda$. For every $\lambda \in (0, 1]$, suppose there are $M, X, Y \in (0, 1)$ with*

$$X = (1 - M)(1 - e^{-G'(\lambda)})^{1/(1-M)} \quad (\text{U-C.C.4})$$

and

$$G(\lambda) + \frac{1}{2}\{\log X + \lambda \log M + \lambda \log Y\} > 0. \quad (\text{U-C.C.5})$$

Assume that on every compact $I \subset (0, 1]$, $\inf_I G' > 0$, $\sup_I M < 1$, and, for every sufficiently small $\delta > 0$, the point

$$p_{\lambda, \delta} = 1 - e^{-G'(\lambda) + 2\delta}, \quad x_{\lambda, \delta} = e^{-\delta}(1 - M)p_{\lambda, \delta}^{1/(1-M)}, \quad (\text{U-C.C.6})$$

paired with Y , lies in $\mathcal{R}_*$. Then, uniformly for $1 \leq \ell \leq k$,

$$\log R(k, \ell) \leq kG(\ell/k) + o(k). \quad (\text{U-C.C.7})$$

Proof. Fix $\epsilon > 0$. The Erdős–Szekeres recursion and Stirling’s formula give $\log R(k, \ell) \leq kH(\ell/k) + o(k)$, uniformly as $\ell/k \rightarrow 0$. Since $H - G = O(\lambda)$, choose $\rho > 0$ so that the desired bound with error ϵk already holds when $\ell \leq \rho k$. It remains to consider $I = [\rho, 1]$.

The hypotheses in the lemma make the logarithm of $x_{\lambda, \delta}/X(\lambda)$ converge uniformly to zero as $\delta \downarrow 0$. Choose one $\delta > 0$ such that, throughout I ,

$$e^{-\epsilon}X(\lambda) \leq x_{\lambda, \delta} < X(\lambda). \quad (\text{U-C.C.8})$$

Uniform continuity of G, G' supplies a finite descending grid $\{\lambda_j\}$ such that every $r \in I$ has a $\lambda_j \in [r, r + h]$ with

$$|G(\lambda_j) - G(r)| \leq \epsilon/2, \quad |G'(\lambda_j) - G'(r)| \leq \delta. \quad (\text{U-C.C.9})$$

Apply the balanced-book corollary of Appendix B at each grid point with $p = p_{\lambda_j, \delta}$, $\mu = M(\lambda_j)$, $x = x_{\lambda_j, \delta}$, and $y = Y(\lambda_j)$. The strict book hypothesis follows from (U-C.C.6), and the assumed interior condition supplies the other hypothesis. Since the grid is finite, the largest of the resulting thresholds is finite.

Now let $N \geq e^{k(G(r) + \epsilon)}$, where $r = \ell/k \in I$. If the red-edge density is at least $1 - e^{-G'(r) + \delta}$, it is at least $p_{\lambda_j, \delta}$ by (U-C.C.9). Equations (U-C.C.5), (U-C.C.8), (U-C.C.9), $\lambda_j k \geq \ell$, and $\log(MY) < 0$ give

$$\begin{aligned} N &\geq e^{k(G(\lambda_j) + \epsilon/2)} \\ &\geq (e^{-\epsilon}X(\lambda_j))^{-k/2} (M(\lambda_j)Y(\lambda_j))^{-\ell/2} \\ &\geq x_{\lambda_j, \delta}^{-k/2} (M(\lambda_j)Y(\lambda_j))^{-\ell/2}. \end{aligned}$$

The balanced-book corollary therefore gives a red K_k or a blue K_ℓ .

Otherwise the average blue degree is greater than $e^{-G'(r) + \delta}(N - 1)$, so some vertex has, for all sufficiently large k , at least

$$\exp\{kG(r) + \epsilon k - G'(r) + \delta/2\} \quad (\text{U-C.C.10})$$

blue neighbours. Induct on ℓ , starting with the already settled range $\ell \leq \rho k$. For $r - 1/k \geq \rho/2$, the mean-value theorem gives

$$k\{G(r) - G(r - 1/k)\} = G'(\xi), \quad G'(r) - G'(\xi) \leq k^{-1} \sup_{[\rho/2, 1]} |G''| \leq \delta/2$$

once k is large. Thus (U-C.C.10) is at least the induction threshold $e^{k(G((\ell-1)/k) + \epsilon)}$. Its blue neighbourhood contains a red K_k , or a blue $K_{\ell-1}$ which extends with the chosen vertex. This completes the induction and the uniform ϵ -form of (U-C.C.7). $\square$

Lemma C.2 (Elementary first prior). *Let*

$$c^{(0)} = \left(-\frac{1}{4}, \frac{2}{25}, \frac{2}{25}\right) \quad \text{and} \quad F_0(\lambda) = h(\lambda) + e^{-\lambda} \left(-\frac{1}{4}\lambda + \frac{2}{25}\lambda^2 + \frac{2}{25}\lambda^3\right).$$

Then, uniformly for $1 \leq \ell \leq k$,

$$\log R(k, \ell) \leq kF_0(\ell/k) + o(k),$$

and the colour-swapped statement holds as well.

Proof. The first JSON file stores the tuple $c^{(0)}$ above and the exact splice

$$\lambda_* = 0.00500000000000000001.$$

For every $0 < \lambda \leq 1$, set

$$M = \lambda e^{-\lambda}, \quad X = (1 - M)(1 - e^{-F'_0(\lambda)})^{1/(1-M)}, \quad Y = 1 - X.$$

The elementary Ramsey recursion gives $(X, Y) \in \mathcal{R}$. For the strictly smaller first coordinate $x_{\lambda, \delta} < X$ used in Lemma C.1, choose z with $x_{\lambda, \delta} < z < X$. Then $(z, 1 - z) \in \mathcal{R}$, while $x_{\lambda, \delta} < z$ and $Y = 1 - X < 1 - z$; hence $(x_{\lambda, \delta}, Y) \in \mathcal{R}_*$.

On $(0, \lambda_*]$, the analytic small-ratio routine proves $F_0, F'_0 > 0$, bounds $e^{-F'_0}$ above and below by positive multiples of λ , and returns an outward-rounded Arb enclosure whose lower endpoint satisfies

$$\frac{1}{\lambda} \left[F_0(\lambda) + \frac{1}{2} \{ \log X + \lambda \log M + \lambda \log Y \} \right] > 0.00330.$$

This is the same cancellation argument recorded in obligation (iv), now applied to the printed polynomial $c^{(0)}$; it covers the open endpoint at zero analytically.

The interval $[\lambda_*, 1]$ is a union of 4,096 closed rational base cells. At 256-bit precision, with recursive bisection if an inclusion is not immediately decisive, outward-rounded Arb inclusions prove on every cell that $F_0, F'_0 > 0$, $0 < M, X, Y < 1$, and

$$F_0(\lambda) + \frac{1}{2} \{ \log X + \lambda \log M + \lambda \log Y \} > 10^{-6}.$$

Thus all hypotheses of Lemma C.1 hold for $G = F_0$; also $F_0 - h = O(\lambda)$ at zero. That lemma gives the claimed uniform prior rate, without using any later target as an assumption. $\square$

Proposition C.3 (Rate-certificate correctness). *Assume the repaired book induction in Appendix B and Arb's containment semantics. If the first certificate satisfies (i)–(v), its target obeys the uniform Ramsey-rate bound (U3.3). If a finite chain is accepted with the exact handoffs in (v), every target in the chain obeys that bound; in particular the sixth target is F_6 .*

Proof. On a closed record interval, (U-C.C.2) has strict positive margin in both colour orientations. Together with the already proved uniform prior rate, this allows X, Y_i to be enlarged slightly while retaining both exponent bounds; hence $(X, Y_i) \in \mathcal{R}_*$. Decreasing the first coordinate to $x_{\lambda, \delta}$ preserves interior membership. The finitely many records also give $\sup M_i < 1$, and the verifier proves $F, F' > 0$ and (U-C.C.3), which are exactly (U-C.C.4)–(U-C.C.5).

On $(0, \lambda_*]$, the elementary witness has $Y = 1 - X$. Since $x_{\lambda, \delta} < X$, choose z strictly between them. The elementary recursion places $(z, 1 - z)$ in $\mathcal{R}$, while $x_{\lambda, \delta} < z$ and $Y < 1 - z$; therefore the perturbed point lies in $\mathcal{R}_*$. Obligation (iv) proves the remaining analytic hypotheses on the whole open

interval, including a positive normalized main slack rather than a sampled extrapolation. Because every target has the form $H(\lambda) + e^{-\lambda}P(\lambda)$ with $P(0) = 0$, its difference from H is $O(\lambda)$. Thus Lemma C.1 proves one accepted link.

For the first link, Lemma C.2 proves the prior. Exact entrywise target-to-prior equality then permits ordinary induction through the six links, without a rounded or refitted intermediate rate. $\square$

The six accepted certificate identities, in order, are

1. `certificate-higher-order-quintic-v1.json`, 2664bc421cd0cb7489289caa283a3a3f22830580f7c21bf0e6cbe092910bc277;
2. `certificate-higher-order-quintic-chain-v2.json`, b5b595b4dc9d1bdc2b5714f68fef99ec1c566bbd612d35ca19100d173d41c4d;
3. `certificate-higher-order-sextic-chain-v3.json`, 2052952c3af98074d5442fb736c7a2952146e92051a43fab84e75f099d9e00d7;
4. `certificate-higher-order-octic-chain-v4.json`, 09253d757516dc258396f82c09a442aaaf9d499640c1cc99b5767e6d51d35942;
5. `certificate-higher-order-decic-chain-v5.json`, 4ec4dbbe190a08134efe68c3a76ce4b4fb3aabdad230af703c19a28a03ecf9a9;
6. `certificate-higher-order-tetradecic-chain-v6.json`, 8689e3f80f363ac4b2dada31d6d71d682288eac24ca5e128e22aecffcd4bde8.

The canonical replay invokes `verify_chain_arb.py` with `--require-paper-chain`. Before doing any interval work, that mode requires these six filenames in this order, recomputes and compares all six SHA-256 values, and requires the final target tuple to equal the fourteen terminating decimals displayed below, entry by entry.

C.2. Claim. Let

$$H(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda$$

and interpret every displayed terminating decimal exactly. Define

$$\begin{aligned} P_6(\lambda) = & -0.250000000000000\lambda + 0.019891840059418\lambda^2 \\ & -0.012275954247190\lambda^3 + 0.144110816398083\lambda^4 \\ & + 0.006277913420654\lambda^5 - 0.066101623491668\lambda^6 \\ & - 0.002287675993602\lambda^7 - 0.058238059505066\lambda^8 \\ & + 0.030675864198693\lambda^9 + 0.052472201910597\lambda^{10} \\ & + 0.043300454861853\lambda^{11} - 0.042529975074653\lambda^{12} \\ & - 0.055263639426635\lambda^{13} + 0.036695886931268\lambda^{14}. \end{aligned}$$

For $F_6(\lambda) = H(\lambda) + P_6(\lambda)e^{-\lambda}$, the exact six-stage chain and the locally closed combinatorial descent prove the local computer-assisted theorem

$$R(k, k) \leq (3.7806984277961236987518816497522162 \dots)^{k+o(k)}.$$

The status is **PROVABLE AS STATED** under the same local proof boundary as the fifth-stage theorem. This is not a published result, an explicit finite- k bound, or a proof-assistant formalization. Independent external human review of this sixth link remains pending.

C.3. Frozen protocol (declared before target search). This is a bounded attempt to add **one** descent after the frozen fifth-stage certificate. It will not continue to a seventh stage.

The only admissible prior is the exact ten-entry target tuple in `certificate-higher-order-decic-chain-v5.json`, whose SHA-256 is `4ec4dbbe190a08134efe68c3a76ce4b4fb3aabdad230af703c19a28a03ecf9a9`. No rounded, refitted, or independently retyped prior is admissible. Before search, both existing 256-bit Arb implementations must prove this exact prior strictly concave on all of $(0, 1]$.

The construction search is limited to target degrees 10 through 14. A candidate will be frozen only if all of the following predeclared gates hold:

1. its base is at least 0.000001 below the frozen fifth-stage base;
2. dense floating replay plus local scalar refinement finds unbuffered continuous slack at least 0.0001500000000000;
3. the exact certificate has at most 131,072 cells and passes the complete six-stage `verify_chain_arb.py` replay;
4. the separately written `verify_region_direct_arb.py` gives both exact Arb exponent margins at least 0.0000005;
5. exact **Decimal** comparison proves the P5 target tuple equals the P6 prior tuple entry by entry; and
6. `audit_tests.py` passes unchanged.

A certificate that merely has positive margins but misses one of these gates will be reported as a frontier/FAIL, not as a new Ramsey bound. Floating optimization and interpolation are construction aids only and have no role in the proof.

C.4. Pre-search prior audit. The certificate hash was recomputed before optimization and equals the frozen hash above. On its exact target decimals the two independent 256-bit Arb implementations returned

```
verify_arb.prove_prior_concavity:      P5'' <=
↪ -0.35642515827888227
verify_region_direct_arb.prove_concavity: P5'' <=
↪ -0.3574394324533535
```

Thus the strict-concavity gate passed before any sixth-stage search. The frozen prior base, computed in Arb rather than binary floating point, is `3.7807566104095593763172345562324210302278058291660...`

C.5. Bounded degree-10–14 frontier. The first bounded pass used the exact P5 tuple above, a requested construction margin of 0.00015, and no target degree outside 10–14. Its diagnostic frontier was

target degree	diagnostic base	sampled	
		minimum slack	disposition
10	3.78075661040956	0.000148091758	no accepted feasible step
11	3.78075661040956	0.000148091758	no accepted feasible step
12	3.78075661040956	0.000148091758	no accepted feasible step
13	3.78069741660317	0.000150109936	frontier only
14, raw	3.78069363690183	0.000150096701	rejected after dense leak

The raw degree-14 point looked best on the construction mesh, but a denser mesh found slack only

0.00014795745008933103 near $\lambda = 0.3386221294$,

so it failed the predeclared gate and was not certified. No degree was added and no search range was enlarged. A single guarded degree-14 point was then selected within the already declared range. A broad 20,001-point scan (geometric near the splice and linear thereafter), followed by bounded scalar refinement around every sampled local minimum, returned

minimum unbuffered floating slack:

0.00015473769852614172 at $\lambda \approx 0.33943703594667307$

next three refined local minima:

0.00015519556902354736 at $\lambda \approx 0.9859202132743512$

0.00015519607550973370 at $\lambda \approx 0.6439420227559093$

0.00015519870501323751 at $\lambda \approx 0.8750494757168542$

minimum sampled F6: 0.030260793830868504 at $\lambda = 0.005$

minimum sampled F6': 0.7658393690544814 at $\lambda = 1$

floating base: 3.780698427796123

Thus the unique frozen candidate cleared both the 0.00015 continuous construction floor and the 0.000001 base-improvement floor. The eventual exact Arb base improvement over P5 is

0.0000581826134356775653529064802047549...

These floating data only selected the candidate. In particular, they are not used to cover the continuum in the proof.

C.6. Exact replay and claim boundary.

C.6.1. *Frozen certificate and exact handoff.* Only the guarded degree-14 target displayed in the Claim was frozen:

```
file: certificate-higher-order-tetradecic-chain-v6.json
SHA-256:
↪ 8689e3f80f363ac4b2dada31d6d71d682288eac24ca5e128e22aecffcd4bde8
cells: 131072
stored left endpoint: 0.00500000000000000001
stored right endpoint: 1
region buffer: 0.0000800000000000000007
constructor predicted worst buffered slack: 0.00011568648667514836
```

The certificate's ten-entry prior tuple was compared with the P5 target tuple using arbitrary-precision `Decimal`, not binary floating point:

```
P5 target == P6 prior: True
```

Thus the sixth link has no rounded or refitted handoff.

C.6.2. *Complete six-stage 256-bit Arb replay.* `verify_chain_arb.py` replayed the elementary prior and all six exact links and returned

```
PASS: verified 6-stage Ramsey-rate certificate chain
```

The new stage's strict lower margins and endpoint value were

```
target small slack/lambda:
0.0037785734384423962376319094335140413...
```

```
standard Ramsey-region margin:
7.464714995786726e-05 at cell 8
```

```
swapped Ramsey-region margin:
5.766703065384951e-05 at cell 131071
```

```
large-regime main slack:
4.818471567626107e-05 at cell 131071
```

```
F6(1):
1.3299087618219930723187812632036105226...
```

```
exp(F6(1)):
3.7806984277961236987518816497522162752...
```

Every proof-critical margin is strictly positive. The analytic small regime and the 131,072 exact cells cover all $0 < \lambda \leq 1$.

C.6.3. *Separately written direct region replay.* `verify_region_direct_arb.py` does not import the primary verifier and directly minimizes both continuous exponent slacks over the complete prior-ratio interval. It returned

```
PASS: independent direct two-sided Ramsey-region replay
prior P5'' worst certified upper: -0.35743943245335352
```

```
worst standard direct exponent slack:    1.0628096485437162e-06 at
↪ cell 0
worst swapped direct exponent slack:    5.766703065384951e-05 at
↪ cell 131071
```

Both independently replayed margins exceed the predeclared 5e-7 floor. `audit_tests.py` also passed the derivative-sign regression, arbitrary-degree polynomial-derivative regression, and known Horizon union-gap regression.

Consequently every predeclared gate passed and the boxed local asymptotic bound follows. The search stops here: no seventh-stage search was performed.

C.7. Reproduction commands. Run from `routes/upper` with the repository environment that supplies `python-flint`:

```
../../.venv/bin/python verify_chain_arb.py --require-paper-chain \
certificate-higher-order-quintic-v1.json \
certificate-higher-order-quintic-chain-v2.json \
certificate-higher-order-sextic-chain-v3.json \
certificate-higher-order-octic-chain-v4.json \
certificate-higher-order-decic-chain-v5.json \
certificate-higher-order-tetradecic-chain-v6.json

../../.venv/bin/python verify_region_direct_arb.py \
certificate-higher-order-tetradecic-chain-v6.json

../../.venv/bin/python audit_tests.py
```

C.8. Claim boundary.

1. The sixth certificate proves one finite descent only; it supplies no convergence theorem for an infinite chain.
2. The inherited asymptotic descent has non-effective $o(k)$ terms, so this is not an explicit finite- k inequality.
3. The numerical proof imports Arb containment semantics through `python-flint` at 256-bit precision.
4. The local combinatorial proof and interval replay have not been formalized in a proof assistant, and this sixth link has not yet received independent external human review or a DOI-bearing archival deposit.

APPENDIX D. DETAILED ONE-STEP LEDGER ESTIMATES

This appendix records the estimates used in Sections 13 and 14 in a form independent of the exploratory calculation log. Throughout, C is fixed and sufficiently large, $w > \omega_0$, and the target set exposed after column i is $V_i = \{i + 1, \dots, r\}$, of size $k = r - i$.

D.1. The Hessian on the entire cutoff box. Recall

$$F_i(b) = \sum_{j \in V_i} \log \Phi(-b_j) - \frac{1}{d} \sum_{\{j,q\} \subset V_i} T_{jq} m(b_j) m(b_q).$$

Since $(\log \Phi(-x))' = -m(x)$, the entries of the Hessian are

$$\partial_{jj}^2 F_i(b) = -m'(b_j) - \frac{m''(b_j)}{d} \sum_{q \in V_i \setminus \{j\}} T_{jq} m(b_q), \quad (\text{L-A.A.1})$$

$$\partial_{jq}^2 F_i(b) = -\frac{T_{jq}}{d} m'(b_j) m'(b_q), \quad j \neq q. \quad (\text{L-A.A.2})$$

The inverse Mills ratio satisfies $m'' \geq 0$. Hence the interaction term in (L-A.A.1) can only strengthen the diagonal of $-\nabla^2 F_i$. On $[c-w, c+w]^{V_i}$, monotonicity gives

$$m'(b_j) \geq j_-(w), \quad |m'(b_j) m'(b_q)| \leq j_+(w)^2. \quad (\text{L-A.A.3})$$

The residual recursion and $r/\sqrt{d} \leq 1/D$ imply

$$T_{\max} \leq \frac{t_0 + \theta r}{1 - \rho}, \quad (\text{L-A.A.4})$$

$$\frac{1}{d} \max_j \sum_{q \neq j} T_{jq} \leq \frac{m_0 + \theta/D}{D(1 - \rho)}. \quad (\text{L-A.A.5})$$

Combining (L-A.A.1)–(L-A.A.5) with Gershgorin's theorem proves

$$-\nabla^2 F_i(b) \succeq \left[j_-(w) - j_+(w)^2 \frac{m_0 + \theta/D}{D(1 - \rho)} \right] I = \mu_w I. \quad (\text{L-A.A.6})$$

This is a full-box statement, not an expansion valid only at the central cutoff.

D.2. Boundary merge with a controlled residual. The gradient at $c\mathbf{1}$ is

$$\partial_j F_i(c\mathbf{1}) = -m_0 - \frac{m_0 m'_0}{d} \sum_{q \in V_i \setminus \{j\}} T_{jq}. \quad (\text{L-A.A.7})$$

The defining recursion

$$T_{ij} = m_0 \sqrt{d} + \frac{m_0 m'_0}{\sqrt{d}} \sum_{q \in V_i \setminus \{j\}} T_{jq} + e_{ij} \quad (\text{L-A.A.8})$$

therefore gives

$$\partial_j F_i(c\mathbf{1}) = -\frac{T_{ij} - e_{ij}}{\sqrt{d}}. \quad (\text{L-A.A.9})$$

For $\delta_j = b_j - c = \sqrt{d} h_{ij} + \varepsilon_{ij}$, with $|\varepsilon_{ij}| \leq d^{-1/5}$, the tangent inequality from (L-A.A.6) gives

$$\begin{aligned} F_i(b) &\leq F_i(c\mathbf{1}) - \sum_{j \in V_i} T_{ij} h_{ij} + \sum_{j \in V_i} \left(\frac{e_{ij} \delta_j}{\sqrt{d}} - \frac{\mu_w \delta_j^2}{2} \right) + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j \in V_i} T_{ij} \\ &\leq F_i(c\mathbf{1}) - \sum_{j \in V_i} T_{ij} h_{ij} + \sum_{j \in V_i} \frac{e_{ij}^2}{2\mu_w d} + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j \in V_i} T_{ij}. \end{aligned} \quad (\text{L-A.A.10})$$

The second line is the scalar square completion $xy - \mu_w y^2/2 \leq x^2/(2\mu_w)$. Thus every boundary edge merges with its full coefficient T_{ij} ; only the explicit residual is paid.

D.3. Mass identities. Let $\Delta_{ij} = T_{ij} - t_0$. Backward induction in the first endpoint gives, for every $j < q$,

$$A_0(r-j-1) + \theta(j-1) \leq \Delta_{jq} \leq \frac{A_0 + \rho\theta}{1-\rho}(r-j-1) + \theta(j-1). \quad (\text{L-A.A.11})$$

For S_i and R_i as in Section 14, this implies

$$S_i \geq a_-(r-2) \binom{k}{2}, \quad R_i \leq a_+(r-2)(k-1), \quad \frac{R_i}{S_i} \leq \frac{2a_+}{a_-k}. \quad (\text{L-A.A.12})$$

The exact accumulated sums are obtained by exchanging the order of summation. The future-degree part of (L-A.A.11) counts each ordered quadruple twice, while the residual part contributes

$$K_r := \sum_{i=1}^{r-1} (r-i)(i-1)^2 = \binom{r}{3} + 2 \binom{r}{4}. \quad (\text{L-A.A.13})$$

Consequently

$$\sum_{i=1}^{r-1} S_i \geq 2A_0 \binom{r}{4} + \theta K_r, \quad \sum_{1 \leq i < j \leq r} e_{ij}^2 = \theta^2 K_r. \quad (\text{L-A.A.14})$$

No asymptotic replacement is used in (L-A.A.13)–(L-A.A.14).

D.4. Three-factor Hölder comparison. Fix a red-admissible $\mathcal{F}_{i-1}$ -history for target r , condition further on A_i^R , and write

$$X_j = -\frac{m(b_j)}{\sqrt{d}} + \frac{\zeta_j}{\sqrt{d}}, \quad u_j^0 = \frac{t_0}{d} \sum_{q \neq j} m(b_q), \quad \delta u_j = \frac{1}{d} \sum_{q \neq j} \Delta_{jq} m(b_q). \quad (\text{L-A.A.15})$$

The variables ζ_j are conditionally independent, centered and 1-subgaussian. The centered exponent is exactly

$$\sum_j (u_j^0 + \delta u_j) \zeta_j - \frac{t_0}{d} \sum_{j < q} \zeta_j \zeta_q - \frac{1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q. \quad (\text{L-A.A.16})$$

Use conjugate exponents

$$Q_0 = \frac{d}{4t_0k}, \quad Q_1 = \frac{d}{4R_i}, \quad P^{-1} = 1 - Q_0^{-1} - Q_1^{-1}. \quad (\text{L-A.A.17})$$

They are all greater than one, with $P \leq 2$, because

$$\frac{4m_0}{D} + \frac{4a_+}{D^2} < \frac{1}{2}. \quad (\text{L-A.A.18})$$

Hölder gives one linear-CGF factor and two quadratic factors. The uniform t_0 factor is the frozen HMS factor. For the increment, use

$$-\sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \leq \frac{1}{2} \sum_j R_j \zeta_j^2, \quad \frac{Q_1 R_j}{2d} \leq \frac{1}{8}. \quad (\text{L-A.A.19})$$

The square-MGF estimate $\log \mathbb{E} e^{x\zeta_j^2} \leq (16/3)x$ for $0 \leq x \leq 1/8$ therefore gives

$$\frac{1}{Q_1} \sum_j \frac{16}{3} \frac{Q_1 R_j}{2d} = \frac{16}{3} \frac{S_i}{d}. \quad (\text{L-A.A.20})$$

For the linear factor, let $P^0 = P_{k,d}^0$. The source exact-CGF deficit on this identical history is

$$\mathfrak{d}_i(b) = \sum_j \left[(u_j^0)^2 - \frac{1}{P^0} K_{b_j}(P^0 u_j^0) \right] \geq \mathfrak{d}_{k,w,d}. \quad (\text{L-A.A.21})$$

Thus it remains only to bound the change from (P^0, u^0) to $(P, u^0 + \delta u)$. Since $0 \leq K_b'' \leq 1$,

$$\frac{K_b(P(u + \delta)) - K_b(Pu)}{P} \leq 2u\delta + \delta^2, \quad (\text{L-A.A.22})$$

$$0 \leq \frac{\partial}{\partial P} \frac{K_b(Pu)}{P} \leq u^2. \quad (\text{L-A.A.23})$$

Furthermore

$$P - P^0 \leq \frac{16R_i}{d}, \quad \sum_j \delta u_j \leq \frac{2M_w S_i}{d}, \quad \max_j \delta u_j \leq \frac{M_w R_i}{d}, \quad (\text{L-A.A.24})$$

and

$$\sum_j (u_j^0)^2 \leq \frac{2m_0^2 M_w^2 S_i}{a_- d}. \quad (\text{L-A.A.25})$$

Substitution in (L-A.A.22)–(L-A.A.24) gives, in order,

$$\frac{4m_0 M_w^2 S_i}{D} \frac{1}{d}, \quad \frac{2M_w^2 a_+ S_i}{D^2} \frac{1}{d}, \quad \frac{32m_0^2 M_w^2 a_+ S_i}{a_- D^2} \frac{1}{d}. \quad (\text{L-A.A.26})$$

These are precisely the three summands of $\hat{L}_w(C) S_i/d$. Combining them with (L-A.A.20) and the central deterministic gain $-m_0^2 S_i/d$ gives

$$-\left(m_0^2 - \frac{16}{3} - \hat{L}_w(C)\right) \frac{S_i}{d} = -\hat{B}_w(C) \frac{S_i}{d}. \quad (\text{L-A.A.27})$$

Equation (L-A.A.21) is charged once, (L-A.A.27) pays only the new $\Delta = T - t_0$ factor and the change of exponent, and (L-A.A.10) pays the residual gradient. This is the promised same-history separation.

D.5. One-step conclusion and extraction. Combining (L-A.A.10), (L-A.A.21), and (L-A.A.27) proves (L2.37) with the value λ_i in (L6.2). The weighted reverse-propagation Lemma 11.3, followed by (L-A.A.14), then gives the empty-state bound

$$P_{R,r}^* \leq \mathcal{H}_C(r) \exp \left[-\mathcal{D}_{C,w,d}(r) - \frac{1}{d} \left\{ 2\hat{B}_w A_0 \binom{r}{4} + E_w K_r \right\} + J_{r,d} \right]. \quad (\text{L-A.A.28})$$

where

$$0 \leq J_{r,d} \leq \frac{m_0 + \theta/D}{1 - \rho} d^{-1/5} \binom{r}{2} = o_C(r^2). \quad (\text{L-A.A.29})$$

If extraction deletes u vertices from a retained set of size ℓ , then the old exact-CGF restoration cost is at most

$$\mathcal{D}_{C,w,d}(\ell) - \mathcal{D}_{C,w,d}(\ell - u) \leq \frac{m_0^4}{D^2} \ell u \leq \ell u, \quad (\text{L-A.A.30})$$

and the loss in the new deterministic exponent is at most

$$\frac{1}{d} \left[2\hat{B}_w A_0 u \binom{\ell}{3} + E_w \left(u \binom{\ell}{2} + 2u \binom{\ell}{3} \right) \right] \leq \left[\frac{\hat{B}_w A_0 + E_w}{3D^2} + \frac{E_w}{2D^2 \ell} \right] \ell u. \quad (\text{L-A.A.31})$$

For sufficiently large C and then sufficiently large ℓ , the bracket in (L-A.A.31) is

$$O(1/\log C) + O_C(\ell^{-1}) < 1.$$

Together with the source-envelope restoration $[\log(1/p) + 1]\ell u$, the total cost is at most $[\log(1/p) + 3]\ell u$. The pinned failure factor therefore leaves $\eta_C = 10 \log(10/p) - \log(1/p) - 3 > 0$, and

$$\sum_{u=1}^{\ell} \binom{\ell}{u} e^{-\eta_C \ell u} = (1 + e^{-\eta_C \ell})^{\ell} - 1 = o(1). \quad (\text{L-A.A.32})$$

This proves that (L-A.A.28) survives perfect-sequence extraction with its entire old and new exponent intact.

APPENDIX E. PINNED SOURCE INTERFACE AND QUANTIFIER LEDGER

This appendix records the exact external boundary of the lower-bound argument. It separates version-pinned conditional probability statements from the locally proved weighted reverse-propagation lemma and imports neither that lemma, the first-moment conversion, nor the Ramsey conclusion proved in Section 16.

E.1. Content-addressed source snapshots. The source files are Hunter–Milojević–Sudakov, arXiv:2512.17718v2, file `arXiv_version.tex`, SHA-256 `b72958ac35554eccb94dedab5800349d2c021af7d60f767125cb46998e0fd54a` and Lin–Niu, arXiv:2605.25843v2, file `off-diagonal-Ramsey-R.tex`, SHA-256

`dbf1bfa2c7603c81b2e56a97028b138ccf9a9299a3d1f20c0b08cf489136f429`

The repository manifest records the versioned retrieval URLs and independently checks these hashes.

E.2. Source-line map for the source interface.

Pinned source lines	Exact use in this paper
HMS 145–170, 443–498	The Gaussian graph, Bartlett representation, column exposure, edge events, perfect events, and the unweighted projection state. These are the probability-space and history definitions in (L2.24)–(L2.33).
HMS 955–971, 1192–1218	Optimized red-perfectness and the exact cutoff approximation. Cauchy–Schwarz gives $\sqrt{d} h_{ij} \leq \alpha_R^2 \ell / \sqrt{d} \leq \omega_0$, while the diagonal approximation contributes $d^{-1/5}$. This is (S1).
HMS 330–354, 878–935	The subgaussian definition, square-MGF lemma, pre-Cauchy centered-linear allocation, and centered quadratic estimate. These give the quantified statements (L2.35)–(L2.36), the uniform t_0 factor in (L5.12), and the square estimate in (L5.14); no final Cauchy coarsening is imported.
Lin–Niu 359–493, 499–537	Monotonicity of upper-truncated Gaussian variance and the exact positive-direction CGF. After replacing their upper cutoff by $-b$, these give (L2.4)–(L2.6) and Lemma 11.1.
HMS 1192–1289	The red law of total probability, cutoff conditioning, projection merge, and scalar reverse-induction allocation. Equation (L2.36a) is the frozen scalar baseline. The indicator-kernel conjugation and weighted propagation (S3) are proved locally from (L2.33), (L2.39), and (L2.40); neither the nonuniform one-step estimate nor its propagation is attributed verbatim to HMS.
HMS 385–410 and 955–971	For every specified retained set, the red-perfectness failure factor is $(p/10)^{10\ell u}$. This is the sole external extraction input (L2.41); the restoration estimates and binomial sum are proved in (L6.5)–(L6.11).
HMS 977–1181, 1294–1312, together with 385–410	The unchanged blue reverse induction, its perfect bound, and the same greedy extraction argument, yielding the raw probability inequality (L2.42). The first-moment threshold (L7.4) is derived locally.

The separately fetched Yang–Mao v1 member has SHA-256

155b7104ec5b6935a576ae9f2b161976a966b0b46bd2b69153c0934ca688da2a

agreeing with the repository manifest. It is not an input to the lower-bound component, so no Yang–Mao line range is invoked here; assigning one would misstate the lower component’s dependency boundary.

E.3. Why the cutoff window is ω_0 . On the event B_r^R ,

$$|b_{ij} - c| \leq \sqrt{d}|h_{ij}| + d^{-1/5} \leq \frac{\alpha_R^2 \ell}{\sqrt{d}} + d^{-1/5} \leq \omega_0 + d^{-1/5}. \quad (\text{L-B.B.1})$$

For every fixed $w > \omega_0$, choosing $\ell \geq \ell_0(C, w)$ makes the last expression at most w . The proof is first carried out for this open range of w , and only after the fixed- C , $\ell \rightarrow \infty$ limit is taken do we let $w \downarrow \omega_0$. No finite- ℓ history is claimed to lie in the closed boundary window. This also explains why p , rather than p_C , occurs in α_R and ω_0 .

E.4. The old exact-CGF entry is an input formula, not a conclusion.

At a step with k future vertices, define

$$b_j = b_{ij}, \quad M_j = \sum_{q \neq j} m(b_q), \quad u_j^0 = \frac{m_0 M_j}{\sqrt{d}}. \quad (\text{L-B.B.2})$$

Stopping the HMS centered-moment calculation before its final Cauchy coarsening gives the allocation $\sum_j (u_j^0)^2$. Applying the exact positive-direction CGF with $P_{k,d}^0$ gives

$$\sum_j \frac{1}{P_{k,d}^0} K_{b_j}(P_{k,d}^0 u_j^0). \quad (\text{L-B.B.3})$$

Their difference is the same-history quantity $\mathfrak{d}_i(b)$ in (L5.16). The cutoff monotonicities give (L5.17), and summing its deterministic lower bound gives (L2.15). With $k/\ell \rightarrow x$ and $\sqrt{d}/\ell \rightarrow D$, the corresponding Riemann sum is (L2.17). Hence $G_*(C)$ is defined and charged before the controlled residual is added; it is not imported through a final Ramsey theorem.

E.5. Raw ledgers and the local crossing calculation. The red interface before the new two deficits is the explicit envelope

$$\mathcal{H}_C(r) = p^{(r)} \exp \left[-\vartheta_C \frac{a^3}{p^3 \sqrt{d}} \binom{r}{3} + \frac{a}{p} d^{-1/5} \binom{r}{2} + r \right]. \quad (\text{L-B.B.4})$$

The blue interface is the explicit probability estimate (L2.42). Dividing their logarithms by ℓ^2 , and only then applying $\binom{n}{k} \leq n^k$, gives the red and blue thresholds (L7.2) and (L7.4). The comparison is not assumed: the source asymptotics imply

$$\frac{B_R(C)}{2} = \frac{1}{12} + o(1), \quad \frac{CB_B(C)}{2} = \frac{1}{2} + o(1), \quad (\text{L-B.B.5})$$

whereas

$$G_*(C) + \widehat{H}_*(C) = O(1/\log C) = o(1). \quad (\text{L-B.B.6})$$

Thus the blue-minus-red gap is $5/12 + o(1)$, proving locally that red is the active constraint for sufficiently large fixed C .

E.6. Order of quantifiers and proof status. The absolute constant K in (L2.10) is fixed before selecting the large- C threshold, but its numerical value is not provided by the pinned source. At theorem level the quantifiers therefore begin in the order

$$\text{fix } K, \quad \exists C_0 = C_0(K) \forall C \geq C_0 \exists \ell_0(C) \forall \ell \geq \ell_0(C), \quad (\text{L-B.B.7})$$

followed by $\ell \rightarrow \infty$. Within the proof, the auxiliary w is fixed in an admissible right-neighborhood of ω_0 during this limit and is then sent down to ω_0 . Only after that fixed- C theorem is established is $C \rightarrow \infty$ used to obtain the coefficient $1/64$. No explicit value of $C_0(K)$ or $\ell_0(C)$ is asserted.

The resulting theorem is source-relative: its local residual algebra, three-factor bridge, weighted propagation, reverse-induction charge, extraction sum, first moments and red–blue crossing are proved here. Items (S1), (S2), (S4), and (S5) are version-pinned source hypotheses; (S3) is Lemma 11.3, proved locally rather than attributed to either source. The arithmetic checker corroborates the finite formulas but does not replace the four imported probabilistic interfaces or the local proof.

APPENDIX F. CONTENT-ADDRESSED ARTIFACT LEDGER

The digests in the two artifact tables below are SHA-256 identities of the compressed files. Their byte counts refer to those same compressed files and therefore permit a check before decompression. Digests explicitly introduced as uncompressed identities or as metadata-record identities have the separate meaning stated in their surrounding text.

Table 7: Budget-five proof pairs.

branch	artifact	bytes	SHA-256
0	CNF	837880	d64f009f840597e266ccfb3b31c88f7e87eb760 d1fc551b7ee08a0854749ba66
0	DRAT	13125926	60687241833df8ae957b3b4fcb02e6e9f7595c6 e3dcbee8b77f02dad5443f913
1	CNF	1797965	d6d321a4ac1aee5e99c48843a8cda9395271e8 9555cea328e29bcda4e70ef15
1	DRAT	37102450	33a9e037e19beeab8dd2af4dca5ce9010d4906b 39dc3e5cc8fd62823f518497a
2	CNF	1363549	8d8a543bb3cc17014941fee432673a1dc2b99c8 cb4885d4d91bba9005f3b10a7
2	DRAT	21678734	159865da9ad23eea298ebd4e9406f2f37ecc601 5a593975e9d63e3b08a859897

The corresponding uncompressed identities, independently recomputed during review, are as follows:

Branch 0 CNF: c907dc96c1ff3378d345ead0ea22456eccef50f321cc53b6cf7dda46784d6389
Branch 0 DRAT: 517d076efd026eab6eeb89583c0226e713be968120344e6b82c3471e0efe6b5d
Branch 1 CNF: 5ba541f58ea42d7fb73ff51a962e98153eece0d4a8bf13c1bea3a86ed0d50429
Branch 1 DRAT: 84a2236b7692207f59c3d2dcb3521b2f2b947428470384ac643c34a7bfbf69
Branch 2 CNF: 441b70e023f4a9fa1837f16b1502ef3b54a2f00b6143fd5e7de0131ac95beebf
Branch 2 DRAT: 4ff3a43bda7774c09695417a9f9684e8a856fb46bf62225a08e746fbf37d1ad6

Table 8: Exact-six proof pairs.

branch	artifact	bytes	SHA-256
0	CNF	17600735	47052de808b98598f2f144251a6ad73c1415dd00fef9e04b99ffe2f176229fbf
0	DRAT	656144450	4d948ab34f1d2475af6efe943e0a4899827c78b96c2bc3835b5161f189c26bb5
1	CNF	5052085	a438c9fef4d99ae516829e4405da8c9f079398501361f9c5f69f46a494ffad14
1	DRAT	433552903	cfe694fd728903ee9ac9f08a66ade64faf7e68763246d1d52ca29a85727494ec
2	CNF	1205208	c51fed43eeca3becf64bcdfe747a4fdf3e516ce0e6a02d7d867941b87a99c06
2	DRAT	152784369	c9648a4e38c17dcdfe97d873095eda002abc689b3b3bd93f02fc8a369198c4b9

For the exact-six formulas, the frozen cut-bank identities are

Branch 0, 251771 masks: 91b5709248ff641a315f5a0389b4f3fde3d38514f3b1a8b31b6cad31224f250b
Branch 1, 63943 masks: f8064f0f226659b4750382f781ba9a862bcc19eb49006eb8cdd2afc002ed227c
Branch 2, 5422 masks: aaa79a2f25d4c0d61c4467f1d0a721fd670a624e0391d96b2122723f97ba4598

The exact-six proof records have SHA-256 identities

Branch 0: 20729b2a818aaca35cdee6e9deeb5a0e6e4bbd6dc5ca11e5abb6dfe9409b035f
Branch 1: 4a3e7b4c729f696ff304c5f71621e018786b552146784a2141529a4c6f028666
Branch 2: ee0c73b80407a4dd48ca69bd1a666f391bfd5729b9a56e062deef2a6f62018de

The source files use the following release-asset names:

```

r3_18_budget5_branch_0_twostage.{cnf,drat}.gz
r3_18_budget5_branch_1.{cnf,drat}.gz
r3_18_budget5_branch_2.{cnf,drat}.gz
r3_18_budget6_branch_0_universal_union.{cnf,drat}.gz
r3_18_budget6_branch_1.{cnf,drat}.gz
r3_18_budget6_branch_2.{cnf,drat}.gz

```

APPENDIX G. INDEPENDENT VERIFICATION PROTOCOL

This appendix states a reviewer-oriented protocol. It is intentionally more specific than the convenience wrapper in the repository.

G.1. Seed verification. Parse the 100×100 matrix, reject any non-binary, asymmetric, or non-zero-diagonal input, and verify its SHA-256. Count undirected edges by the strict upper triangle. Enumerate every vertex triple and record all triangles. Run the exact independent-set verifier at target 18. Repeat the structural checks for the 99-vertex principal submatrix and compare it byte-for-byte with its separately stored certificate. A successful outcome must report 827 edges and exactly the triangle $\{97, 98, 99\}$ for H , and 809 edges, no triangle, and no independent 18-set for G_0 .

G.2. Formula reconstruction. For each exact-six branch i :

1. Assign primary identifiers $1, \dots, 4950$ to pairs in lexicographic order.
2. Generate the 161700 triangle clauses in lexicographic triple order and compare them to the DIMACS prefix.
3. Form the 826 residual deletion literals in the lexicographic order inherited from the primary pair list. Without calling PySAT's `CardEnc`, generate the four Sinz clause families in Lemma 21.1. Generate the equality block as at-most 821 on the complemented literals followed by at-most 5 on the original literals. Compare all 16420 clauses and the final variable identifier 13160.
4. Compare the fixed negative unit for f_i .
5. For each remaining clause, require 153 distinct, positive primary literals. Invert the pair map, recover exactly 18 vertices, and verify that the clause is the full $\binom{18}{2}$ pair family of those vertices.
6. Compare the recovered ordered masks with the frozen cut bank and check that no unparsed clause remains.

The analogous budget-five reconstruction replaces equality at five by an at-most-four counter. It has 8238 total variables and 7394 counter clauses in each branch. The three hitting-bank sizes are 287, 15872, and 8704. The executable audit is `routes/finite/check_r3_18_seqcounter.py`; its schema-only mode is part of the lightweight tier, and the complete mode compares the six actual DIMACS blocks during `finite-heavy`.

For each branch-1 singleton instance, independently reconstruct the common 718452-clause relaxation including all degree counters, append exactly one negative primary-variable unit, and require 154190 variables and 718453

clauses. The four units must be $-1085, -1672, -1675, -1680$. Separately reconstruct the unit-free common CNF, parse a complete assignment of all 154190 variables, evaluate every clause, and run an exact independent-set verifier on the recovered graph.

G.3. DRAT replay. Use one of the two manifest-authenticated `drat-trim` binaries built from source commit `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`. Verify the adjacent source marker and the exact platform-binary hash. The repository helper

```
bash scripts/build_drat_trim.sh \
  /absolute/new/checker-directory
```

performs a source checkout for non-normative manual comparison; an arbitrary fresh build is not accepted by the strict core-proof gate. For each of the ten proof pairs in Tables 7, 8 and 12, check compressed byte counts and SHA-256 values, test gzip integrity, and decompress into a new temporary directory. Then invoke

```
/path/to/drat-trim formula.cnf proof.drat
```

and require both exit status zero and an exact output line `s VERIFIED`. The checker may print timing diagnostics after that status line. Delete the temporary raw files only after recording the uncompressed hashes.

For the complete package, the repository offers integrated semantic and proof replay:

```
.venv/bin/python reproduce.py finite-heavy \
  --artifact-dir /path/to/release-assets \
  --drat-trim /path/to/audited-manifest-binary
```

Time limits may need to be increased on slower hardware. A timeout is UNKNOWN, never a failed or successful proof.

G.4. Logical audit checklist. A reviewer can check the mathematical implication independently of the code:

1. the three f_i are precisely the edges of a seed triangle;
2. every triangle-free final graph omits at least one f_i ;
3. original nonedges are primary variables but not deletion literals;
4. every installed C_S is entailed by $\alpha(F) < 18$;
5. omitting uninstalled C_S clauses makes the CNF weaker, so an UNSAT result remains sufficient;
6. the budget-five refutations force any putative budget-six candidate to have exactly six deletions;
7. after choosing one absent f_i , exactly five residual deletion literals are true; and
8. checked UNSAT of all three exact-six branches contradicts the model provided by any candidate graph;
9. checked UNSAT of each singleton-augmented common relaxation proves the corresponding positive edge consequence; and

10. the checked common SAT model establishes only non-vacuity, while its explicit independent 18-set prevents it from being reported as a repair.

APPENDIX H. FROZEN EXACT-SEVEN FIRST-ROUND RECORD

All three endpoints are UNKNOWN. The branch records are content-addressed as follows.

Table 9: Exact-seven bounded-search records.

branch	result JSON SHA-256	checkpoint SHA-256
0	97e777da32b82a83b8e999c9af1f92 7c87b67689b8fb72f8b31ded07b39a 2f7f	918dc1cc80860d97b3209f6d969a15 6b9df5fb8b4df7f0fa910e72fdc577 8098
1	81734a23298ea9c3364a047c3b7b10 910bcd44faf9ff55c3a251fd709f6a c64f	8491cbd5e073170b6f7a503efb3d6a f6c077b6d72defbf9cd6feb6f06e85 fc7f
2	92e2c948892d6f298e0ca3a5055cf3 1cf3eea361b6eee8197ba536922e43 7554	9e8ea23904378ec1bfc6e4c486fd32 f10ccc99c1646855c7589c21cc3c9b 9f5c

The frozen-state checker, its unit tests, and the branch generator have respective SHA-256 identities

checker: 8c1f2629193de8b16e018f036ce462b8e18110b9f4f1c13b3162b25fe415d6f4
tests: 46b0324d67baaf66bae49027bf4ddbfe3aa08684d3cbe364326bc4400914c987
generator: 1 3 a 7 c f e b 5 6 4 2 9 7 f c a d 2 e a 2 c 6 c a 8 3 1 a 6 b
 f4b4755e0223abcf9d1bd1178103d0de

Each branch loaded the common universal bank in under six seconds and then spent approximately 300 seconds in its first solver call. The registered endpoints were

branch	fixed absent	solver	endpoint
0	(97, 98)	CaDiCaL	UNKNOWN
1	(97, 99)	Glucose 4.2	UNKNOWN
2	(98, 99)	MapleChrono	UNKNOWN

The state checker verifies formula dimensions, the universal bank digest, the exact counter truth table, exclusion of original nonedges from the counter, and the three timeout records. It does not transform those timeout records into a satisfiability or unsatisfiability claim.

H.1. Deletion-first branch-1 follow-up. The historical ascending implementation used for the first nine deletion-first endpoints has SHA-256 identities

runner: 36dc3b53941605bc4ec132b70b4f61c5afbbcd13742fe9056a38ddb1683e5a0

tests: ed62d2318e64d6731bca343e829f2e65c058f8f484a72b027b2a87da9f2fd76f

The order-aware implementation used for the subsequent seven gates has runner and test identities

runner: 22ddb00aa93727aa51d2edd6d09b0e536a230e0cc17426004858e98620d6c4f8

tests: b65ebc6f265de6552cdf6f8c09586f4c7c2ae01db0292908ae7998beea75a35c

The sanitized summary identity for all sixteen endpoints is updated in the project-relative machine record named below. The runner’s structural fingerprint is `e 9 1 f 0 7 d b 0 f 0 1 b 4 4 2 6 3 f a 7 9 0 9 2 2 e 4 6 e f b a149b2f461d7afb77b53f847b9641839`. The project-relative machine summary and human-readable command ledger are `routes/finite/r3_18_budget7_benders_pilot_summary.json` and `routes/finite/R3_18_BUDGET7_BENDERS_PILOT_2026-08-30.md`.

Table 10: Historical ascending branch-1 deletion-first endpoints.

run	endpoint	sec.	cuts	unproved NG
g42_c0	subproblem	3.680	0	0
	oracle limit			
g42_c4096	subproblem	9.700	16	0
	oracle limit			
m22_c0	subproblem	15.256	2816	0
	oracle limit			
m22_c4096	subproblem	47.856	1039	0
	oracle limit			
m22_c16384	global wall	180.026	0	0
maple_c4096	subproblem	62.784	768	2
	oracle limit			
g42+maple	subproblem	121.884	10780	0
	oracle limit			
maple+maple	subproblem	197.868	13362	2
	oracle limit			
m22+maple	global wall	300.095	12800	0

Table 11: Order-aware branch-1 gates.

run	endpoint	sec.	models	cuts
g42_bidir60	global wall	64.077	1	512
maple_bidir60	global wall	60.165	0	0
g42_bidir10	global wall	120.218	5	2560
m22_bidir10	global wall	120.038	1	512
maple_bidir10	global wall	120.059	4	2048
g42_batch4096	global wall	180.200	4	16384

run	endpoint	sec.	models	cuts
maple_batch4096	global wall	180.089	3	12288

In Table 10, the final column counts only solver-trusted fixed-deletion UNSAT endpoints retained by the search. No DRAT or LRAT proof was emitted for either endpoint, and no ordered gate added a new no-good. All rows in Tables 10 and 11 therefore have global status UNKNOWN, and none is used in the proof of the finite theorem.

On the two reconstructed deep supports, reverse vertex order returned 512 validated independent 18-sets in 11,261 and 10,057 nodes, about 0.006 seconds each; the ascending order had reached its resource limit. Each nonempty ordered gate likewise returned its requested cut batch in at most 0.075 seconds, but none reached a fixed-deletion subproblem. The same oracle-cap and order/batch/master matrix was therefore stopped rather than extrapolated.

H.2. Proof-carrying singleton consequences. The follow-up common relaxation Ψ_1 has 154190 variables and 718452 clauses before candidate units. Its formula fingerprint is `a6f920afa451174ed05932174951481a63647a2ba603c762379f9357323fa5e2`. It contains 161700 triangle clauses, 19680 exact-six counter clauses, one fixed branch unit, 285300 degree-cap clauses, and the ordered bank of 251771 universal independent-18 clauses. No learned core clause is installed. Appending one candidate-deletion unit gives 718453 clauses.

The proof-run exporter is frozen by Git commit `408e3bb` and SHA-256 `5e05f75affc5c6c5323fb2f1bddd3097c7a851c00330fd558f1f731e978936cf`. The final four-singleton machine summary has SHA-256 `8ae4f0d9ea919c915ba642eb03ec3d418b4419a89781377509acbe925d89555a`. The strongest released antichain consists of the following four singleton refutations; the earlier pair refutations are valid but dominated and are not included as normative release assets.

Table 12: Branch-1 singleton-proof payloads.

label	assumed deleted edge	CNF bytes	DRAT bytes
K_a	(11, 62)	18914252	707650273
K_b	(18, 61)	18914251	855530402
K_c	(18, 64)	18914252	1117799611
K_d	(18, 69)	18914248	811556002

The corresponding compressed CNF identities are, in table order,

K_a: `1ec7a15b9afd7c7c0944bbc663db2522b47d6f4f12fe98f23501d1006fabe411`
K_b: `1df74150cf9678e977364d4df782e179d62dff811b940d9fc0ea35c30cda845b`
K_c: `95efbf95802cf05e7ce5d676c35470989b126b6145eb00f7e67a546948bde07e`
K_d: `b852ca92b33ac461d425f62309486e30645f36f7736bd18b2b5e15b90fb27483`

The compressed DRAT identities are, in the same order,

K_a: c71be4804965a66e2783c61ae4711e1dc5faeb13a42ecab50672dd0efdd8818c
 K_b: c34b97f66e3513c8261e1424f66b9e2299a49b9f38a6d953ade17e6c4e266bfa
 K_c: 375abf0884e415c2f4fea15d1971b818b2535996f017269fee742e90376f7fc3
 K_d: 34853e20f7055e9a1e33b2668437046414e37a06821a82526eb4c822e61ab886

CaDiCaL was pinned to source commit c60730422e758ef1cebe7aeddff2dda31c996bf04; drat-trim was pinned to 2e3b2dc0ecf938addbd779d42877b6ed69d9a985. Each solver run exited 20 with a complete proof, and each checker run exited zero with an exact s VERIFIED line. A separately compiled arm64 checker from the same source commit performed a separate cross-architecture replay of all four traces. This is a second binary replay, not a second implementation of DRAT semantics.

The PySAT-free semantic checker has SHA-256 2dcc16137a0bd555a2553e9987968fda429e9098ec8c21affb05bfd5a5e03f3b. It parses the seed and ordered bank, reconstructs both counter families without importing the production builder, and compares the complete DIMACS byte stream. Its four audit-record SHA-256 identities are

K_a: 4637a558f46abefcfafcf5940781eb4c9c76556793491e48b1eb5e919d9d6d581
 K_b: c5d1f6823fb08388517c4eb9e0cddc040e27d080168ff4bd513928e095a908a9
 K_c: 3a95e9e412d78c2fb7af1e1ff68828846958a5f969a70996051d2c5dcbbef9bf
 K_d: f38e987e9fecdd7c4ed646cd9c0f50a3838f8ee6cadd436cd4b563d5db431b26c

The common relaxation itself is nonempty. Its deterministic compressed CNF has SHA-256 39249ef8378de3f2ef412e514f6283cdaa032bc17fb25fc3448463e5566f5365; the complete compressed model has SHA-256 9057ab25b785640345e6f724cb1c79313642bb62ec2a19d9f310e25700de024d. An independent audit with SHA-256 7b2af8d92d35721fd6197b3ac3c5d52066bd04faa545e219cc57b94dce035921 evaluated all clauses and recovered the independent 18-set displayed in the main text. Thus this witness proves non-vacuity but is not a target repair.

The two historical fixed-deletion supports in Table 10 delete both (11, 62) and (18, 61). Each is therefore excluded already by either of the corresponding singleton certificates. Their original solver calls emitted no proofs and are not retroactively described as certified; the later common-relaxation proofs are a distinct evidence chain.

The machine-readable proof record is routes/finite/r3_18_budget7_branch1_core_proof_summary.json; the full human ledger is routes/finite/R3_18_BUDGET7_BRANCH1_CORE_PROOFS_2026-08-30.md. The four singleton CNF/DRAT pairs, the common CNF/model pair, and two audited checker binaries are designated large release assets rather than Git objects. Their public-release download and clean-clone replay remain a separate publication gate and do not alter the exact-seven status UNKNOWN.

H.3. Maximal-union construction and bounded endpoint. A standalone degree-initialized one-dimensional Weisfeiler–Leman checker certifies the color-class profile (3,15,100), and hence the triviality of the seed automorphism group. Its script, three-test regression, and audit-record SHA-256 identities are 613c902fe89fedfc2e2132e6a5d7ea182b7cc4b2fd245885861c29f36ba62

f18, 93047e1673d73f6adca0109bb200d524191b9f9671d0ed84b7466203f66cc114, and 8ad9fbec731a3c2f3b3012695022366fc66070265c6f5a27bfc0c66f34e872ce. Thus the branch-1 gate below has no seed-symmetry implication for branches zero and two.

The direct branch-1 maximal-union formula combines the four checked retention units, all 530525 distinct independent-18 masks in the four-family union, and the full 4950-pair maximal-triangle-free selector encoding. Its identity is

variables	639290
clauses	1972360
bytes	408370088
SHA-256	09e1784c3f43c4901dc6f6b4749fc5a74b025b08f74be58133edd1ed1096ebdb

The production formula record and independent reconstruction audit have SHA-256 identities cb3eaa1ed462713e13a951dbc113555ce8f1aa8325c266658d9cc32c8dd72026 and 551e9188a4b7eb42bb40080acb0b2378ca6af69d412fb51f0c964ec2460190d2, respectively. A post-review rebuild retained these two record identities and the exact raw-CNF identity above.

The final generator, independent formula checker, and their 11-test regression have respective SHA-256 identities

generator: c07c76c5c6dde10474c10d48703270b83cff5a8865d223e7674342227dbafaf
4
checker: 9458ae9cdccc380efc1d083d6a6c06c14bd7a49ac728ca43c836ee5b62bacda2
tests: 2a85014c95de3e13e642ee91218cada552f4b4419c04c3e73a8b530465418df1

The tests include command-line regressions proving that authenticated input symlinks are not dereferenced before the non-symlink check.

One Sirius invocation used CaDiCaL 3.0.1 from source commit c60730422e758ef1cebe7aeddff2dda31c996bf04; the executable SHA-256 was bd054bcc5864fd20c9ff117f8fff94f810f5b210014ff33971a6c1db1d0eca45. The external wall was 300 seconds and the wrapper returned 124; the solver result was exactly c UNKNOWN. The incomplete DRAT prefix had 922038097 bytes and SHA-256 a6693fe2b3bab57f9d0e9ebe39bd7baa3883bfe2af15a5eb33b0aded972b5438. It was not replayed and was deleted after hashing. In particular, it is not a partial certificate and supplies no evidence toward unsatisfiability.

The sanitized endpoint record and its fail-closed audit have SHA-256 identities e35f7761f92ddebdb4029553fe7a2828e31c79d68820d37b1aa2bb6ac4f80571 and 4e4e212f0c6bae101db00dfb118f602cc6f874e9d8833f3c43cc946432e2cc54. The endpoint checker and its three-test regression have identities 7c6879bcd9c07ed27ebf25cb601a4ce8c69e5714178e35a9fb286c740b3c2ad0 and 83f545543792e22d9d825ae22ee9b00883172cf902a21d8b7be368cc80e736bd. They enforce the status MAXIMAL_UNION_TIMEOUT_UNKNOWN_NO_CUT; no branch or global conclusion is attached to this endpoint.

APPENDIX I. CAPPED SEARCH AND BATCH UNIFORMITY

We spell out the exact properties of the search primitive used in Lemma 27.2. Start from the uniform superposition on $[M]$ and use the randomized-iteration unknown-solution search of Boyer–Brassard–Høyer–Tapp [BBHT98]. If the marked density is at least λ , truncate the geometric iteration schedule once its maximum iteration count is a sufficiently large constant multiple of $1/\sqrt{\lambda}$. The standard analysis gives a success probability $p_0 > 0$ independent of M and λ , while the total number of membership calls is $O(1/\sqrt{\lambda})$. The truncation is fixed before the oracle outcomes are seen, so the block also terminates when there are no marked states.

For any fixed Grover iteration count, the amplitudes of all marked basis states are equal. Mixing over the iteration count preserves this permutation symmetry. Measure the candidate and evaluate the exact membership predicate; the block succeeds only when this verification accepts. Conditional on acceptance, every marked identity therefore has probability $1/|T|$. A new block reinitializes its registers and random choices, so accepted identities from distinct blocks are independent conditional on the block-success indicators.

For the batch statement, run

$$B = C(m + \log(1/\delta))$$

independent capped blocks, with C an absolute constant chosen using p_0 . The number of accepted blocks stochastically dominates $\text{Bin}(B, p_0)$. A Chernoff lower-tail bound makes the probability of fewer than m acceptances at most δ . Conditional on any success pattern with at least m acceptances, the first m accepted identities are independent and uniform on T . This proves Lemma 27.2.

APPENDIX J. DERIVATION OF THE RANDOMIZED CLASSICAL LOWER BOUND

We derive Proposition 30.2 from the random-Painter weight bound of Conlon–Fox–Grinshpun–He [CFGH20]. Let

$$\beta = 2 - \sqrt{2}, \quad \alpha = 1 - \frac{1}{\sqrt{2}},$$

so that $\beta = 2\alpha$. Set

$$T = \left\lfloor 2^{\beta K - 2} \right\rfloor, \quad c = \lfloor \alpha K \rfloor = \alpha K + \epsilon, \quad -1 < \epsilon \leq 0.$$

Expose $G(M, 1/2)$ adaptively to a deterministic T -query algorithm. The cited random-Painter estimate bounds the probability that the queried edges already contain a fully exposed monochromatic K -set by

$$2 \cdot 2^{-\binom{K}{2} + c(c-1)} (2T)^{K-c}. \tag{6}$$

Because $2T \leq 2^{\beta K-1}$, the base-2 logarithm of Equation (6) is at most

$$1 - \binom{K}{2} + c(c-1) + (K-c)(\beta K-1) = 1 - \frac{K}{2} + \epsilon^2.$$

For $K \geq 8$ this is at most -2 , so the event has probability at most $1/4$.

If the algorithm outputs a K -set that is not already a fully exposed monochromatic set, condition on the transcript. If the exposed internal edges contain both colours, the set cannot be homogeneous. If they contain exactly one colour, at least one unexposed fair edge must match that colour, so the conditional success probability is at most $1/2$. If no internal edge has been exposed, the probability is $2^{1-\binom{K}{2}} \leq 1/2$ for $K \geq 3$. Thus the conditional probability of a homogeneous output is at most $1/2$ in every case relevant here. The total success probability under $G(M, 1/2)$ is at most

$$\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8} < \frac{2}{3}.$$

Yao's minimax principle now implies that any randomized algorithm with worst-case success probability at least $2/3$ needs more than T queries. Since $M = 4^{K-1}$,

$$2^{\beta K} = \Theta(M^{\beta/2}) = \Theta(M^{1-1/\sqrt{2}}),$$

which proves Proposition 30.2.

APPENDIX K. MULTICOLOUR EXTENSION AND EXECUTABLE CHECKS

K.1. A multicolour extension of size-biased survival. The size-biased mechanism is not specific to two colours. Suppose every edge has one of $q \geq 2$ colours, and let

$$r = q(K-2) + 1, \quad B_d = 1 + q + \dots + q^{d-1} = \frac{q^d - 1}{q - 1}.$$

Choose the power of two

$$L_q = 2^{\lceil \log_2(2B_r) \rceil},$$

so $2B_r \leq L_q < 4B_r$. If the $s-1$ non-pivot vertices split into sizes $a_1, \dots, a_q$, the induction in Lemma 29.1 becomes

$$\sum_{j=1}^q (a_j - B_{d-1})_+ \geq (s-1 - qB_{d-1})_+ = (s - B_d)_+,$$

because $B_d = 1 + qB_{d-1}$. Thus, for $d \geq 1$, the d -step survival probability is at least $(s - B_d)_+ / (s-1)$ when $s \geq 2$. Starting from $L_q \geq 2B_r$ gives probability greater than $1/2$, and among the r recorded colours one occurs at least $K-1$ times.

Corollary K.1 (Multicolour size-biased Ramsey search). *For $q, K \geq 2$, every q -edge-coloured complete graph on at least L_q vertices admits a bounded-error quantum algorithm that outputs a monochromatic K -clique using*

$$O\left(r^2 \sqrt{L_q} \log(2r) \log \frac{1}{\eta}\right)$$

colour-oracle queries in the worst case, where r , B_r , and L_q are as defined above.

K.2. Executable audit. The standard-library script `experiments/quantum_ramsey/implicit_majority_audit.py` is invoked from the repository root by

```
python3 \
  experiments/quantum_ramsey/implicit_majority_audit.py \
  --self-check --max-k 12 --simulate-k 3 \
  --trials 1000 --seed 20260831
```

It verifies the dyadic error schedule, prefix-product inequality, and variable recurrence with exact rational arithmetic through $K = 64$. It also compares the size-biased survival lemma with an exact dynamic program over all split trees through nine levels, emits the Jain-et-al./Liu–Zhandry exponent translation, checks conditional marked-item uniformity by an explicit 16-dimensional Grover state-vector calculation, and runs deterministic-seed diagnostics on complete, empty, parity, fixed random, and independently resampled random graphs.

The end-to-end diagnostic replaces quantum search by an ideal uniform marked-item sampler; only the isolated capped-search primitive receives the small state-vector check. These checks test the combinatorial invariant, error schedule, and sampler symmetry. They are not a simulation of the full algorithm, a hardware experiment, or an empirical proof of quantum speedup.

Graph family on 16 vertices	failed/trials	minimum final $ S_3 $
complete	0/1000	13
empty	0/1000	13
parity	0/1000	6
one fixed $G(16, 1/2)$	0/1000	2
fresh $G(16, 1/2)$ per trial	0/1000	2

The isolated 16-state calculation mixed Grover iteration counts from zero through four. Across marked-set sizes 1, 2, 3, 4, 8, 15, its block-success probability ranged from 0.4026746749 to 0.5973253251, and the largest conditional deviation from the uniform distribution on marked items was zero to machine precision. For the exhaustive split-tree checks, the nine-level binary instance had exact worst survival 0.5086135566 versus the proved lower bound 513/1024, and the four-level ternary instance had exact worst survival 0.7155732457 versus 88/127. Zero observed failures in the table are a regression result, not an estimate that the failure probability is zero.

As a separate pointer for the hypergraph open problem in Section 7, higher-order symmetry representations have been useful in a different hypergraph algorithmic setting [ZXD24]. Whether such structure can be exposed with sublinear coherent queries in a Ramsey problem is open.

REFERENCES

- [BBHT98] Michel Boyer, Gilles Brassard, Peter Høyer, and Alain Tapp. Tight bounds on quantum searching. *Fortschritte der Physik*, 46(4–5):493–506, 1998.
- [BCM⁺13] Zhengbing Bian, Fabian Chudak, William G. Macready, Geordie Rose, and Frank Gaitan. Experimental determination of Ramsey numbers. *Physical Review Letters*, 111:130505, 2013.
- [BFF⁺24] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, Nils Froleyks, and Florian Pollitt. CaDiCaL 2.0. In *Computer Aided Verification—CAV 2024*, volume 14681 of *Lecture Notes in Computer Science*, pages 133–152. Springer, 2024.
- [BGSP12] Gunnar Brinkmann, Jan Goedgebeur, and Jan-Christoph Schlage-Puchta. Ramsey numbers $R(K_3, G)$ for graphs of order 10. *The Electronic Journal of Combinatorics*, 19(4):P36, 2012.
- [BMA24] Mathias Barreto, Olivier Marchal, and Julyan Arbel. Optimal sub-gaussian variance proxy for truncated gaussian and exponential random variables. *arXiv preprint*, 2024.
- [CFGH20] David Conlon, Jacob Fox, Andrey Grinshpun, and Xiaoyu He. Online Ramsey numbers and the subgraph query problem. In *Building Bridges II: Mathematics of László Lovász*, volume 28 of *Bolyai Society Mathematical Studies*, pages 159–194. Springer, 2020.
- [CGMS23] Marcelo Campos, Simon Griffiths, Robert Morris, and Julian Sahasrabudhe. An exponential improvement for diagonal Ramsey. *arXiv preprint*, 2023.
- [Dör05] Sebastian Dörn. Quantum complexity bounds for independent set problems. *arXiv preprint*, 2005.
- [Erd47] Paul Erdős. Some remarks on the theory of graphs. *Bulletin of the American Mathematical Society*, 53:292–294, 1947.
- [ES35] Paul Erdős and George Szekeres. A combinatorial problem in geometry. *Compositio Mathematica*, 2:463–470, 1935.
- [GC12] Frank Gaitan and Lane Clark. Ramsey numbers and adiabatic quantum computing. *Physical Review Letters*, 108:010501, 2012.
- [GNNW24] Parth Gupta, Ndiame Ndiaye, Sergey Norin, and Louis Wei. Optimizing the CGMS upper bound on Ramsey numbers. *arXiv preprint*, 2024.
- [Heu16] Marijn J. H. Heule. The DRAT format and DRAT-trim checker. *arXiv preprint*, 2016.
- [HMS25] Zach Hunter, Aleksa Milojević, and Benny Sudakov. Gaussian random graphs and Ramsey numbers. *arXiv preprint*, 2025.
- [IMaMS18] Alexey Ignatiev, António Morgado, and Jo ao Marques-Silva. PySAT: A Python toolkit for prototyping with SAT oracles. In *Theory and Applications of Satisfiability Testing—SAT 2018*, volume 10929 of *Lecture Notes in Computer Science*, pages 428–437. Springer, 2018.
- [JLRX24] Siddhartha Jain, Jiawei Li, Robert Robere, and Zhiyang Xun. On pigeon-hole principles and Ramsey in TFNP. In *65th Annual IEEE Symposium on Foundations of Computer Science*, 2024.
- [Joh17] Fredrik Johansson. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017.
- [KNY17] Ilan Komargodski, Moni Naor, and Eylon Yogev. White-box vs. black-box complexity of search problems: Ramsey and graph property testing. In *58th*

- Annual IEEE Symposium on Foundations of Computer Science*, 2017. ECCC Report TR17-015.
- [LMS12] Troy Lee, Frédéric Magniez, and Miklos Santha. A learning graph based quantum query algorithm for finding constant-size subgraphs. *Chicago Journal of Theoretical Computer Science*, 2012(10), 2012.
- [LN26] Qizhong Lin and Lin Niu. Sharper Ramsey lower bounds from refined gaussian estimates. *arXiv preprint*, 2026.
- [LZ19] Qipeng Liu and Mark Zhandry. On finding quantum multi-collisions. In *Advances in Cryptology—CRYPTO 2019*, volume 11692 of *Lecture Notes in Computer Science*. Springer, 2019.
- [Mon18] Ashley Montanaro. Quantum walk speedup of backtracking algorithms. *Theory of Computing*, 14(15):1–24, 2018.
- [MSX25] Jie Ma, Wujie Shen, and Shengjie Xie. An exponential improvement for Ramsey lower bounds. *arXiv preprint*, 2025.
- [NRT26] Ansh Nagda, Prabhakar Raghavan, and Abhradeep Thakurta. Reinforced generation of combinatorial structures: Ramsey numbers, 2026.
- [PM25] Joel E. Pion and Susan M. Mniszewski. Toward computing bounds for Ramsey numbers using quantum annealing. *Quantum Information Processing*, 24(7):221, 2025.
- [Rad26] Stanisław P. Radziszowski. Small Ramsey numbers. *The Electronic Journal of Combinatorics, Dynamic Survey DS1*, 2026. Revision 18, 24 April 2026.
- [Sin05] Carsten Sinz. Towards an optimal CNF encoding of Boolean cardinality constraints. In *Principles and Practice of Constraint Programming—CP 2005*, volume 3709 of *Lecture Notes in Computer Science*, pages 827–831. Springer, 2005.
- [Wan15] Hefeng Wang. Determine Ramsey numbers on a quantum computer. *arXiv preprint*, 2015.
- [WHH14] Nathan Wetzler, Marijn J. H. Heule, and Warren A. Hunt, Jr. DRAT-trim: Efficient checking and trimming using expressive clausal proofs. In *Theory and Applications of Satisfiability Testing—SAT 2014*, volume 8561 of *Lecture Notes in Computer Science*, pages 422–429. Springer, 2014.
- [Xin26] Cheng Xin. Reproducible ramsey analysis across asymptotic and finite regimes: Research software and proof certificates. Public GitHub source repository, 2026. Designated evidence tag evidence-2026-08-30; 26 proof, model, and checker assets are content-addressed by artifacts/MANIFEST.tsv.
- [YM26] Gang Yang and Yaping Mao. New upper bound for multicolor Ramsey numbers. *arXiv preprint*, 2026.
- [ZXD24] Simon Zhang, Cheng Xin, and Tamal K. Dey. Expressive higher-order link prediction through hypergraph symmetry breaking. *Transactions on Machine Learning Research*, November 2024.